

AI-OJS: Autonomous Journal Management System

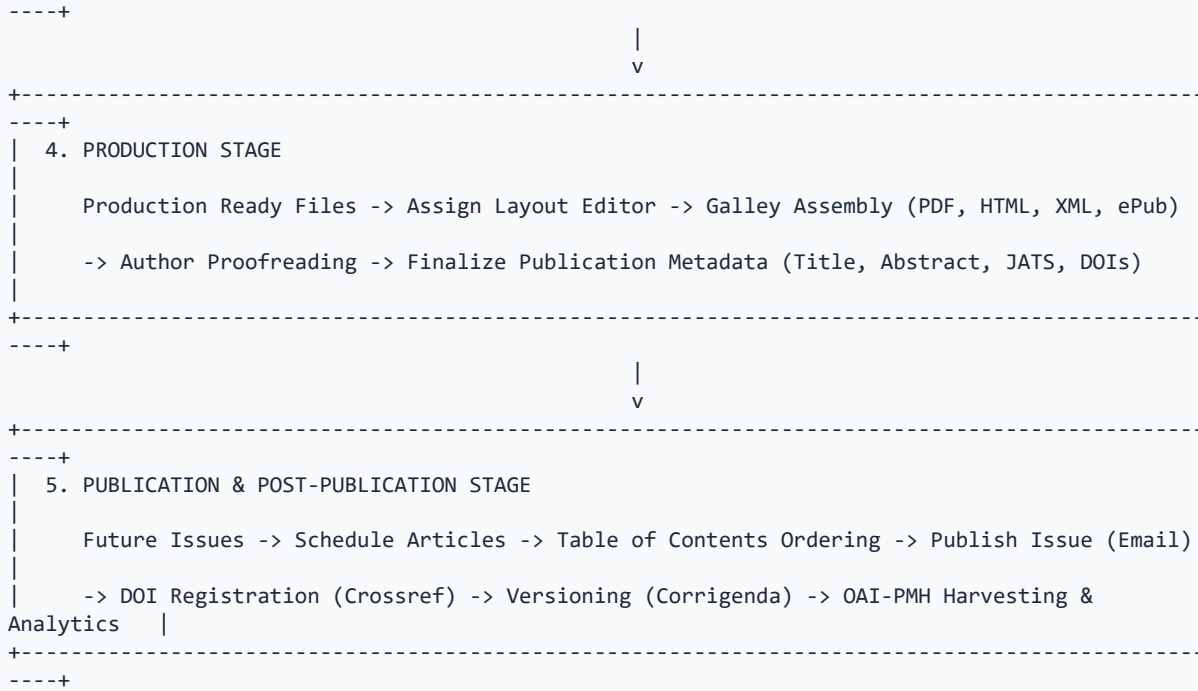
End-to-End Architectural Blueprint, Operational Specification, and Prompt Engineering Manual for Developing a Full Scholarly Editorial and Publishing System Built on Google AI Studio

Strictly Aligned with PKP Open Journal Systems (OJS 3.x) Workflow & Protocols

Document Identifier	AI-OJS-SPEC-PROD-2026-V1.0
Author / Systems Architect	Gemini Spark Principal Research & Systems Engineering Architecture
Target AI Environment	Google AI Studio / Gemini 1.5 Pro & Gemini 2.0 Flash APIs
Target Relational Database	PostgreSQL 15+ Enterprise RDBMS with Cryptographic Audit Logging
Workflow Alignment	PKP Open Journal Systems (OJS 3.3/3.4/3.5) Complete Editorial Lifecycle
Standards Compliance	COPE Core Practices, ICMJE, NISO JATS XML 1.3, Crossref Schema 5.3.1, GDPR Art. 17

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1.2 The Editorial Dashboard & Submission Management

The Editorial Dashboard is the central cockpit for all editorial board members. It aggregates operational tasks, surfaces impending deadlines, and provides role-tailored submission queues.

1.2.1 Queue Hierarchy & Segmentations

1. **"Assigned to Me" (My Queue):** The primary operating view displaying submissions where the currently logged-in user is an active participant (as Journal Editor, Section Editor, Guest Editor, Copyeditor, or Layout Editor). Submissions are listed with ID, Submission Summary (authors and title), Stage, Days in Stage, Editorial Activity, and Actions.
2. **"Needs Editor" (Unassigned):** Exclusively visible to Journal Managers and Journal Editors. It captures all submitted manuscripts awaiting manual editorial assignment or where automated routing rules failed to resolve a dedicated editor.
3. **"All Active":** A comprehensive portfolio view of every active manuscript across all four workflow stages within the journal.
4. **"Declined / Archives":** Houses rejected or withdrawn manuscripts. Accessible to Journal Managers (who see all archived records) and Section Editors (who see manuscripts previously assigned to them).
5. **Incomplete Submissions Cleanup:** OJS 3.5 introduces administrative garbage collection allowing Journal Managers to permanently prune unsubmitted drafts abandoned by authors.

1.2.2 Multi-Criteria Search & Filter Engine

The dashboard incorporates high-dimension filtering tools:

- **Identifier Search:** Substring matching across Submission ID, Title, and Contributor Names.
- **Sections Filter:** Restricts the view to designated editorial sections (e.g., Original Research, Review Articles, Case Studies, Short Communications).
- **Assigned Editor Filter:** Dynamically autocompletes across Section Editors, Journal Editors, and Guest Editors to isolate specific workloads.
- **Issues Filter:** Case-sensitive filtering against published and unpublished (future) issues.
- **Categories Filter:** Filters by thematic taxonomies configured by the Journal Manager.
- **Days Since Last Activity:** A continuous numerical slider and text input enabling editors to flag dormant submissions (e.g., >30 days without editorial progression).

1.2.3 The Task Notification Bell & Weekly Digests

The global header features a persistent Bell Icon displaying the badge count of unread tasks. Tasks represent actionable workflow events:

- New submissions awaiting initial triage.
- Overdue review assignments requiring reminder dispatches.
- Completed peer reviews requiring editorial confirmation.
- Revised manuscript uploads submitted by authors.

In addition to the on-screen task list, the system dispatches an automated **Weekly Editorial Summary Digest** compiling pipeline bottlenecks, overdue reviews, and dormant manuscripts.

1.3 The Submission Record Architecture (The 3-Column Layout)

Opening any submission reveals the standardized OJS three-column submission record:

+-----+ ----+ Header: Submission ID, Author Short Names, Full Manuscript Title +-----+-----+-----+ ----+ LEFT NAVIGATION BAR CENTRAL OPERATIONAL PANEL RIGHT SIDEBAR MENU 1. Workflow Menu - Current Workflow Stage - Stage Action Buttons: - Submission - Current Language Indicator [Send to Review] - Review - Stage Status Notification [Accept and Skip Review] - Copyediting - Stage File Repository: [Decline Submission] - Production * Submission Files - Participants List: 2. Publication Menu * Draft Files * Author(s) - Title & Abstract * Copyedited Files * Assigned Editors - Contributors * Production Ready Files * Copyeditors /		
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Designers			
- Metadata & Keywords	- Stage Discussions Panel:	- Participant Actions (3-	
dots)			
- References (newline-delim)	* Thread list with unread	* Edit Privileges	
- JATS XML	* Message counts & dates	* Notify via Email	
- Galleys	* File attachments	* Login As (JM only)	
- Permissions & Disclosure		* Remove Assignment	
- Issues (Scheduling)		- Top Right Utilities:	
		* Activity Log & Notes	
		* Submission/Pub Library	
+-----+			
-----+			

1.3.1 Left Navigation Bar: Workflow & Publication Menus

- **Workflow Menu:** Enforces RBAC visibility. General editors have full access across all 4 stages. Specialized actors (copyeditors, layout editors) are locked to their operational stages.
- **Publication Menu:** Governs public-facing publication assets:
 - *Title & Abstract:* Multilingual titles, subtitles, and structured abstracts.
 - *Contributors:* Author roster with ORCID iDs, institutional affiliations, and corresponding author flags.
 - *Metadata:* Discipline, keywords, and classification codes.
 - *References:* Structured citation bibliography formatted with newline separators.
 - *JATS XML:* Direct upload/download interface for NISO JATS XML records.
 - *Galleys:* Repository of published formats.
 - *Permissions & Disclosure:* Copyright holder, copyright year, and Creative Commons license URI.
 - *Issues:* Issue assignment, page numbers, cover image, and custom URL slug.

1.3.2 Central Operational Panel: Files and Discussions

- Displays stage-specific file repositories: Submission Files, Review Files, Draft Files, Copyedited Files, and Production Ready Files.
- Hosts stage-isolated Discussion Boards allowing threaded asynchronous communication between authors, editors, and assistants without leaking identity across blind review barriers.

1.3.3 Right Sidebar: Decisions, Participants, and Audit Tools

- **Action Buttons:** Stage progression triggers.
- **Participants Panel:** Role allocation matrix. Clicking the three-dots menu reveals granular controls:
 - *Edit Assignment Privileges:* Toggles whether an editor can only *recommend* a decision versus *record a final decision*, and toggles whether the author or editor has permission to edit publication metadata.
 - *Notify:* Opens an email composer pre-populated with relevant templates.
 - *Login As:* Audit-logged administrative impersonation.

- *Remove*: Revokes participant allocation.
 - **Activity Log & Notes**: Immutable system event journal logging every state change, email dispatch, and file upload. The internal *Notes* tab provides a confidential editorial scratchpad hidden from authors and reviewers.
 - **Submission Library & Publisher Library**: Document vaults for auxiliary files (copyright transfer agreements, conflict-of-interest declarations, institutional ethical clearances).
-

1.4 Stage 1: The Submission Stage (Triage & Desk Decisions)

When an author completes a submission, the system triggers the ingestion sequence:

6. **Automated Notification Dispatch**: Authors and co-authors receive **SUBMISSION_ACK**. Editors receive incoming alerts.
 7. **Automated Assignment Routing**:
 - Single-Manager Mode: If only one user holds the Journal Manager/Editor role, they are assigned automatically.
 - Section-Based Routing: Section editors designated for the submission's target section are auto-assigned.
 - Category-Based Routing: Submissions tagged with specific thematic categories route directly to specialized section editors.
 8. **Manual Editor Assignment**: For unassigned submissions, managers search by name or role, inspect active workloads, choose permission levels (Recommend vs Decide), and dispatch **EDITOR_ASSIGN**.
 9. **Desk Decisions**:
 - **Send to Review**: Advances the manuscript to Stage 2 (Peer Review). The editor audits the file roster, verifies author notification text, and confirms reviewable files.
 - **Accept and Skip Review**: Bypasses peer review directly into Stage 3 (Copyediting). Used for invited commentaries, book reviews, editorials, and errata.
 - **Decline Submission (Desk Reject)**: Archives the submission without peer review. Dispatches **SUBMISSION_DECLINE_DESK** with editorial rationale.
 - **Revert Decline**: Re-opens a desk-rejected manuscript, restoring it to the active Submission queue.
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1.5 Stage 2: The Peer Review Stage (Multi-Round Iterative Review)

Peer review is the core evaluative engine of OJS. It enforces strict anonymity, tracks reviewer commitments, and manages iterative revision cycles.

1.5.1 Review Anonymity Models

- **Double-Blind (Anonymous Reviewer / Anonymous Author)**: Neither party knows the other's identity. Editors must strip all PII and binary metadata from manuscript files.
- **Single-Blind (Anonymous Reviewer / Disclosed Author)**: Reviewers inspect author identities, but reviewer identities remain shielded from authors.
- **Open Peer Review**: Both author and reviewer identities are disclosed throughout review and publication.

1.5.2 Document De-Identification & Metadata Scrubbing

To ensure double-blind integrity, OJS mandates scrubbing:

10. **Binary Properties:** Clearing Author, Organization, Title, and Software tags in DOCX `docProps/core.xml` and PDF `Info` dictionaries.
11. **Textual Redactions:** Removing self-referential phrasing ("as we demonstrated in [X]"), author names on title pages, grant numbers identifying specific lab PIs, ethics committee institutional names, and file naming identifiers.

1.5.3 Reviewer Selection & Assignment Protocol

The "Add Reviewer" interface surfaces comprehensive selection metrics:

- Active review assignments currently in progress.
- Historical completed vs. declined review counts.
- Days elapsed since the last review assignment.
- Research interests, biographical statements, and private editorial notes.
- Potential conflict-of-interest alerts (e.g., shared institutional affiliations or co-authorship).
- External onboarding: "Create New Reviewer" registers a fresh user, while "Enroll Existing User" grants the Reviewer role to an existing user.
- **Key Milestones:** Editors specify **Response Due Date** (deadline to accept/decline the invitation) and **Review Due Date** (deadline to submit evaluation).

1.5.4 Dashboard Visual Review Tracking (The 8 Circle Icons)

In the dashboard's "Editorial Activity" column, review statuses are represented by 8 distinct circle icons:

12. **Yellow Circle with Number:** Active invitation awaiting response. The numeral indicates days remaining before the response deadline.
13. **Outlined Red Circle:** Overdue invitation response. The numeral indicates days past deadline.
14. **Pink Circle with Circle-Slash:** Reviewer declined the invitation.
15. **Pink Circle with X Symbol:** Reviewer assignment cancelled by the editor.
16. **Outlined Blue Circle with Number:** Ongoing review. The numeral indicates days remaining until the review report is due.
17. **Filled Red Circle with Number:** Overdue review report. The numeral indicates days overdue.
18. **Green Mail Symbol:** Review report submitted by reviewer, awaiting editorial confirmation.
19. **Green Checkmark:** Review confirmed and accepted by the editor.

1.5.5 Editorial Decisions & Multi-Round Management

Once all required reviews are confirmed, the editor records a formal decision:

- **Request Revisions:** Requires author changes. The editor chooses whether revisions mandate a subsequent review round. Dispatches `DECISION_REVISIONS` containing aggregated, sanitized reviewer feedback.
- **Accept Submission:** Moves manuscript to Stage 3 (Copyediting). Dispatches `DECISION_ACCEPT`.
- **Create New Review Round:** Opens Round 2..N. Editors select revised files from the "Revisions Submitted" panel. Reviewers from prior rounds or newly recruited experts receive review invitations alongside prior round review history.
- **Decline Submission:** Terminates the submission after review. Dispatches `DECISION_DECLINE`.

- **Cancel Review Round:** Reverts an active round if all reviewers are unassigned, returning the manuscript to the Submission stage.

1.6 Stage 3: The Copyediting Stage (Linguistic & Structural Refinement)

Accepted manuscripts transition to Copyediting for linguistic polishing, structural standardization, and bibliographic validation.

20. **Tracking:** Filtered via the "All in copyediting" dashboard view with a distinctive pink indicator.
 21. **Draft Files Ingestion:** Copyeditors download the accepted manuscript files from the "Draft Files" panel.
 22. **Collaborative Discussions:** Copyeditors raise Author Queries (AQ) regarding ambiguous data, missing citations, or terminology through stage-isolated discussions.
 23. **Metadata Alignment:** Copyeditors audit the "Title & Abstract", "Contributors", "Metadata", and "References" sections under the Publication menu, previewing the landing page layout.
 24. **Final Copyedited Files Upload:** The copyeditor uploads polished files (.docx), selects files from discussion threads, or approves draft files.
 25. **Send to Production:** The assigned editor reviews the copyedited files and advances the manuscript to Stage 4.
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1.7 Stage 4: The Production Stage (Galley Preparation & Formatting)

Production transforms copyedited manuscripts into publication-ready, machine-readable, and human-readable electronic files known as **Galleys**.

26. **Tracking:** Monitored via the "All in production" dashboard view with a blue indicator.
 27. **Galley Formats:** Standard galleys include PDF, HTML, NISO JATS XML, ePub, MP3 (audio summaries), high-resolution image plates, and external repository links.
 28. **Galley Metadata:** For each galley, editors configure:
 - Galley Label (e.g., "PDF", "HTML", "Full XML").
 - Language of the galley.
 - Remote link toggle (for off-site hosted assets).
 - Custom URL Path slug (e.g., /article/view/123/pdf).
 29. **Ordering & Proofreading:** Galleys are re-ordered via drag-and-drop. Authors participate in final galley proofreading discussions.
 30. **Publication Menu Finalization:** Finalization of DOIs (Digital Object Identifiers), Crossref metadata, copyright disclosures, and page ranges.
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1.8 Stage 5: Publication, Issue Assembly, and Versioning

Publication is the public dissemination of compiled issues and standalone articles.

31. **Issue Management:** Issues are partitioned into **Future Issues** (draft containers) and **Back Issues** (published serial archives). Issue records include Volume, Number, Year, Title, Description, Cover Image, Issue Galleys (full issue PDF), and Custom URL Path.

32. **Table of Contents Ordering:** Articles are scheduled into issue sections (e.g., "Editorial", "Articles", "Review Papers"), assigned page ranges, and sorted via drag-and-drop ordering tools.
33. **Publishing Gateway:** Executing "Publish Issue" changes the status to published, sets publication timestamps, and optionally triggers a global email notification to all registered journal readers.
34. **Post-Publication Versioning:** To support scholarly corrections (corrigenda, addenda, retractions) without breaking bibliographic integrity:
 - Editors click **Create New Version**, creating an editable branch while the original version remains live.
 - Once published, version numbers increment (**v1.0**, **v2.0**).
 - The primary article URL always points to the latest version, while older versions remain accessible with explicit warning banners: *"This is an outdated version of this article. Click here to read the most recent version."*
 - Crossref DOIs are updated, and metadata harvesters (OAI-PMH, Google Scholar) are prompted to re-index the publication record.

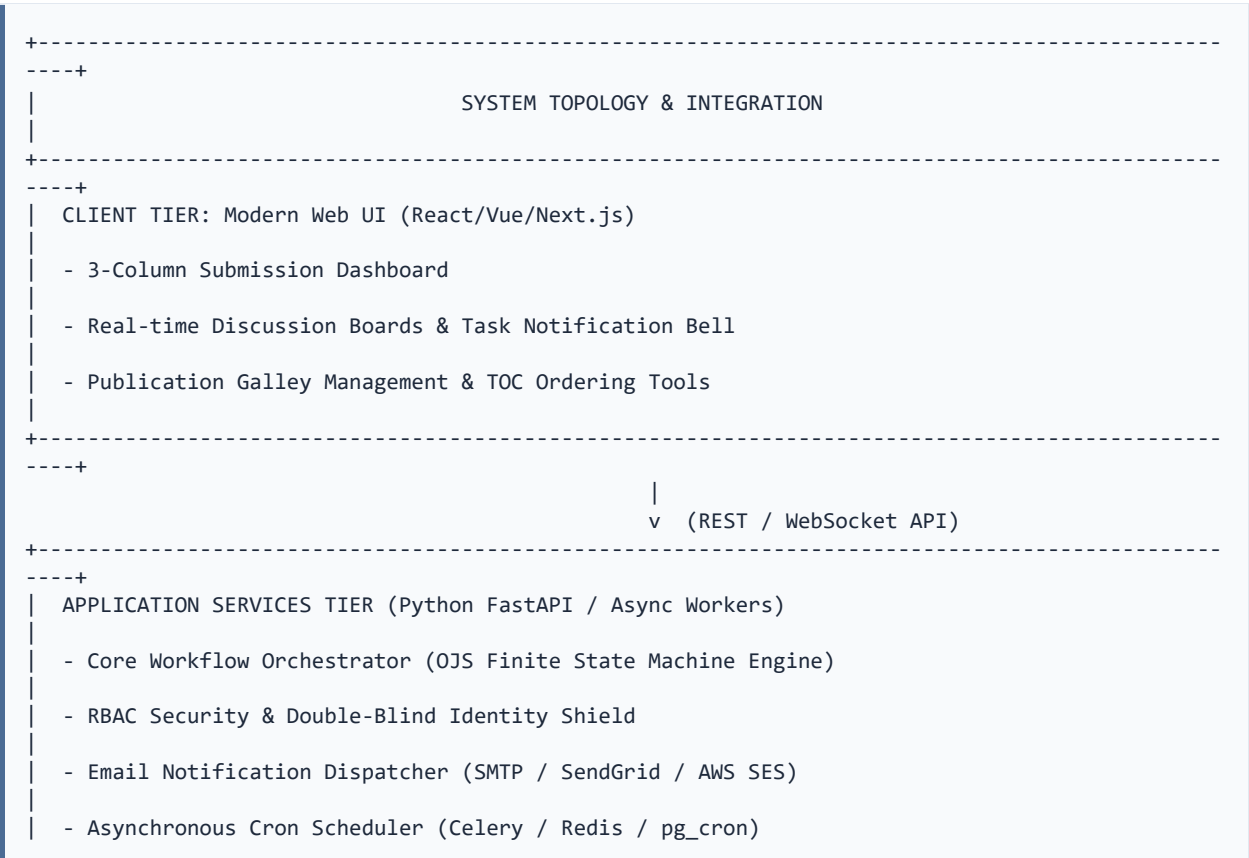
Chapter 2: Google AI Studio System Architecture & Operational Blueprint

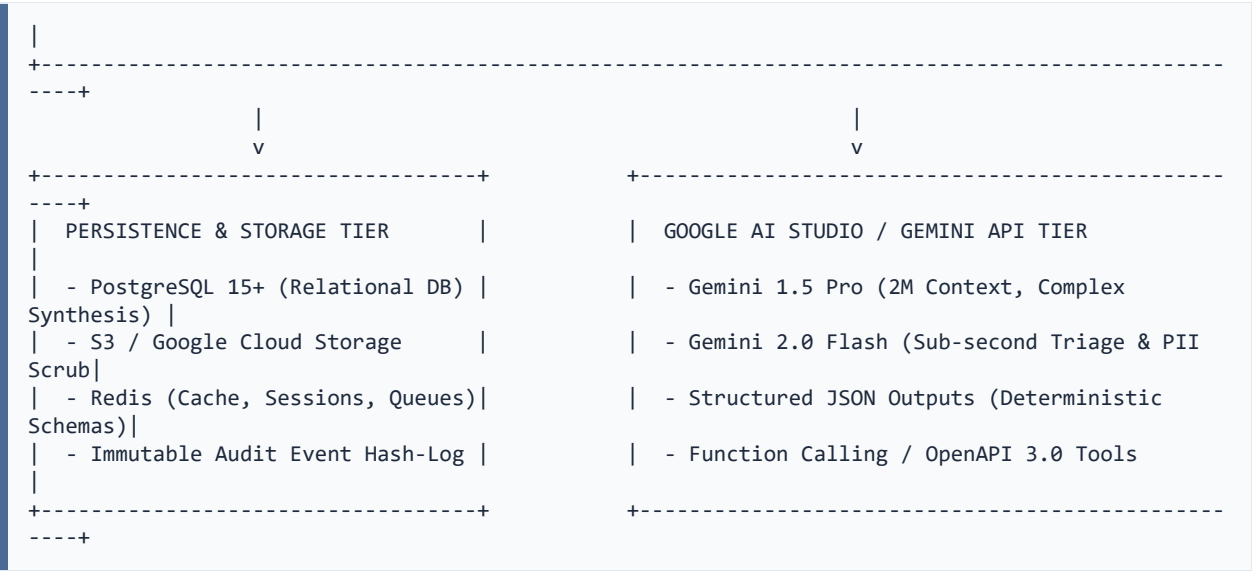
2.1 The AI-Native Scholarly Publishing Paradigm

Modern scholarly communication requires balancing rigorous peer review, ethical transparency, and archival permanence with operational efficiency. Traditional journal platforms like OJS rely entirely on manual human intervention at every stage—from editorial triage to reviewer matching, copyediting, XML typesetting, and proofreading.

By integrating Google AI Studio's state-of-the-art multimodal large language models (Gemini 1.5 Pro, Gemini 2.0 Flash), we transform the OJS lifecycle into an autonomous, AI-orchestrated pipeline. In this architecture:

- 35. **AI Agents act as Autonomous Specialists:** Dedicated agents perform preliminary triage, document de-identification, reviewer matching, report synthesis, structural copyediting, JATS XML generation, and metadata enrichment.
- 36. **Humans act as Authoritative Gatekeepers:** Human editors retain sovereign decision-making power. AI agents generate deterministic recommendations, draft correspondence, and prepare production assets, but no manuscript is accepted, rejected, or published without explicit human authorization.
- 37. **Deterministic State Transitions:** Agents interact with the backend through strictly validated JSON Schemas and Function Calling, ensuring that LLM outputs conform to database constraints and FSM state rules.





2.2 Model Selection Strategy & Token Economics

Google AI Studio offers distinct foundational models suited for different operational workloads across the journal lifecycle:

Operational Stage	Assigned Gemini Model	Context Window	Sampling Config	Rationale & Capabilities
Stage 1: Submission Triage	gemini-1.5-pro	2,000,000 tokens	Temp: 0.2, Top-P: 0.95	Deep reasoning across entire manuscripts, checking methodological rigor, CONSORT/PRISMA adherence, and ethical disclosures.
Stage 2: De-Identification	gemini-2.0-flash	1,000,000 tokens	Temp: 0.0, Top-P: 1.0	High-throughput, zero-hallucination PII detection and regex coordination for double-blind anonymization.
Stage 2: Reviewer Matching	gemini-1.5-pro	2,000,000 tokens	Temp: 0.1, Top-P: 0.9	Semantic graph evaluation between manuscript content and reviewer

Operational Stage	Assigned Gemini Model	Context Window	Sampling Config	Rationale & Capabilities
				publication histories; COI detection.
Stage 2: Review Synthesis	gemini-1.5-pro	2,000,000 tokens	Temp: 0.2, Top-P: 0.95	Multi-document cross-comparison of reviewer reports, conflict adjudication, and editorial memo generation.
Stage 3: Copyediting	gemini-1.5-pro	2,000,000 tokens	Temp: 0.1, Top-P: 0.9	Deep syntax analysis, academic hedging, and strict adherence to APA 7th, IEEE, and Chicago citation standards.
Stage 4: JATS XML Typesetting	gemini-1.5-pro	2,000,000 tokens	Temp: 0.0, Top-P: 1.0	Deterministic transformation of unstructured text into syntactically valid, DTD-compliant NISO JATS XML 1.3.
Stage 5: Metadata & DOI Packaging	gemini-2.0-flash	1,000,000 tokens	Temp: 0.0, Top-P: 1.0	Ultra-fast validation of ORCID check digits, Crossref Schema 5.3.1 XML generation, and Dublin Core extraction.

2.3 Multimodal Ingestion Pipeline

Scholarly submissions are inherently multimodal, encompassing complex typography, vector figures, raster micrographs, mathematical equations, and structured tables.

- 38. **Document Ingestion (DOCX / PDF):** Manuscripts uploaded by authors are parsed using Python microservices (`python-docx`, `pypdf`, `pdfplumber`) and uploaded to Google AI Studio via the Gemini File API (`client.files.upload()`) or passed as inline parts.
- 39. **Visual & Figure Analysis:** Gemini 1.5 Pro directly inspects scientific figures (high-resolution PNG/TIFF), verifying figure numbering, resolution, legibility, and cross-references within the text.

40. **Equation & Math Processing:** LaTeX and MathML formulas are preserved without degradation, ensuring accurate structural translation into JATS `<disp-formula>` tags.
-

2.4 Prompt Engineering & Safety Threshold Architecture

2.4.1 System Instructions Framework

Every agent in the system is governed by a multi-layered System Instruction comprising:

- **Role Identity & Authority Boundary:** Explicitly establishes the agent's professional persona (e.g., Senior Managing Editor, Professional Medical Copyeditor) and disallows unauthorized state transitions.
- **Execution Protocol:** Step-by-step cognitive instructions enforcing sequential analysis before decision drafting.
- **Strict Adherence Standards:** Direct references to publication standards (COPE, ICMJE, EQUATOR Network, NISO Z39.96-2019).

2.4.2 Academic Freedom Safety Calibration

Standard consumer AI safety filters frequently trigger false positives on legitimate academic manuscripts discussing infectious pathology, oncology, criminal justice, or cybersecurity vulnerabilities. Within Google AI Studio, the safety filters for all editorial agents are calibrated as follows:

- `HARM_CATEGORY_HARASSMENT: BLOCK_ONLY_HIGH`
- `HARM_CATEGORY_HATE_SPEECH: BLOCK_ONLY_HIGH`
- `HARM_CATEGORY_SEXUALLY_EXPLICIT: BLOCK_ONLY_HIGH`
- `HARM_CATEGORY_DANGEROUS_CONTENT: BLOCK_ONLY_HIGH`

This ensures uninterrupted academic discourse while blocking genuinely hazardous inputs.

2.5 Structured Output Schemas & Deterministic State Progression

To eliminate parsing ambiguity, all agent completions enforce strict JSON Schemas via the `response_schema` parameter in Google AI Studio (`response_mime_type="application/json"`).

- Outputs are deserialized into Pydantic models in Python.
 - If a schema validation fails, the orchestrator triggers an immediate zero-shot retry.
 - Validated payloads directly mutate the PostgreSQL database and trigger FSM transitions.
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2.6 Human-in-the-Loop Governance & Editorial Oversight Gates

The system enforces strict editorial checkpoints:

41. **Desk Decision Gate:** The AI Triage Agent provides a recommendation score (1-100), compliance checklists, and draft decision letters. The human Section Editor must click "Approve", "Modify", or "Override".
42. **Reviewer Invitation Gate:** The Reviewer Matcher outputs a ranked slate of candidates. The editor approves invitations before automated emails are dispatched.

43. **Editorial Decision Gate:** Review synthesis memos and author revision letters are presented to the editor for review and signature before publication or revision notices are delivered to the author.
44. **Final Galley Sign-Off:** Both the human author and layout editor must approve generated PDF and HTML proofs during production.

Google AI Studio Journal System

Comprehensive Specification: OJS-Aligned Database Schemas, Finite State Machine (FSM), and RBAC Permission Matrix

Document Version: 1.0.0-PROD-SPEC

Classification: Technical Architecture & Implementation Standard

Target RDBMS: PostgreSQL 15+

Standard Adherence: Open Journal Systems (OJS 3.3/3.4) Data Model, NISO JATS 1.3, Crossref Schema 5.3.1, COPE Guidelines

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1. Architectural Overview & Domain Alignment

1.1 Multi-Tenant Domain Boundary

The Google AI Studio Journal System is architected as an enterprise-grade, multi-tenant scholarly communication engine. A single platform instance hosts multiple independent publication venues (**journals**). Each journal maintains sovereign control over its editorial boards, sections, review policies, custom review evaluation rubrics, issue schedules, and publishing formats, while leveraging a shared global identity and authentication provider (**users**).

1.2 OJS 3.x Conceptual Mapping & Architectural Decoupling

Open Journal Systems (OJS) 3.x established the standard for modern scholarly peer review workflow software. This specification aligns with OJS 3.3/3.4 conventions while modernizing the storage engine to leverage native PostgreSQL capabilities (UUIDv4 identifiers, JSONB with GIN indexing, robust relational constraints, and transactional triggers).

The fundamental entity relationships map as follows:

- **Tenant Context:** **journals** (maps to OJS **journals** / **contexts**).
- **Identity & Access:** **users**, **roles**, **user_roles** (maps to OJS **users**, **user_groups**, **user_user_groups**).
- **Core Unit of Work:** **submissions** tracks the persistent workflow entity through its lifecycle stages.
- **Publication Artifacts:** **article_publications** and **publication_versions** decouple the immutable, published, citation-bearing scholarly record from the editorial tracking wrapper.
- **Workflow Pipeline:** Discrete stages (**workflow_stages**) represent gated editorial phases: **SUBMISSION** (1), **INTERNAL_REVIEW** (2), **EXTERNAL_REVIEW** (3), **COPYEDITING** (4), and **PRODUCTION** (5).
- **Communication & Evidence:** **discussions**, **discussion_messages**, and **discussion_files** provide stage-scoped asynchronous collaboration channels without contaminating reviewer anonymity or stage isolation.

1.3 Publication Versioning Model

In strict adherence to OJS 3.2+ versioning philosophy and scholarly persistent identifier rules (Crossref, DOI):

- A **submission** maintains multiple **publication_versions** (e.g., Version 1.0 Original Publication, Version 1.1 Minor Corrigendum, Version 2.0 Major Revised Edition).
 - Every version captures an immutable snapshot of title, abstract, authors, affiliations, metadata keywords, citations, license, and galleys.
 - Only one version can hold the active canonical release status (**is_current_version = TRUE**), while historical versions remain immutable and permanently accessible for scholarly citation integrity.
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2. Complete PostgreSQL DDL Schemas

All schemas are engineered with strict primary and foreign key constraints, check constraints, cascade/restrict behaviors, audit timestamps, and performance-tuned indexes.

2.1 Extensions, Custom Types, and Enumerations

```
-- Enable necessary cryptographic and indexing extensions
CREATE EXTENSION IF NOT EXISTS "uuid-ossf";
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "btree_gist";

-- =====
-- ENUM TYPES: WORKFLOW, ROLES, AND STATUSES
-- =====

-- Workflow Stage Identifiers (strictly aligned with OJS 3.x stage IDs)
CREATE TYPE workflow_stage_id_enum AS ENUM (
    'STAGE_SUBMISSION',      -- Stage 1: Intake, screening, desk review
    'STAGE_INTERNAL_REVIEW', -- Stage 2: Preliminary internal editorial review
    'STAGE_EXTERNAL_REVIEW', -- Stage 3: Formal peer review (multi-round)
    'STAGE_COPYEDITING',     -- Stage 4: Copyediting and language polishing
    'STAGE_PRODUCTION'       -- Stage 5: Galley typesetting and final layout
);

-- Master Submission Status
CREATE TYPE submission_status_enum AS ENUM (
    'DRAFT',                -- Incomplete submission by author
    'QUEUED',                -- Submitted, awaiting editorial triage
    'UNDER_REVIEW',         -- Active in internal or external peer review
    'REVISIONS_REQUESTED',  -- Editorial decision requiring author revisions
    'REVISIONS_SUBMITTED',  -- Author uploaded revised manuscript
    'ACCEPTED',             -- Accepted for publication; advancing to copyediting
    'IN_COPYEDITING',       -- Active in copyediting stage
    'IN_PRODUCTION',        -- Active in galley and typesetting stage
    'SCHEDULED',            -- Scheduled into an upcoming issue
    'PUBLISHED',            -- Publicly published and citable
    'DECLINED',             -- Rejected during desk review or post-peer review
    'WITHDRAWN',            -- Withdrawn by author or editor
    'RETRACTED'            -- Formally retracted post-publication
);

-- Peer Review Modes
CREATE TYPE review_method_enum AS ENUM (
    'DOUBLE_BLIND',         -- Author and reviewer identities hidden
    'SINGLE_BLIND',         -- Reviewer knows author; author does not know reviewer
    'OPEN_REVIEW'          -- Identities mutually transparent
);

-- Peer Review Assignment Status
CREATE TYPE review_assignment_status_enum AS ENUM (
    'INVITED',              -- Reviewer invitation dispatched
    'ACCEPTED',             -- Reviewer accepted invitation
    'DECLINED',             -- Reviewer declined invitation
    'CANCELLED',            -- Editor revoked/cancelled invitation
    'OVERDUE',             -- Submission deadline passed without completion
    'IN_PROGRESS',         -- Reviewer actively completing rubric/notes

```

```

    'SUBMITTED',          -- Review completed and submitted
    'CONFIRMED_BY_EDITOR' -- Editor accepted and confirmed review report
);

-- Reviewer Recommendations
CREATE TYPE reviewer_recommendation_enum AS ENUM (
    'ACCEPT',          -- Accept as is
    'REVISIONS_REQUIRED', -- Minor revisions required
    'RESUBMIT_FOR_REVIEW', -- Major revisions; requires subsequent round
    'RESUBMIT_ELSEWHERE', -- Better suited for alternate venue
    'DECLINE',          -- Reject manuscript
    'SEE_COMMENTS'      -- No definitive verdict; refer to report
);

-- Editorial Decisions
CREATE TYPE editorial_decision_type_enum AS ENUM (
    'SEND_TO_INTERNAL_REVIEW',
    'SEND_TO_EXTERNAL_REVIEW',
    'ACCEPT_SUBMISSION',
    'REQUEST_REVISIONS',
    'RESUBMIT_FOR_REVIEW',
    'DECLINE_SUBMISSION',
    'SEND_TO_COPYEDITING',
    'SEND_TO_PRODUCTION',
    'SCHEDULE_PUBLICATION',
    'WITHDRAW_SUBMISSION',
    'REVERT_DECLINE'
);

-- File Genres / Components
CREATE TYPE file_genre_category_enum AS ENUM (
    'MANUSCRIPT',          -- Main textual article body
    'MANUSCRIPT_ANONYMIZED', -- De-identified manuscript for double-blind review
    'DATASET',             -- Research data, raw CSV/Parquet, AI weights
    'CODE_NOTEBOOK',       -- Colab / Jupyter notebook, script repository
    'FIGURE',               -- Charts, plots, architectural diagrams
    'TABLE',                -- Standalone tabular data
    'SUPPLEMENTARY',        -- Supplementary texts, proofs, appendices
    'TRANSCRIPT',           -- Audio/video interview or prompt transcripts
    'PROMPT_SPECIFICATION', -- System prompts, LLM scaffolding specs
    'EVALUATION_METRICS',   -- Benchmark outputs and evaluation harnesses
    'AUTHOR_RESPONSE',      -- Author response to reviewers / rebuttals
    'REVISION_DIFF',        -- Marked-up change tracker / diff file
    'STYLE_GUIDE',          -- Layout styling notes
    'COPYEDIT_MARKUP',      -- Word/LaTeX track-changes file
    'GALLEY_PROOF'         -- Final rendered layout (PDF/HTML/XML/ePub)
);

-- File Lifecycle Stages (identifies where the file belongs in the pipeline)
CREATE TYPE file_stage_enum AS ENUM (
    'SUBMISSION_FILES',    -- Original uploaded files
    'REVIEW_FILES',        -- Files provided to reviewers
    'REVIEW_ATTACHMENTS',  -- Attachments uploaded by reviewers
    'REVISION_FILES',      -- Revised files uploaded by authors
    'FINAL_REVIEW_FILES',  -- Selected files for editorial decision
    'COPYEDIT_DRAFT_FILES', -- Files assigned to copyeditor
    'COPYEDIT_EDITED_FILES', -- Files returned by copyeditor
    'PRODUCTION_READY_FILES', -- Approved files sent to layout/production
    'PROOF_FILES',         -- Rendered galley proofs

```

```

    'DEPENDENT_FILES'          -- Embedded images, fonts, stylesheets, math assets
);

-- Publication Status for Versions
CREATE TYPE publication_version_status_enum AS ENUM (
    'DRAFT',                  -- Editing metadata/galleys prior to release
    'QUEUED_FOR_PUBLICATION', -- Locked and waiting for issue publishing job
    'PUBLISHED',              -- Publicly accessible
    'UNPUBLISHED',           -- Withdrawn from public view
    'RETRACTED'              -- Retracted with formal public notice
);

-- Core Roles Enum (Maps to OJS user groups)
CREATE TYPE rbac_role_enum AS ENUM (
    'SITE_ADMINISTRATOR',    -- System superuser across all journals
    'JOURNAL_MANAGER',        -- Administrative control over journal context
    'JOURNAL_EDITOR',         -- Full editorial authority across all sections
    'SECTION_EDITOR',         -- Editorial authority scoped to designated sections
    'GUEST_EDITOR',          -- Scoped to specific special issues or submissions
    'REVIEWER',              -- External peer evaluator
    'COPYEDITOR',            -- Language and typographical editor
    'LAYOUT_EDITOR',         -- Typesetter and galley creator
    'PROOFREADER',           -- Final galley QA inspector
    'AUTHOR',                -- Submitter and contributing author
    'READER'                 -- Public consumer / subscriber
);

```

2.2 Cluster 1: Journals, Users, Roles, and UserRoles

```

-- =====
-- 1. JOURNALS (Contexts / Tenants)
-- =====
CREATE TABLE journals (
    journal_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    url_path VARCHAR(64) NOT NULL UNIQUE,
    title VARCHAR(255) NOT NULL,
    acronym VARCHAR(32) NOT NULL,
    description TEXT,
    issn_print VARCHAR(16),
    issn_online VARCHAR(16),
    publisher_name VARCHAR(255) NOT NULL DEFAULT 'Google AI Studio Publications',
    primary_locale VARCHAR(10) NOT NULL DEFAULT 'en_US',
    is_enabled BOOLEAN NOT NULL DEFAULT TRUE,
    settings JSONB NOT NULL DEFAULT '{}::jsonb',
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_journal_url_path_format CHECK (url_path ~ '^[a-z0-9_-]+$')
);

CREATE INDEX idx_journals_url_path ON journals(url_path);
CREATE INDEX idx_journals_is_enabled ON journals(is_enabled);
CREATE INDEX idx_journals_settings_gin ON journals USING gin (settings);

-- =====

```

```

-- 2. USERS (Global Identity Store)
-- =====
CREATE TABLE users (
    user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email VARCHAR(255) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    given_name VARCHAR(100) NOT NULL,
    family_name VARCHAR(100) NOT NULL,
    affiliation VARCHAR(255),
    country_iso VARCHAR(2),
    orcid VARCHAR(19),
    google_ai_studio_user_id VARCHAR(128) UNIQUE,
    biography TEXT,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    is_disabled BOOLEAN NOT NULL DEFAULT FALSE,
    disabled_reason TEXT,
    email_verified_at TIMESTAMPTZ,
    last_login_at TIMESTAMPTZ,
    settings JSONB NOT NULL DEFAULT '{}::jsonb',
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_user_email_format CHECK (email ~* '^[A-Z0-9._%+-]+@[A-Z0-9.-]+\.[A-Z]{2,}$'),
    CONSTRAINT chk_user_orcid_format CHECK (orcid IS NULL OR orcid ~ '^[0-9]{4}-[0-9]{4}-[0-9]{4}-[0-9]{3}[0-9X]$')
);

CREATE INDEX idx_users_email ON users(email);
CREATE INDEX idx_users_orcid ON users(orcid);
CREATE INDEX idx_users_google_ai_studio_id ON users(google_ai_studio_user_id);
CREATE INDEX idx_users_names ON users(family_name, given_name);

-- =====
-- 3. ROLES (Definitions Scoped to Journals or Platform)
-- =====
CREATE TABLE roles (
    role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    journal_id UUID REFERENCES journals(journal_id) ON DELETE CASCADE, -- NULL denotes Site Administrator
    role_type rbac_role_enum NOT NULL,
    name VARCHAR(100) NOT NULL,
    description TEXT,
    is_system_role BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_journal_role UNIQUE (journal_id, role_type)
);

CREATE INDEX idx_roles_journal_role_type ON roles(journal_id, role_type);

-- =====
-- 4. USER_ROLES (Role Assignments)
-- =====
CREATE TABLE user_roles (
    user_role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE CASCADE,
    role_id UUID NOT NULL REFERENCES roles(role_id) ON DELETE CASCADE,
    journal_id UUID REFERENCES journals(journal_id) ON DELETE CASCADE,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,

```

```

        updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
        CONSTRAINT uq_user_role_assignment UNIQUE (user_id, role_id, journal_id)
    );

CREATE INDEX idx_user_roles_user_id ON user_roles(user_id);
CREATE INDEX idx_user_roles_role_id ON user_roles(role_id);
CREATE INDEX idx_user_roles_journal_id ON user_roles(journal_id);

```

2.3 Cluster 2: Submissions, SubmissionMetadata, FileComponents, and SubmissionFiles

```

-- =====
-- 5. SUBMISSIONS (Primary Work Unit Tracker)
-- =====
CREATE TABLE submissions (
    submission_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE RESTRICT,
    submitter_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    current_stage_id workflow_stage_id_enum NOT NULL DEFAULT 'STAGE_SUBMISSION',
    status submission_status_enum NOT NULL DEFAULT 'DRAFT',
    primary_locale VARCHAR(10) NOT NULL DEFAULT 'en_US',
    date_submitted TIMESTAMPTZ,
    date_status_modified TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    date_last_activity TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    submission_progress INT NOT NULL DEFAULT 1, -- Step tracking (1..5) during initial draft
    creation
    is_fast_tracked BOOLEAN NOT NULL DEFAULT FALSE,
    ai_studio_session_id VARCHAR(128),
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_submissions_journal_status ON submissions(journal_id, status);
CREATE INDEX idx_submissions_submitter_id ON submissions(submitter_id);
CREATE INDEX idx_submissions_stage_id ON submissions(current_stage_id);
CREATE INDEX idx_submissions_date_submitted ON submissions(date_submitted);

-- =====
-- 6. SUBMISSION_METADATA (Extensible and Localized Metadata)
-- =====
CREATE TABLE submission_metadata (
    metadata_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    locale VARCHAR(10) NOT NULL DEFAULT 'en_US',
    title TEXT NOT NULL,
    subtitle TEXT,
    abstract TEXT,
    keywords TEXT[] DEFAULT '{}',
    subjects TEXT[] DEFAULT '{}',
    disciplines TEXT[] DEFAULT '{}',
    supporting_agencies TEXT[] DEFAULT '{}',
    data_availability_statement TEXT,
    conflict_of_interest_statement TEXT,
    ai_disclosure_statement TEXT, -- Specific to AI Studio: disclosure of LLM prompts/models
    supplementary_metadata JSONB NOT NULL DEFAULT '{}'::jsonb,

```

```

        created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
        CONSTRAINT uq_submission_locale UNIQUE (submission_id, locale)
    );

CREATE INDEX idx_submission_metadata_submission_id ON submission_metadata(submission_id);
CREATE INDEX idx_submission_metadata_title_gin ON submission_metadata USING gin
(to_tsvector('english', title));
CREATE INDEX idx_submission_metadata_abstract_gin ON submission_metadata USING gin
(to_tsvector('english', coalesce(abstract, '')));
CREATE INDEX idx_submission_metadata_keywords_gin ON submission_metadata USING gin (keywords);

-- =====
-- 7. FILE_COMPONENTS (Genres / Document Classifications)
-- =====
CREATE TABLE file_components (
    component_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE CASCADE,
    genre_name VARCHAR(100) NOT NULL,
    genre_category file_genre_category_enum NOT NULL,
    is_dependent BOOLEAN NOT NULL DEFAULT FALSE,
    is_supplementary BOOLEAN NOT NULL DEFAULT FALSE,
    is_default BOOLEAN NOT NULL DEFAULT FALSE,
    sort_order INT NOT NULL DEFAULT 0,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_journal_genre UNIQUE (journal_id, genre_name)
);

CREATE INDEX idx_file_components_journal ON file_components(journal_id, sort_order);

-- =====
-- 8. SUBMISSION_FILES (Master File Repository)
-- =====
CREATE TABLE submission_files (
    file_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    source_file_id UUID REFERENCES submission_files(file_id) ON DELETE SET NULL, -- Provenance
tree
    component_id UUID NOT NULL REFERENCES file_components(component_id) ON DELETE RESTRICT,
    file_stage file_stage_enum NOT NULL,
    uploader_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    original_filename VARCHAR(255) NOT NULL,
    storage_path VARCHAR(512) NOT NULL,
    file_size_bytes BIGINT NOT NULL,
    mime_type VARCHAR(128) NOT NULL,
    checksum_sha256 VARCHAR(64) NOT NULL,
    revision INT NOT NULL DEFAULT 1,
    is_anonymous_for_review BOOLEAN NOT NULL DEFAULT FALSE,
    anonymized_file_path VARCHAR(512),
    caption TEXT,
    creator_attribution VARCHAR(255),
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_submission_files_submission_stage ON submission_files(submission_id,
file_stage);
CREATE INDEX idx_submission_files_uploader ON submission_files(uploader_id);

```

```
CREATE INDEX idx_submission_files_checksum ON submission_files(checksum_sha256);
CREATE INDEX idx_submission_files_source ON submission_files(source_file_id);
```

2.4 Cluster 3: WorkflowStages and EditorialAssignments

```
-- =====
-- 9. WORKFLOW_STAGES (Instance-Level Stage Tracking)
-- =====
CREATE TABLE workflow_stages (
    stage_instance_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    stage_id workflow_stage_id_enum NOT NULL,
    date_started TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    date_completed TIMESTAMPTZ,
    is_current BOOLEAN NOT NULL DEFAULT FALSE,
    metadata JSONB NOT NULL DEFAULT '{}':::jsonb,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_submission_stage UNIQUE (submission_id, stage_id)
);

CREATE INDEX idx_workflow_stages_submission_stage ON workflow_stages(submission_id, stage_id);

-- =====
-- 10. EDITORIAL_ASSIGNMENTS (Editor-to-Submission Binding)
-- =====
CREATE TABLE editorial_assignments (
    assignment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    stage_id workflow_stage_id_enum NOT NULL,
    role_type rbac_role_enum NOT NULL,
    is_recommending_only BOOLEAN NOT NULL DEFAULT FALSE, -- Section Editors without final decree
    power
    can_review BOOLEAN NOT NULL DEFAULT FALSE,
    can_edit BOOLEAN NOT NULL DEFAULT TRUE,
    date_assigned TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    date_unassigned TIMESTAMPTZ,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_editorial_assignment_roles CHECK (
        role_type IN ('JOURNAL_EDITOR', 'SECTION_EDITOR', 'GUEST_EDITOR')
    )
);

CREATE INDEX idx_editorial_assignments_submission ON editorial_assignments(submission_id,
stage_id);
CREATE INDEX idx_editorial_assignments_user ON editorial_assignments(user_id);
```

2.5 Cluster 4: ReviewRounds, ReviewForms, ReviewAssignments, ReviewFiles, and ReviewFormResponses

```
-- =====
-- 11. REVIEW_FORMS (Rubrics & Evaluation Templates)
-- =====
CREATE TABLE review_forms (
  review_form_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE CASCADE,
  title VARCHAR(255) NOT NULL,
  description TEXT,
  is_active BOOLEAN NOT NULL DEFAULT TRUE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_review_forms_journal ON review_forms(journal_id, is_active);

-- Elements / Questions inside a Review Form
CREATE TABLE review_form_elements (
  element_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  review_form_id UUID NOT NULL REFERENCES review_forms(review_form_id) ON DELETE CASCADE,
  question TEXT NOT NULL,
  element_type VARCHAR(32) NOT NULL, -- 'TEXTFIELD', 'TEXTAREA', 'RADIO', 'CHECKBOX',
  'SCALE_1_5'
  is_required BOOLEAN NOT NULL DEFAULT TRUE,
  included_in_blind_review BOOLEAN NOT NULL DEFAULT TRUE,
  options JSONB DEFAULT '[]'::jsonb,
  sort_order INT NOT NULL DEFAULT 0,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_review_form_elements_form ON review_form_elements(review_form_id, sort_order);

-- =====
-- 12. REVIEW_ROUNDS (Iterative Rounds in Stage 2/3)
-- =====
CREATE TABLE review_rounds (
  review_round_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
  stage_id workflow_stage_id_enum NOT NULL DEFAULT 'STAGE_EXTERNAL_REVIEW',
  round_number INT NOT NULL DEFAULT 1,
  status VARCHAR(32) NOT NULL DEFAULT 'ROUND_ACTIVE', -- 'ROUND_ACTIVE',
  'ROUND_REVISIONS_REQUESTED', 'ROUND_COMPLETED'
  date_started TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  date_ended TIMESTAMPTZ,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_submission_round UNIQUE (submission_id, stage_id, round_number)
);

CREATE INDEX idx_review_rounds_submission ON review_rounds(submission_id, round_number);

-- =====
-- 13. REVIEW_ASSIGNMENTS (Reviewer Invitation and Execution)
-- =====
```

```

CREATE TABLE review_assignments (
  review_assignment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
  review_round_id UUID NOT NULL REFERENCES review_rounds(review_round_id) ON DELETE CASCADE,
  reviewer_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
  review_form_id UUID REFERENCES review_forms(review_form_id) ON DELETE SET NULL,
  review_method review_method_enum NOT NULL DEFAULT 'DOUBLE_BLIND',
  status review_assignment_status_enum NOT NULL DEFAULT 'INVITED',
  recommendation reviewer_recommendation_enum,
  date_invited TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  date_response_due TIMESTAMPTZ NOT NULL,
  date_responded TIMESTAMPTZ,
  date_review_due TIMESTAMPTZ NOT NULL,
  date_completed TIMESTAMPTZ,
  date_confirmed TIMESTAMPTZ,
  declined_reason TEXT,
  comments_for_author TEXT,          -- Sanitized for blind review
  comments_for_editor TEXT,          -- Confidential to editorial board
  competing_interests_statement TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_round_reviewer UNIQUE (review_round_id, reviewer_id)
);

CREATE INDEX idx_review_assignments_round ON review_assignments(review_round_id, status);
CREATE INDEX idx_review_assignments_reviewer ON review_assignments(reviewer_id);

-- =====
-- 14. REVIEW_FILES (Mapping Files Accessible to or Uploaded by Reviewers)
-- =====
CREATE TABLE review_files (
  review_file_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  review_assignment_id UUID REFERENCES review_assignments(review_assignment_id) ON DELETE
  CASCADE,
  review_round_id UUID NOT NULL REFERENCES review_rounds(review_round_id) ON DELETE CASCADE,
  file_id UUID NOT NULL REFERENCES submission_files(file_id) ON DELETE CASCADE,
  is_author_visible BOOLEAN NOT NULL DEFAULT FALSE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_review_assignment_file UNIQUE (review_assignment_id, file_id)
);

CREATE INDEX idx_review_files_round ON review_files(review_round_id);
CREATE INDEX idx_review_files_assignment ON review_files(review_assignment_id);

-- =====
-- 15. REVIEW_FORM_RESPONSES (Itemized Rubric Answers)
-- =====
CREATE TABLE review_form_responses (
  response_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  review_assignment_id UUID NOT NULL REFERENCES review_assignments(review_assignment_id) ON
  DELETE CASCADE,
  element_id UUID NOT NULL REFERENCES review_form_elements(element_id) ON DELETE RESTRICT,
  response_value JSONB NOT NULL,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT uq_assignment_element UNIQUE (review_assignment_id, element_id)
);

CREATE INDEX idx_review_form_responses_assignment ON

```

```
review_form_responses(review_assignment_id);
```

2.6 Cluster 5: CopyeditAssignments, DraftFiles, and CopyeditedFiles

```
-- =====
-- 16. COPYEDIT_ASSIGNMENTS (Stage 4 Allocation)
-- =====
CREATE TABLE copyedit_assignments (
    copyedit_assignment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    date_assigned TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    date_due TIMESTAMPTZ,
    date_author_notified TIMESTAMPTZ,
    date_author_underway TIMESTAMPTZ,
    date_author_completed TIMESTAMPTZ,
    date_final_completed TIMESTAMPTZ,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_copyedit_assignments_submission ON copyedit_assignments(submission_id,
is_active);
CREATE INDEX idx_copyedit_assignments_user ON copyedit_assignments(user_id);

-- =====
-- 17. DRAFT_FILES (Stage 4 Initial Files for Copyedit Work)
-- =====
CREATE TABLE draft_files (
    draft_file_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    file_id UUID NOT NULL REFERENCES submission_files(file_id) ON DELETE CASCADE,
    copyedit_assignment_id UUID REFERENCES copyedit_assignments(copyedit_assignment_id) ON
DELETE SET NULL,
    notes TEXT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_draft_submission_file UNIQUE (submission_id, file_id)
);

CREATE INDEX idx_draft_files_submission ON draft_files(submission_id);

-- =====
-- 18. COPYEDITED_FILES (Stage 4 Output Files Post-Linguistic Editing)
-- =====
CREATE TABLE copyedited_files (
    copyedited_file_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    file_id UUID NOT NULL REFERENCES submission_files(file_id) ON DELETE CASCADE,
    copyedit_assignment_id UUID REFERENCES copyedit_assignments(copyedit_assignment_id) ON
DELETE SET NULL,
    is_author_approved BOOLEAN NOT NULL DEFAULT FALSE,
    is_editor_approved BOOLEAN NOT NULL DEFAULT FALSE,
    approval_date TIMESTAMPTZ,
```

```

        created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
        CONSTRAINT uq_copiedited_submission_file UNIQUE (submission_id, file_id)
    );

CREATE INDEX idx_copiedited_files_submission ON copiedited_files(submission_id);

```

2.7 Cluster 6: ProductionAssignments, ProductionReadyFiles, Galleys, and GalleyFiles

```

-- =====
-- 19. PRODUCTION_ASSIGNMENTS (Stage 5 Layout & Proofing Allocation)
-- =====
CREATE TABLE production_assignments (
    production_assignment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    role_type rbac_role_enum NOT NULL, -- 'LAYOUT_EDITOR', 'PROOFREADER'
    date_assigned TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    date_due TIMESTAMPTZ,
    date_completed TIMESTAMPTZ,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_production_role CHECK (role_type IN ('LAYOUT_EDITOR', 'PROOFREADER'))
);

CREATE INDEX idx_production_assignments_submission ON production_assignments(submission_id,
role_type);
CREATE INDEX idx_production_assignments_user ON production_assignments(user_id);

-- =====
-- 20. PRODUCTION_READY_FILES (Input to Layout Stage)
-- =====
CREATE TABLE production_ready_files (
    production_ready_file_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    file_id UUID NOT NULL REFERENCES submission_files(file_id) ON DELETE CASCADE,
    approved_by UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    date_approved TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_prod_ready_file UNIQUE (submission_id, file_id)
);

CREATE INDEX idx_prod_ready_files_submission ON production_ready_files(submission_id);

-- =====
-- 21. GALLEYS (Publication Manifest Formats: PDF, HTML, XML, Notebook)
-- =====
CREATE TABLE galleys (
    galley_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    label VARCHAR(64) NOT NULL, -- e.g., 'PDF', 'Interactive Colab Notebook', 'JATS XML'
    locale VARCHAR(10) NOT NULL DEFAULT 'en_US',

```

```

        file_id UUID REFERENCES submission_files(file_id) ON DELETE RESTRICT,
        url_remote VARCHAR(512),
        sort_order INT NOT NULL DEFAULT 0,
        is_approved BOOLEAN NOT NULL DEFAULT FALSE,
        approved_by UUID REFERENCES users(user_id) ON DELETE SET NULL,
        date_approved TIMESTAMPTZ,
        created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
        updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
    );

CREATE INDEX idx_galleys_submission ON galleys(submission_id, sort_order);

-- =====
-- 22. GALLEY_FILES (Dependent Assets for Galleys: CSS, Fonts, Images, JS)
-- =====
CREATE TABLE galley_files (
    galley_file_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    galley_id UUID NOT NULL REFERENCES galleys(galley_id) ON DELETE CASCADE,
    file_id UUID NOT NULL REFERENCES submission_files(file_id) ON DELETE CASCADE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_galley_file UNIQUE (galley_id, file_id)
);

CREATE INDEX idx_galley_files_galley ON galley_files(galley_id);

```

2.8 Cluster 7: Issues, IssueSections, IssueGalleys, ArticlePublications, and PublicationVersions

```

-- =====
-- 23. ISSUES (Serial Containers)
-- =====
CREATE TABLE issues (
    issue_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE CASCADE,
    volume INT,
    number VARCHAR(32),
    year INT NOT NULL,
    title VARCHAR(255),
    description TEXT,
    special_issue_theme VARCHAR(255),
    is_published BOOLEAN NOT NULL DEFAULT FALSE,
    date_published TIMESTAMPTZ,
    doi VARCHAR(128) UNIQUE,
    url_path VARCHAR(64) UNIQUE,
    cover_image_path VARCHAR(512),
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT chk_issue_year CHECK (year >= 1900 AND year <= 2100)
);

CREATE INDEX idx_issues_journal_published ON issues(journal_id, is_published, year, volume);

-- =====
-- 24. ISSUE_SECTIONS (Editorial Sections: Articles, Reviews, Benchmarks)

```

```

-- =====
CREATE TABLE issue_sections (
    section_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE CASCADE,
    title VARCHAR(255) NOT NULL,
    abbrev VARCHAR(32),
    policy TEXT,
    word_count_limit INT,
    sort_order INT NOT NULL DEFAULT 0,
    is_abstract_required BOOLEAN NOT NULL DEFAULT TRUE,
    is_peer_reviewed BOOLEAN NOT NULL DEFAULT TRUE,
    is_open_access BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_journal_section_title UNIQUE (journal_id, title)
);

CREATE INDEX idx_issue_sections_journal ON issue_sections(journal_id, sort_order);

-- =====
-- 25. ISSUE_GALLEYS (Entire Issue Composite PDFs / Volumes)
-- =====
CREATE TABLE issue_galleys (
    issue_galley_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    issue_id UUID NOT NULL REFERENCES issues(issue_id) ON DELETE CASCADE,
    label VARCHAR(64) NOT NULL, -- 'Complete Issue PDF'
    locale VARCHAR(10) NOT NULL DEFAULT 'en_US',
    storage_path VARCHAR(512) NOT NULL,
    file_size_bytes BIGINT NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_issue_galleys_issue ON issue_galleys(issue_id);

-- =====
-- 26. ARTICLE_PUBLICATIONS (Submission Publication Record)
-- =====
CREATE TABLE article_publications (
    publication_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    section_id UUID NOT NULL REFERENCES issue_sections(section_id) ON DELETE RESTRICT,
    issue_id UUID REFERENCES issues(issue_id) ON DELETE SET NULL,
    doi VARCHAR(128) UNIQUE,
    url_path VARCHAR(64) UNIQUE,
    sequence INT NOT NULL DEFAULT 0, -- Order within issue section
    pages VARCHAR(64), -- Page numbering, e.g., '12-28' or article e-number
    date_published TIMESTAMPTZ,
    license_url VARCHAR(255) DEFAULT 'https://creativecommons.org/licenses/by/4.0/',
    copyright_holder VARCHAR(255),
    copyright_year INT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_article_publications_submission ON article_publications(submission_id);
CREATE INDEX idx_article_publications_issue_section ON article_publications(issue_id,
section_id, sequence);
CREATE INDEX idx_article_publications_doi ON article_publications(doi);

```

```

-- =====
-- 27. PUBLICATION_VERSIONS (OJS 3.2+ Immutable Version Snapshots)
-- =====
CREATE TABLE publication_versions (
    version_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    publication_id UUID NOT NULL REFERENCES article_publications(publication_id) ON DELETE
    CASCADE,
    version_number INT NOT NULL DEFAULT 1,
    version_label VARCHAR(32) NOT NULL DEFAULT '1.0',
    version_status publication_version_status_enum NOT NULL DEFAULT 'DRAFT',
    is_current_version BOOLEAN NOT NULL DEFAULT FALSE,
    change_summary TEXT, -- Changelog / Corrigendum notes
    version_metadata JSONB NOT NULL, -- Complete serialized snapshot of authors, title,
    abstract, citations
    date_published TIMESTAMPTZ,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_publication_version UNIQUE (publication_id, version_number)
);

CREATE INDEX idx_publication_versions_pub ON publication_versions(publication_id,
version_number);
CREATE INDEX idx_publication_versions_current ON publication_versions(publication_id) WHERE
is_current_version = TRUE;

```

2.9 Cluster 8: Discussions, DiscussionMessages, DiscussionFiles, and StageNotifications

```

-- =====
-- 28. DISCUSSIONS (Stage-Scoped Communication Threads)
-- =====
CREATE TABLE discussions (
    discussion_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    stage_id workflow_stage_id_enum NOT NULL,
    topic VARCHAR(255) NOT NULL,
    initiator_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    is_closed BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_discussions_submission_stage ON discussions(submission_id, stage_id,
is_closed);

-- Participant access control to discussions
CREATE TABLE discussion_participants (
    participant_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    discussion_id UUID NOT NULL REFERENCES discussions(discussion_id) ON DELETE CASCADE,
    user_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_discussion_user UNIQUE (discussion_id, user_id)
);

```

```

CREATE INDEX idx_discussion_participants_user ON discussion_participants(user_id);

-- =====
-- 29. DISCUSSION_MESSAGES (Individual Thread Messages)
-- =====
CREATE TABLE discussion_messages (
    message_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    discussion_id UUID NOT NULL REFERENCES discussions(discussion_id) ON DELETE CASCADE,
    sender_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    message_body TEXT NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_discussion_messages_thread ON discussion_messages(discussion_id, created_at
ASC);

-- =====
-- 30. DISCUSSION_FILES (Attachments in Discussion Threads)
-- =====
CREATE TABLE discussion_files (
    discussion_file_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    message_id UUID NOT NULL REFERENCES discussion_messages(message_id) ON DELETE CASCADE,
    file_id UUID NOT NULL REFERENCES submission_files(file_id) ON DELETE CASCADE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT uq_message_file UNIQUE (message_id, file_id)
);

CREATE INDEX idx_discussion_files_message ON discussion_files(message_id);

-- =====
-- 31. STAGE_NOTIFICATIONS (Event Broadcasts & Action Triggers)
-- =====
CREATE TABLE stage_notifications (
    notification_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE CASCADE,
    submission_id UUID REFERENCES submissions(submission_id) ON DELETE CASCADE,
    stage_id workflow_stage_id_enum,
    recipient_id UUID NOT NULL REFERENCES users(user_id) ON DELETE CASCADE,
    notification_type VARCHAR(64) NOT NULL, -- e.g., 'NOTIFICATION_REVIEW_ASSIGNMENT',
    'NOTIFICATION_REVISION_DUE'
    title VARCHAR(255) NOT NULL,
    message_content TEXT NOT NULL,
    is_read BOOLEAN NOT NULL DEFAULT FALSE,
    read_at TIMESTAMPTZ,
    action_url VARCHAR(512),
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_stage_notifications_recipient ON stage_notifications(recipient_id, is_read);
CREATE INDEX idx_stage_notifications_submission ON stage_notifications(submission_id);

```

2.10 Cluster 9: ActivityLogs, EditorialNotes, SubmissionLibraries, and PublisherLibraries

```

-- =====
-- 32. ACTIVITY_LOGS (Immutable Audit Trail)
-- =====
CREATE TABLE activity_logs (
    log_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE RESTRICT,
    submission_id UUID REFERENCES submissions(submission_id) ON DELETE CASCADE,
    user_id UUID REFERENCES users(user_id) ON DELETE SET NULL,
    event_type VARCHAR(64) NOT NULL, -- 'SUBMISSION_SUBMITTED', 'DECISION_RENDERED',
    'FILE_UPLOADED', 'VERSION_PUBLISHED'
    message TEXT NOT NULL,
    payload JSONB NOT NULL DEFAULT '{}'::jsonb,
    ip_address INET,
    user_agent VARCHAR(512),
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_activity_logs_submission ON activity_logs(submission_id, created_at DESC);
CREATE INDEX idx_activity_logs_journal ON activity_logs(journal_id, event_type);
CREATE INDEX idx_activity_logs_user ON activity_logs(user_id);
CREATE INDEX idx_activity_logs_payload_gin ON activity_logs USING gin (payload);

-- =====
-- 33. EDITORIAL_NOTES (Confidential Internal Scratchpad)
-- =====
CREATE TABLE editorial_notes (
    note_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    author_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    title VARCHAR(255),
    note_text TEXT NOT NULL,
    is_confidential_to_chief_editors BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_editorial_notes_submission ON editorial_notes(submission_id);

-- =====
-- 34. SUBMISSION_LIBRARIES (Submission-Specific Context Documents)
-- =====
CREATE TABLE submission_libraries (
    library_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    submission_id UUID NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
    uploader_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
    document_name VARCHAR(255) NOT NULL,
    document_type VARCHAR(64) NOT NULL, -- 'DATA_SHARING_AGREEMENT', 'IRB_APPROVAL',
    'COPYRIGHT_TRANSFER'
    storage_path VARCHAR(512) NOT NULL,
    checksum_sha256 VARCHAR(64) NOT NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_submission_libraries_submission ON submission_libraries(submission_id);

```

```

=====
-- 35. PUBLISHER_LIBRARIES (Journal-Wide Guidelines and Templates)
-- =====
CREATE TABLE publisher_libraries (
  publisher_library_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  journal_id UUID NOT NULL REFERENCES journals(journal_id) ON DELETE CASCADE,
  uploader_id UUID NOT NULL REFERENCES users(user_id) ON DELETE RESTRICT,
  title VARCHAR(255) NOT NULL,
  category VARCHAR(64) NOT NULL, -- 'AUTHOR_GUIDELINES', 'CONTRACT_TEMPLATES',
  'CALL_FOR_PAPERS'
  storage_path VARCHAR(512) NOT NULL,
  is_public BOOLEAN NOT NULL DEFAULT FALSE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP
);

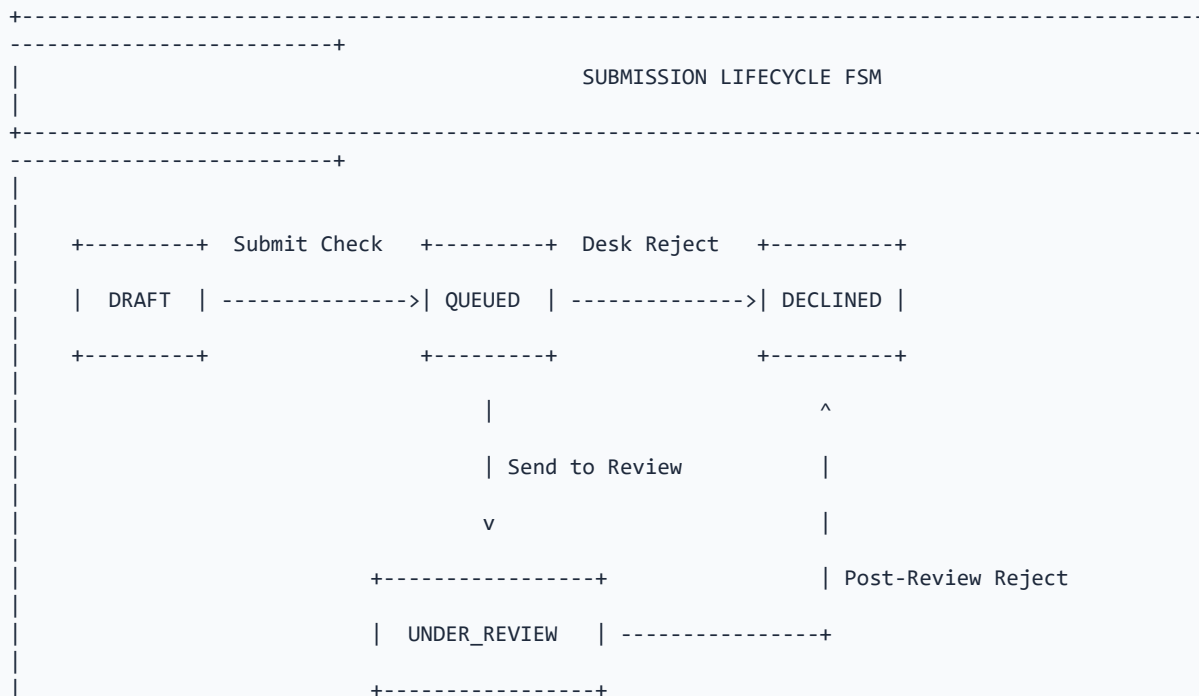
CREATE INDEX idx_publisher_libraries_journal ON publisher_libraries(journal_id, is_public);

```

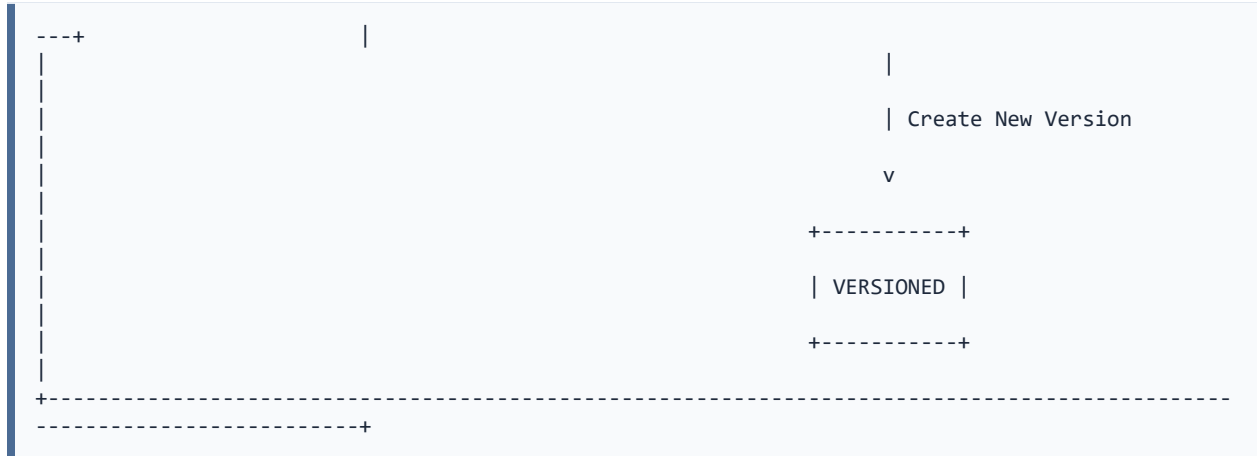
3. Complete Finite State Machine (FSM) Specification

The lifecycle of scholarly contributions in the Google AI Studio Journal System is governed by an integrated, multi-level Finite State Machine. The system combines:

49. A **Macro-Level Submission Lifecycle FSM** governing stage progression.
50. Micro-level sub-state machines within individual stages (notably the iterative **Peer Review Round FSM**, the **Reviewer Assignment Sub-FSM**, the **Copyediting Reconciliation Sub-FSM**, and the **Publication Versioning FSM**).







3.1 Global Submission Lifecycle FSM

- **`DRAFT`**: Author initializes submission, drafts metadata, uploads preliminary files. Editable solely by author.
- **`QUEUED`**: Submission submitted by author. System validates presence of title, abstract, primary manuscript file, and ethics checklists. Awaiting editorial assignment.
- **`UNDER\_REVIEW`**: Editorial team initiates review round. Evaluated under double-blind, single-blind, or open review.
- **`REVISIONS\_REQUESTED`**: Editorial decision rendered requiring author action. Author uploaded files transition to **REVISION_FILES**.
- **`ACCEPTED`**: Formal editorial acceptance rendered. Manuscript passes technical validation.
- **`IN\_COPYEDITING`**: Professional linguistic polish, markup reconciliation, and author copyedit signoff.
- **`IN\_PRODUCTION`**: Typesetting, generation of galleys (PDF/HTML/XML/Notebook), proofreading queries.
- **`SCHEDULED`**: Galley proofs verified; linked to a specific volume/issue/section with DOI minted.
- **`PUBLISHED`**: Publicly accessible, indexed, citable, and archived.
- **`DECLINED`**: Manuscript rejected with formal decision letter. Immutable terminal state unless explicitly un-declined by Journal Editor.
- **`WITHDRAWN`**: Formally retracted by author or editorial board prior to publication.
- **`RETRACTED`**: Formal post-publication invalidation with linked retraction notice per COPE guidelines.

3.2 Peer Review Workflow Stage FSM (Iterative Multi-Round Engine)

The peer review stage is recursive and iterative. Each round comprises:

51. **Round Initialization**: **review_rounds** record incremented (**round_number** = 1, 2, ...).

52. **Reviewer Assignment Sub-FSM**:

- State **INVITED**: Invitation email dispatched with review token and abstract.
- State **ACCEPTED**: Reviewer commits to deadline. Full manuscript and review form unlocked.
- State **DECLINED**: Reviewer rejects invitation; prompt triggers editor to find alternate.
- State **IN_PROGRESS**: Reviewer saves intermediate rubric drafts.

- State **SUBMITTED**: Reviewer submits rubric, confidential comments, and author-facing notes.
- State **CONFIRMED_BY_EDITOR**: Editor reviews report, sanitizes any leaked identity strings, and confirms acceptance of review.

53. Round Deliberation & Decision:

- Guard condition: Minimum required confirmed reviews reached (typically ≥ 2).
- Allowed Decisions:
- **ACCEPT_SUBMISSION** \$

→ Advances to **ACCEPTED** \$

→ **STAGE_COPYEDITING**.

- **REQUEST_REVISIONS** (Minor) \$

→ Submission status **REVISIONS_REQUESTED**.

- **RESUBMIT_FOR_REVIEW** (Major) \$

→ Submission status **REVISIONS_REQUESTED**; author revisions trigger a new **review_round**.

- **DECLINE_SUBMISSION** \$

→ Transitions submission to **DECLINED**.

3.3 Copyediting Workflow Stage FSM

54. **Intake**: Copyeditor assigned via **copyedit_assignments**. Accepted review files copied into **COPYEDIT_DRAFT_FILES**.

55. **Copyeditor Revision**: Copyeditor marks up manuscript, generates queries, and uploads **COPYEDIT_EDITED_FILES**.

56. **Author Review Query**: Author receives notification, inspects track changes, addresses queries via stage **discussions**, and uploads reconciled version.

57. **Final Copyeditor Approval**: Copyeditor inspects author responses and marks assignment complete.

58. **Editorial Promotion**: Editor promotes approved copyedited files to **PRODUCTION_READY_FILES**.

3.4 Production & Galley Preparation FSM

59. **Layout Assignment**: Layout editor assigned via **production_assignments**.

60. **Galley Typesetting**: Production-ready files compiled into publication formats (JATS 1.3 XML, PDF, Responsive HTML, Google Colab Notebook). Files registered in **galleys**.

61. **Proofreading Round**: Proofreader and Author assigned to review galleys. Discrepancies resolved via stage discussion threads.

62. **Galley Signoff**: Proofreader, Author, and Layout Editor confirm approvals.

63. **Issue Scheduling**: Journal Editor binds submission to **article_publications**, assigns issue volume/number, and sets publication sequence.

3.5 Publication, Versioning, and Retraction FSM

64. **Release:** When the parent **issue** is published (or continuous publishing triggered), publication version 1.0 is set to **PUBLISHED**, and DOI is submitted to Crossref.
65. **Versioning (Post-Publication Updates):**
- Trigger: Author/Editor submits minor corrigendum or major update.
 - Guard: Publication must currently be **PUBLISHED**.
 - Action: System creates a new draft version (**publication_versions** with incremented **version_number**, e.g., 2).
 - Once validated, new version is published (**is_current_version** = **TRUE**), setting previous version **is_current_version** = **FALSE**. Both versions remain permanently accessible with unique version DOIs (e.g., **10.1234/aij.2026.01.v1** and **10.1234/aij.2026.01.v2**).
66. **Retraction Protocol:**
- Trigger: Investigation confirms misconduct, ethical breach, or fatal scientific error.
 - Action: Status transitioned to **RETRACTED**. Galley files watermarked with "RETRACTED", and formal retraction notice published linking to original DOI.

3.6 Exhaustive State Transition Matrix Table

#	Current State	Event / Trigger Action	Guard Conditions	Authorized Roles	Target State	Emitted Events & Notifications
T01	DRAFT	submit_article	Required metadata complete; \$\\ge 1\$ manuscript file uploaded; checklists confirmed	Author	QUEUED	SUBMISSION_SUBMITTED dispatched to Journal Editor & Submitters
T02	QUEUED	assign_editor	Target editor active in journal; no conflict of interest	Journal Manager, Journal Editor	QUEUED	EDITOR_ASSIGNED dispatched to assigned editor

| T03 | QUEUED | send_to_external_review | \$\\ge 1\$ editor assigned; anonymized manuscript file ready if double-blind | Journal Editor, Section Editor | UNDER_REVIEW | WORKFLOW_STAGE_TRANSITION (Stage 1 \$\\rightarrow\$ 3); ROUND_INITIALIZED |

T04	`QUEUED`	`desk_decline`	Rejection letter provided	Journal Editor, Section Editor	`DECLINED`	`SUBMISSION_DECLINED` sent to author; submission locked
T05	UNDER_REVIEW	invite_reviewer	Reviewer has no COI with author; active review round exists	Journal Editor, Section Editor	UNDER_REVIEW	REVIEWER_INVITED containing review token & abstract
T06	UNDER_REVIEW	reviewer_accepted	Assignment status is INVITED ; token valid	Reviewer	UNDER_REVIEW	REVIEW_INVITATION_ACCEPTED ; full files unlocked
T07	UNDER_REVIEW	reviewer_decline	Assignment status is INVITED	Reviewer	UNDER_REVIEW	REVIEW_INVITATION_DECLINED ; editor alerted
T08	UNDER_REVIEW	submit_review	Rubric completed; recommendation selected; competing interests declared	Reviewer	UNDER_REVIEW	REVIEW_SUBMITTED ; editor notified for confirmation
T09	UNDER_REVIEW	confirm_review	Review text validated and sanitized	Journal Editor, Section Editor	UNDER_REVIEW	REVIEW_CONFIRMED ; reviewer acknowledged
T10	UNDER_REVIEW	request_revisions	Stage 1\$ confirmed review; editorial decision letter compiled	Journal Editor, Section Editor	REVISIONS_REQUESTED	REVISIONS_REQUIRED sent to author; stage round flagged

T04	`QUEUED`	`desk_decline`	Rejection letter provided	Journal Editor, Section Editor	`DECLINED`	`SUBMISSION_DECLINED` sent to author; submission locked
T11	REVISIONS_REQUESTED	submit_revisions	\$\ge 1\$ revision file uploaded; author response statement provided	Author	UNDER_REVIEW	REVISIONS_SUBMITTED; editor alerted to evaluate revision
T12	UNDER_REVIEW	initiate_new_round	Prior round completed with RESUBMIT_FOR_REVIEW; revised files linked	Journal Editor, Section Editor	UNDER_REVIEW	NEW_REVIEW_ROUND_STARTED (Round \$N+1\$)
T13	UNDER_REVIEW	accept_post_review	\$\ge 1\$ confirmed review recommending acceptance	Journal Editor, Section Editor	ACCEPTED	SUBMISSION_ACCEPTED sent to author; formal acceptance letter logged
T14	UNDER_REVIEW	decline_post_review	\$\ge 1\$ confirmed review; rejection rationale documented	Journal Editor, Section Editor	DECLINED	SUBMISSION_DECLINED sent to author; round closed

| T15 | ACCEPTED | send_to_copyediting | Acceptance decision recorded; clean manuscript file selected | Journal Editor, Section Editor | IN_COPYEDITING | STAGE_TRANSITION (Stage 3 \$\rightarrow\$ 4); copyeditor invited |

T16	`IN_COPYEDITING`	`assign_copyeditor`	Target copyeditor active	Journal Editor, Section Editor	`IN_COPYEDITING`	`COPYEDITOR_ASSIGNED`
T17	IN_COPYEDITING	submit_copyedited_file	Copyedited file uploaded; author instructions detailed	Copyeditor	IN_COPYEDITING	COPYEDIT_AUTHOR_NOTIFICATION dispatched
T18	IN_COPYEDITING	author_signoff_copyedit	Author responded to queries; final markup reconciled	Author	IN_COPYEDITING	COPYEDIT_AUTHOR_COMPLETED sent to copyeditor & editor
T19	IN_COPYEDITING	complete_copyediting	Final edited file confirmed; copyedit assignment signed off	Journal Editor, Copyeditor	IN_COPYEDITING	COPYEDITING_COMPLETED; ready for production promotion

| T20 | IN_COPYEDITING | send_to_production | \$ge 1\$ production-ready file approved | Journal Editor, Section Editor | IN_PRODUCTION | STAGE_TRANSITION (Stage 4 \$ightarrow\$ 5); layout team assigned |

T21	`IN_PRODUCTION`	`upload_galley_proof`	Galley format defined (PDF/HTML/XML); valid file linked	Layout Editor, Journal Editor	`IN_PRODUCTION`	`GALLEY_CREATED`; proofreading requests enabled
T22	IN_PRODUCTION	proofreading_signoff	Proofreader and Author approve layout without	Proofreader, Author	IN_PRODUCTION	PROOFREADING_COMPLETED; all galleys approved

T21	`IN_PRODUCTION`	`upload_galley_proof`	Galley format defined (PDF/HTML/XML); valid file linked	Layout Editor, Journal Editor	`IN_PRODUCTION`	`GALLEY_CREATED`; proofreading requests enabled
corrections						
T23	IN_PRODUCT	schedule_for_issue	All galleys approved; target issue exists; section assigned	Journal Editor, Journal Manager	SCHEDULED	ARTICLE_SCHEDULED_FOR_ISSUE ; metadata locked
T24	SCHEDULED	publish_issue	Issue metadata finalized; scheduled date reached	Journal Manager, Journal Editor	PUBLISHED	ISSUE_PUBLISHED ; DOIs minted to Crossref; indexers pinged
T25	PUBLISHED	create_new_version	Valid corrigendum/amendment request; reason stated	Journal Editor, Journal Manager	PUBLISHED	PUBLICATION_VERSION_FORKED ; draft version created
T26	PUBLISHED	retract_publication	Formal COPE retraction notice approved; legal grounds verified	Journal Editor, Journal Manager	RETRACTED	ARTICLE_RETRACTED ; landing page modified; Crossref updated

4. Granular RBAC Permission Matrix

4.1 System Roles and Persona Definitions

- **Site Administrator (`SITE\_ADMINISTRATOR`)**** Global superuser across all journals on the platform. Manages server configuration, tenant provisioning, system health, and global taxonomy.
- **Journal Manager (`JOURNAL\_MANAGER`)**** Executive administrator of a specific journal. Controls journal policies, sections, review forms, issue creation, and role assignments within the journal.
- **Journal Editor (`JOURNAL\_EDITOR`)**** Chief editorial authority. Assigns section editors, renders binding editorial decisions (Accept, Decline, Revise), commissions peer reviews, and schedules issues.
- **Section Editor (`SECTION\_EDITOR`)**** Manages submissions assigned to specific subject sections. Selects reviewers, manages review rounds, and makes recommendations or binding decisions (based on permissions).
- **Guest Editor (`GUEST\_EDITOR`)**** Scoped editorial authority limited strictly to specific

special issues or explicitly assigned submissions.

6. **Reviewer** (`REVIEWER`): External specialist conducting peer reviews. Access is strictly scoped to assigned submissions during active review rounds.
7. **Copyeditor** (`COPYEDITOR`): Professional linguistic editor responsible for grammar, reference consistency, styling, and author queries during Stage 4.
8. **Layout Editor** (`LAYOUT\_EDITOR`): Production specialist responsible for converting copyedited manuscripts into publication galley (PDF, HTML, XML, Jupyter Notebooks) during Stage 5.
9. **Proofreader** (`PROOFREADER`): Quality assurance specialist who inspects final galley against original copyedited files to catch typographical or rendering defects.
10. **Author** (`AUTHOR`): Scholar who creates, drafts, submits, revises, and inspects manuscripts throughout the lifecycle.

4.2 Comprehensive Capability & Action Matrix

The following matrix defines the access level for each role across all functional resources in the platform:

- **C**: Create / Upload / Initiate
- **R**: Read / Inspect / Download
- **U**: Update / Edit / Modify
- **D**: Delete / Remove
- **A**: Assign / Delegate / Invite
- **E**: Execute Decision / Transition / Sign Off
- **-**: No Access (Strictly Blocked)

Functional Domain	Specific Action / Capability	Site Admin	Journal Mgr	Journal Editor	Section Editor	Guest Editor	Reviewer	Copyeditor	Layout Editor	Proofreader	Author
Platform Management	Provision & Configure Journals	C,R,U,D	-	-	-	-	-	-	-	-	-
	Global System Monitoring & Backups	R,U,E	-	-	-	-	-	-	-	-	-
Journal Configuration	Manage Journal Metadata & Settings	R,U	C,R,U	R	-	-	-	-	-	-	-
	Define Sections & Publication Policies	R,U	C,R,U,D	R,U	-	-	-	-	-	-	-
	Configure Review Forms & Rubrics	R,U	C,R,U,D	C,R,U	-	-	-	-	-	-	-
User & Role Administration	Assign Manager/Editor Roles	C,R,U,D	C,R,U,D	-	-	-	-	-	-	-	-
	Assign Section/Guest Editors & Copyeditors	C,R,U,D	C,R,U,D	C,R,U,A	-	-	-	-	-	-	-
	Enroll Reviewers in Directory	C,R,U,D	C,R,U,D	C,R,U,D	C,R,U	C,R	-	-	-	-	-
Submission Intake (Stage 1)	Create & Submit New Manuscript	C,R,U	-	-	-	-	-	-	-	-	-
	C,R,U,E (Draft)	-	-	-	-	-	-	-	-	-	-
	Desk Triage & Metadata Screening	R,U	R,U	R,U,E	R,U,E*	R,U,E*	-	-	-	-	R (Own)
	Desk Reject (Decline without Review)	-	-	E	E*	E*	-	-	-	-	-
	Assign Handling Editors	A	A	A	-	-	-	-	-	-	-
Peer Review (Stage 2/3)	Initiate Review Round	-	-	C,E	C,E*	C,E*	-	-	-	-	-
	Search & Select Potential Reviewers	R	R	R	R	R	-	-	-	-	-
	Invite & Manage Review Assignments	A,D	A,D	A,D	A,D*	A,D*	-	-	-	-	-
	Access Un-anonymized Author Data	R	R	R	R*	R*	-	(Blind)	R	R	R (Own)
	Accept / Decline Review Invitation	-	-	-	-	-	E	-	-	-	-
	Submit Completed Review & Rubric	-	-	-	-	-	C,U,E	-	-	-	-
	Sanitize & Confirm Review Report	-	-	U,E	U,E*	U,E*	-	-	-	-	-
	Issue Editorial Decision (Accept/Revise/Reject)	-	-	E	E*	E*	-	-	-	-	-
	Upload Author Revisions & Rebuttals	-	-	-	-	-	-	-	-	-	C,U,E

```

| **Copyediting (Stage 4)** | Assign Copyeditor | - | A | A | A* | A* | - | - | - | - | - |
| | Upload & Markup Draft Files | - | - | R,U | R,U* | R,U* | - | C,R,U | - | - | - |
| | Raise Copyediting Queries to Author | - | - | C,R | C,R | C,R | - | C,R | - | - | R,U |
| | Author Signoff on Copyedit Text | - | - | R | R | R | - | R | - | - | E |
| | Promote to Production-Ready | - | - | E | E* | E* | - | - | - | - | - |
| **Production & Layout (Stage 5)** | Assign Layout Editor & Proofreader | - | A | A | A* | A* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Upload Galley Formats (PDF/HTML/XML/Notebook) | - | - | C,R,U | R | R | - | - | C,R,U | - | - |
|
| | Proofreading Inspection & Query Creation | - | - | R | R | R | - | - | R | C,R,U,E | R,U |
| | Final Galley Signoff & Lock | - | - | E | E* | E* | - | - | E | E | E (Author Galley) |
| **Issue & Publication Management** | Create & Organize Issues & Volumes | - | C,R,U,D | C,R,U,D | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Schedule Article to Issue/Section | - | E | E | - | - | - | - | - | - | - |
| | Publish Issue (Live Release) | - | E | E | - | - | - | - | - | - | - |
| | Mint & Register DOIs with Crossref | E | E | E | - | - | - | - | - | - | - |
| | Create New Publication Version (Corrigenda) | - | C,E | C,E | - | - | - | - | - | - | - |
| | Retract Published Article | E | E | E | - | - | - | - | - | - | - |
| **Communications & Audit** | Participate in Stage Discussions | C,R,U | C,R,U | C,R,U | C,R,U* |
| C,R,U* | - (Scope) | C,R,U* | C,R,U* | C,R,U* | C,R,U* |
| | View Confidential Editorial Notes | R | R | C,R,U,D | C,R,U* | C,R,U* | - | - | - | - | - |
| | View System Activity Audit Logs | R | R | R | R* | R* | - | - | - | - | - |
| | Access Submission & Publisher Libraries | R | C,R,U,D | C,R,U | R | R | - | R | R | R | R |
| (Submission) |

```

**Note on Section/Guest Editors:* Permissions marked with an asterisk (`\*`) are strictly constrained to submissions assigned to their respective section or special issue.

4.3 Anonymity, Blind Review, and Privacy Security Guarantees

- **Double-Blind De-identification Enforcement:****
 - In double-blind review (`review\_method = 'DOUBLE\_BLIND'`), all reviewer queries programmatically strip author given names, family names, affiliations, email addresses, ORCIDs, and submission history.
 - Files presented to reviewers are restricted strictly to files where `is\_anonymous\_for\_review = TRUE`. The system's ingestion pipeline validates that PDF/DOCX metadata (Creator, Author, Company, Title, Document ID) is scrubbed.
 - Reviewer identity is shielded from authors in both double-blind and single-blind modes.
- **Reviewer Sanitization Protocol:****
 - Reviewers submit comments in two segregated fields: `comments\_for\_author` and `comments\_for\_editor`.
 - Before reviewer reports are transmitted to authors upon a revision or accept decision, the editor must execute the `confirm\_review` step, verifying that accidental identity cues or inappropriate remarks have been excised.

4.4 Conflict of Interest (COI) and Stage Access Guards

- **Institutional Co-Affiliation Filter:**** The reviewer assignment engine automatically checks institution names and domains between candidate reviewers and all co-authors of a submission. Discrepancies raise blocking warnings.
- **Co-Authorship Recency:**** Scholars who have co-authored any paper with the submitter within the trailing 36 months are disqualified from reviewing.
- **Author-Editor Exclusion:**** An editor who is listed as an author or co-author on a manuscript submitted to their journal is categorically barred from viewing the submission's editorial notes, reviewer identities, or handling the decision lifecycle. The manuscript is automatically locked and reassigned to an independent handling editor.

****End of Specification****

OJS-Aligned AI Journal System: Email Notification, Discussion & Audit Engine Specification

Document Version: 2.4.0-PROD

Status: Approved for Production Engineering

Standard Compliance: Open Journal Systems (PKP-OJS 3.4+), Committee on Publication Ethics (COPE), ISO/IEC 27001, GDPR Article 6/17 Scholarly Record Exemption

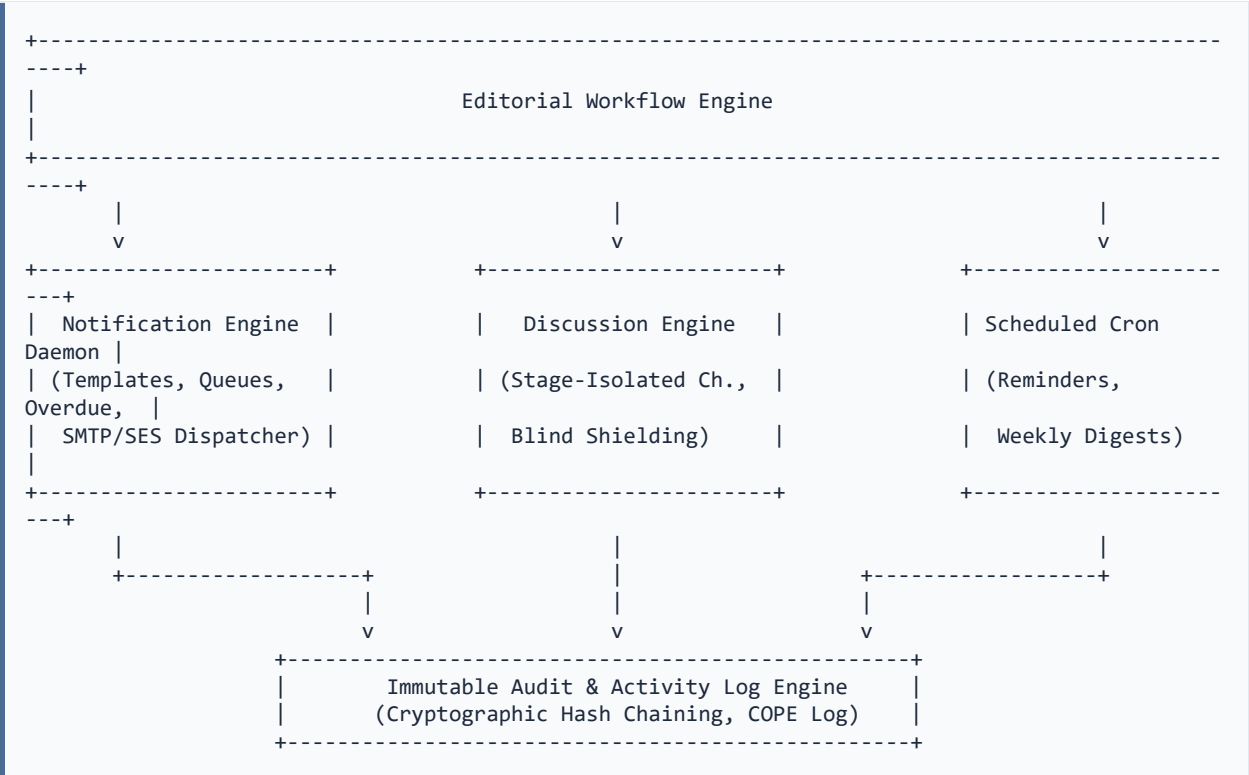
Target Platform: Next-Generation AI-Augmented Editorial Management Engine

1. Executive Summary & Architectural Overview

This technical specification details the complete communication, discussion, scheduling, and audit architecture for the Open Journal Systems (OJS)-aligned AI Journal System. The engine handles the full scholarly publishing lifecycle across five core stages:

- 67. Submission Stage (STAGE_SUBMISSION)
- 68. Review Stage (STAGE_REVIEW)
- 69. Copyediting Stage (STAGE_COPYEDITING)
- 70. Production Stage (STAGE_PRODUCTION)
- 71. Publication & Archival Stage (STAGE_PUBLISHED)

1.1 Architectural High-Level Topology



1.2 Core Design Principles

72. **Strict Stage Isolation:** Communications, discussions, and file references created in one stage do not leak into another without explicit editorial promotion.
 73. **Double-Blind Integrity:** System-level masking removes all identifiable author or reviewer metadata from email headers, body strings, file attachments, and discussion streams.
 74. **Template Rigor:** Every notification is driven by a deterministic template containing structured variables, explicit recipient targeting, and distinct delivery semantics (synchronous transactional dispatch vs. asynchronous cron-pollled queues).
 75. **Non-Repudiation and Audit Immutability:** Every message sent, discussion posted, file versioned, or reminder triggered produces an append-only, cryptographically verifiable audit log entry.
-

2. Complete Catalog of 31 OJS-Aligned Email Notification Templates

All email templates support internationalization (RFC 5646 language tags), plain-text and HTML multipart MIME formatting, and standard variable interpolation via the syntax `{${variableName}}`.

2.1 Variable Reference Dictionary

- `{${journalName}}`: Full title of the academic journal.
- `{${journalInitials}}`: Short code or acronym of the journal.
- `{${journalUrl}}`: Canonical URL of the journal portal.
- `{${editorialContactSignature}}`: Full signature of the editorial office, including contact email and affiliation.
- `{${submissionId}}`: Internal submission integer identifier (e.g., `10482`).
- `{${submissionTitle}}`: Full sanitized manuscript title.
- `{${submissionAbstract}}`: Abstract text (sanitized of markup).
- `{${sectionTitle}}`: Assigned journal section (e.g., "Research Articles", "Review Papers").
- `{${authorName}}`: Primary corresponding author's full formal name.
- `{${authorList}}`: Comma-separated full author list (suppressed during double-blind review contexts).
- `{${editorName}}`: Full name of the assigned Editor-in-Chief or Section Editor.
- `{${reviewerName}}`: Full name of the invited peer reviewer.
- `{${copyeditorName}}`: Full name of the assigned copyeditor.
- `{${productionEditorName}}`: Full name of the assigned layout/production editor.
- `{${responseDueDate}}`: Deadline date for accepting or declining an invitation (e.g., `October 15, 2026`).
- `{${reviewDueDate}}`: Deadline date for review completion.
- `{${revisionDueDate}}`: Deadline date for author revision resubmission.
- `{${copyeditDueDate}}`: Deadline date for copyediting review completion.
- `{${proofreadDueDate}}`: Deadline date for galley proofreading sign-off.
- `{${submissionUrl}}`: Deep link directly into the workflow dashboard for the specific role.
- `{${reviewAssignmentUrl}}`: Direct access link for reviewer (includes secure single-use access token if direct-auth is enabled).
- `{${discussionUrl}}`: Direct link to the stage discussion thread.

- `{peerReviewsText}`: Aggregated, anonymized review reports and editorial comments.
- `{issueIdentification}`: Volume, Issue number, and publication year (e.g., *Vol. 14 No. 2 (2026)*).
- `{articleDoi}`: Registered Digital Object Identifier string.
- `{articleUrl}`: Public landing page URL for the published article.

2.2 Templates 1 to 10: Submission Triage & Review Invitations

Template 1: SUBMISSION_ACK

- **Template Code:** `SUBMISSION_ACK`
- **Sender:** `{journalName} Editorial Office <editorial@{journalDomain}>`
- **Recipient(s):** Primary Submitting Author (`{authorEmail}`)
- **CC:** All Co-Authors (`{coAuthorEmails}`)
- **BCC:** Submission Log Archive (`archive-submissions@{journalDomain}`)
- **Trigger Event & Timing:** Immediate upon author final submission confirmation in `STAGE_SUBMISSION`.
- **Subject:** `[{journalInitials}] Submission Acknowledgement: #{submissionId} - {submissionTitle}`
- **Email Body:**

Dear `{authorName}`,

Thank you for submitting your manuscript titled "`{submissionTitle}`" to `{journalName}`.

Your manuscript has been assigned the tracking identifier: `#{submissionId}`.

Please retain this identifier for all future correspondence with the editorial office. You may check the progress of your submission through our peer review tracking portal at: `{submissionUrl}`

Submission Details:

- Title: `{submissionTitle}`
- Section: `{sectionTitle}`
- Submission Date: `{submissionDate}`

Your manuscript will now undergo initial screening by the Editorial Board to assess compliance with the journal's scope, ethical requirements, and technical guidelines. You will be notified as soon as a Handling Editor is assigned and an initial determination has been reached.

If you have any questions or require modifications to your metadata, please contact the editorial office using our stage discussion engine within the portal or by replying to this notification.

Sincerely,

Editorial Office
`{journalName}`
`{journalUrl}`
`{editorialContactSignature}`

Template 2: SUBMISSION_UNASSIGNED_NOTICE

- **Template Code:** SUBMISSION_UNASSIGNED_NOTICE
- **Sender:** \${journalName} Automated Workflow System <no-reply@\${journalDomain}>
- **Recipient(s):** Editors-in-Chief (\${editorsInChiefEmails})
- **CC:** Managing Editors (\${managingEditorsEmails})
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon completion of SUBMISSION_ACK dispatch.
- **Subject:** [\${journalInitials}] Action Required: New Submission #\${submissionId} Requires Section Editor Assignment
- **Email Body:**

Dear Editorial Leadership,

A new manuscript has been submitted to \${journalName} and currently has no assigned Handling Editor or Section Editor.

Manuscript Details:

- Submission ID: #\${submissionId}
- Section: \${sectionTitle}
- Title: \${submissionTitle}
- Authors: \${authorList}
- Submitting Author: \${authorName} (\${authorEmail})
- Date Submitted: \${submissionDate}

Abstract Preview:

\${submissionAbstract}

Please access the editorial management queue at your earliest convenience to assign a Section Editor or proceed with initial desk triage:

\${submissionUrl}

Automated Workflow Notification

\${journalName}

Template 3: EDITOR_ASSIGN

- **Template Code:** EDITOR_ASSIGN
- **Sender:** \${journalName} Editor-in-Chief <eic@\${journalDomain}>
- **Recipient(s):** Assigned Section Editor (\${editorEmail})
- **CC:** None
- **BCC:** editorial-triage@\${journalDomain}
- **Trigger Event & Timing:** Immediate when Editor-in-Chief or Managing Editor assigns an editor to handle the submission.
- **Subject:** [\${journalInitials}] Assignment as Handling Editor: #\${submissionId} - \${submissionTitle}
- **Email Body:**

Dear Dr. {\$editorName},

You have been assigned as the Handling Editor for the following manuscript submitted to {\$journalName}:

Submission ID: #{\$submissionId}
 Title: {\$submissionTitle}
 Section: {\$sectionTitle}
 Authors: {\$authorList}

Please review the submission files, confirm that you have no conflict of interest with the authors or the work, and proceed with the initial desk evaluation (Send to Peer Review, Request Resubmission, or Decline Submission).

To view the submission and begin editorial management, please navigate to: {\$submissionUrl}

Thank you for your dedicated service to {\$journalName}.

Warm regards,

{\$assigningEditorName}
 Editor-in-Chief, {\$journalName}
 {\$editorialContactSignature}

Template 4: SUBMISSION_DECLINE_DESK

- **Template Code:** SUBMISSION_DECLINE_DESK
- **Sender:** {\$editorName} <{\$editorEmail}>
- **Recipient(s):** Corresponding Author ({\$authorEmail})
- **CC:** Co-Authors ({\$coAuthorEmails})
- **BCC:** decisions-archive@{\$journalDomain}
- **Trigger Event & Timing:** Immediate upon recording a "Desk Reject / Decline Submission" decision prior to peer review.
- **Subject:** [{\$journalInitials}] Editorial Decision: #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear {\$authorName},

Thank you for your submission of manuscript #{\$submissionId}, titled "{\$submissionTitle}", to {\$journalName}.

The Editorial Board has conducted an initial evaluation of your manuscript. Due to the high volume of submissions we receive, we must prioritize papers that closely align with our editorial scope, demonstrate substantial novelty, and meet our rigorous methodological requirements.

Following careful consideration, we regret to inform you that we are unable to accept your

manuscript for publication or send it out for formal peer review.

Specific Editorial Comments:

 {\$editorialComments}

This decision implies no negative judgment on the scientific validity of your research, and we encourage you to consider submitting your work to a journal whose scope is more specifically aligned with your findings.

Thank you for your interest in {\$journalName}, and we wish you success with your ongoing research.

Sincerely,

{\$editorName}
 Handling Editor, {\$journalName}
 {\$editorialContactSignature}

Template 5: REVIEW_REQUEST

- **Template Code:** REVIEW_REQUEST
- **Sender:** {\$editorName} <{\$editorEmail}>
- **Recipient(s):** Invited Reviewer ({\$reviewerEmail})
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate when the Handling Editor selects a reviewer and initiates an invitation.
- **Subject:** [{\$journalInitials}] Peer Review Invitation: #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear Dr. {\$reviewerName},

Given your recognized expertise in this research domain, I would like to invite you to review the manuscript titled "{\$submissionTitle}", which has been submitted to {\$journalName}.

Manuscript Summary:

- Submission ID: #{\$submissionId}
- Title: {\$submissionTitle}
- Section: {\$sectionTitle}
- Review Model: Double-Blind Peer Review

Abstract:

{\$submissionAbstract}

Timeline Expectations:

- Response to this invitation requested by: {\$responseDueDate}
- Review completion requested by: {\$reviewDueDate}

To accept or decline this review request, and to access the manuscript files if accepting, please visit our secure review portal:
 {\$reviewAssignmentUrl}

If you have a potential conflict of interest (financial, professional, or personal), or if you are unable to accept this assignment within the requested timeframe, please decline via the link above or reply to this message. Suggestions for alternative qualified reviewers are always welcomed.

{\$journalName} operates a double-blind review process. Please ensure that all review reports and uploaded files remain entirely free of identifying information.

Thank you for considering this request and for supporting our peer review process.

Sincerely,

{\$editorName}
 Handling Editor, {\$journalName}
 {\$editorialContactSignature}

Template 6: REVIEW_REQUEST_REMIND

- **Template Code:** REVIEW_REQUEST_REMIND
- **Sender:** {\$editorName} <{\$editorEmail}>
- **Recipient(s):** Invited Reviewer ({\$reviewerEmail})
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Scheduled Cron: Triggered 3 days after REVIEW_REQUEST if the reviewer has neither accepted nor declined.
- **Subject:** [{\$journalInitials}] Reminder: Peer Review Invitation for #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear Dr. {\$reviewerName},

This is a gentle reminder regarding our invitation sent on {\$invitationDate} to review the manuscript #{\$submissionId}, titled "{\$submissionTitle}", for {\$journalName}.

The requested deadline to respond to this invitation was {\$responseDueDate}.

We would be very grateful to know whether you are in a position to assist us with this evaluation. Please click one of the options below to indicate your availability:

Accept or Decline Invitation:
 {\$reviewAssignmentUrl}

If you cannot accept this review, kindly notify us so that we may reassign the manuscript without delay. If possible, recommendations of alternate expert reviewers would be deeply appreciated.

Thank you for your time and consideration.

Warm regards,

{ \$editorName}
Handling Editor, { \$journalName}
{ \$editorialContactSignature}

Template 7: REVIEW_RESPONSE_ACCEPT

- **Template Code:** REVIEW_RESPONSE_ACCEPT
- **Sender:** { \$journalName } Workflow System <no-reply@{ \$journalDomain }>
- **Recipient(s):** Handling Editor ({ \$editorEmail })
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon the reviewer clicking "Accept Review" in the portal.
- **Subject:** [{ \$journalInitials }] Reviewer Accepted: # { \$submissionId } - Dr. { \$reviewerName }
- **Email Body:**

Dear Dr. { \$editorName },

Reviewer Dr. { \$reviewerName } has accepted the invitation to review manuscript # { \$submissionId }, titled " { \$submissionTitle } ".

Review Schedule:

- Acceptance Date: { \$acceptanceDate }
- Review Due Date: { \$reviewDueDate }

The reviewer has been granted access to the double-blind masked submission files. You can monitor review progress from your editorial dashboard:
{ \$submissionUrl }

Automated Workflow Notification
{ \$journalName }

Template 8: REVIEW_RESPONSE_DECLINE

- **Template Code:** REVIEW_RESPONSE_DECLINE
- **Sender:** { \$journalName } Workflow System <no-reply@{ \$journalDomain }>
- **Recipient(s):** Handling Editor ({ \$editorEmail })
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon the reviewer clicking "Decline Review" in the portal.

- **Subject:** [{\$journalInitials}] Reviewer Declined: #{\$submissionId} - Dr. {\$reviewerName}
- **Email Body:**

Dear Dr. {\$editorName},

Reviewer Dr. {\$reviewerName} has declined the invitation to review manuscript #{\$submissionId}, titled "{\$submissionTitle}".

Reason Provided by Reviewer:
{\$declineReason}

Suggested Alternative Reviewers:
{\$suggestedReviewers}

Please access the submission portal to invite another reviewer to ensure timely evaluation of this manuscript:
{\$submissionUrl}

Automated Workflow Notification
{\$journalName}

Template 9: REVIEW_REMIND_ONE_WEEK

- **Template Code:** REVIEW_REMIND_ONE_WEEK
- **Sender:** {\$editorName} <{\$editorEmail}>
- **Recipient(s):** Reviewer ({\$reviewerEmail})
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Scheduled Cron: Evaluated daily at 00:00 UTC; triggered when `reviewDueDate - currentDate == 7 days`.
- **Subject:** [{\$journalInitials}] Review Due in One Week: #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear Dr. {\$reviewerName},

This is a courtesy reminder that your peer review for manuscript #{\$submissionId}, titled "{\$submissionTitle}", is due in one week on {\$reviewDueDate}.

To access the manuscript files, evaluation rubric, and online review submission form, please visit:
{\$reviewAssignmentUrl}

If you anticipate needing a modest extension to complete a thorough review, please let us know as soon as possible so that we may adjust our editorial schedule accordingly.

Thank you very much for your critical contribution to the scholarly standards of {\$journalName}.

Warm regards,

```
{ $editorName }
Handling Editor, { $journalName }
{ $editorialContactSignature }
```

Template 10: REVIEW_REMIND_DUE_DATE

- **Template Code:** REVIEW_REMIND_DUE_DATE
- **Sender:** { \$editorName } <{ \$editorEmail }>
- **Recipient(s):** Reviewer ({ \$reviewerEmail })
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Scheduled Cron: Evaluated daily at 00:00 UTC; triggered when `currentDate == reviewDueDate`.
- **Subject:** [{ \$journalInitials }] Review Due Today: # { \$submissionId } - { \$submissionTitle }
- **Email Body:**

Dear Dr. { \$reviewerName },

We are writing to remind you that your peer review for manuscript # { \$submissionId }, titled "{ \$submissionTitle }", is due today, { \$reviewDueDate }.

We appreciate that academic schedules are demanding. If your report is ready, please submit it using our review form at:
{ \$reviewAssignmentUrl }

If you require an extension of a few days, please notify us immediately by replying to this email or via the review portal so we can keep the authors informed of the timeline.

Thank you once again for your assistance.

Sincerely,

```
{ $editorName }
Handling Editor, { $journalName }
{ $editorialContactSignature }
```

2.3 Templates 11 to 20: Review Follow-ups, Completions & Editorial Decisions

Template 11: REVIEW_OVERDUE

- **Template Code:** REVIEW_OVERDUE
- **Sender:** { \$editorName } <{ \$editorEmail }>
- **Recipient(s):** Reviewer ({ \$reviewerEmail })
- **CC:** Handling Editor ({ \$editorEmail })
- **BCC:** None

- **Trigger Event & Timing:** Scheduled Cron: Evaluated daily; triggered when `currentDate > reviewDueDate` (staggered at +3 days, +7 days, and +14 days).
- **Subject:** `[{$journalInitials}] URGENT: Review Overdue for #{$submissionId} - {$submissionTitle}`
- **Email Body:**

Dear Dr. {\$reviewerName},

Our records indicate that your review for manuscript #{\$submissionId}, titled "{\$submissionTitle}", was due on {\$reviewDueDate} and is now {\$daysOverdue} days overdue.

The authors and the editorial board are eagerly awaiting your evaluation in order to reach a decision on this submission.

Please submit your review comments as soon as possible at:
{\$reviewAssignmentUrl}

If unforeseen circumstances prevent you from completing this review within the next 48 hours, please let us know immediately so that we can take alternative steps to avoid further delaying the authors.

We value your time and support of {\$journalName}.

Sincerely,

{\$editorName}
Handling Editor, {\$journalName}
{\$editorialContactSignature}

Template 12: REVIEW_SUBMITTED

- **Template Code:** `REVIEW_SUBMITTED`
- **Sender:** `{$journalName} Workflow System <no-reply@{$journalDomain}>`
- **Recipient(s):** Handling Editor (`{$editorEmail}`)
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon reviewer submitting their final review report.
- **Subject:** `[{$journalInitials}] Review Completed: #{$submissionId} by Dr. {$reviewerName}`
- **Email Body:**

Dear Dr. {\$editorName},

Reviewer Dr. {\$reviewerName} has submitted their review report for manuscript #{\$submissionId}, titled "{\$submissionTitle}".

Review Summary:

- Recommendation: {\$reviewerRecommendation}
- Date Submitted: {\$reviewCompletionDate}
- Reviewer Files Attached: {\$hasReviewerAttachments}

You can inspect the submitted review, evaluate whether additional reviews are needed, or formulate an editorial decision at:
`{ $submissionUrl }`

Automated Workflow Notification
`{ $journalName }`

Template 13: REVIEW_COMPLETE_ACK

- **Template Code:** `REVIEW_COMPLETE_ACK`
- **Sender:** `{ $editorName } <{ $editorEmail }>`
- **Recipient(s):** Reviewer (`{ $reviewerEmail }`)
- **CC:** None
- **BCC:** `reviewer-credits@{ $journalDomain }`
- **Trigger Event & Timing:** Immediate upon editorial confirmation/receipt of the submitted review.
- **Subject:** `[{ $journalInitials }] Acknowledgement and Thank You for Reviewing: #{ $submissionId }`
- **Email Body:**

Dear Dr. `{ $reviewerName }`,

Thank you very much for submitting your detailed peer review for manuscript `#{ $submissionId }`, titled "`{ $submissionTitle }`", for `{ $journalName }`.

Your constructive insights and rigorous evaluation are invaluable in maintaining the high scholarly quality of our publications. We deeply appreciate the time, expertise, and effort you dedicated to this review.

Your contribution has been recorded in our reviewer recognition database. If you wish to claim credit for this review on Web of Science / ORCID, you may download a formal certificate of recognition from your profile dashboard:
`{ $journalUrl }/reviewer/recognition`

Thank you again for your continued support of `{ $journalName }`.

Sincerely,

`{ $editorName }`
 Handling Editor, `{ $journalName }`
`{ $editorialContactSignature }`

Template 14: REVIEW_CANCEL

- **Template Code:** `REVIEW_CANCEL`
- **Sender:** `{ $editorName } <{ $editorEmail }>`
- **Recipient(s):** Reviewer (`{ $reviewerEmail }`)

- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate when the Handling Editor cancels a review assignment.
- **Subject:** [{\$journalInitials}] Review Assignment Cancelled: #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear Dr. {\$reviewerName},

We are writing to inform you that we have cancelled your review assignment for manuscript #{\$submissionId}, titled "{\$submissionTitle}".

Cancellation Reason:
{\$cancellationReason}

You are no longer expected to submit a review for this manuscript. We thank you for your willingness to assist {\$journalName}, and we look forward to collaborating with you on future submissions that match your expertise.

Sincerely,

{\$editorName}
Handling Editor, {\$journalName}
{\$editorialContactSignature}

Template 15: REVIEW_REINSTATE

- **Template Code:** REVIEW_REINSTATE
- **Sender:** {\$editorName} <{\$editorEmail}>
- **Recipient(s):** Reviewer ({\$reviewerEmail})
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate when the Handling Editor reinstates a previously cancelled review assignment.
- **Subject:** [{\$journalInitials}] Review Assignment Reinstated: #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear Dr. {\$reviewerName},

Your review assignment for manuscript #{\$submissionId}, titled "{\$submissionTitle}", has been reinstated in the editorial system of {\$journalName}.

Updated Review Schedule:
- New Due Date: {\$reviewDueDate}

You may access the manuscript files and the online review portal at:
{\$reviewAssignmentUrl}

If this updated timeline poses any scheduling conflicts, please contact us immediately so we can accommodate your availability.

Thank you for your ongoing commitment to {\$journalName}.

Sincerely,

{\$editorName}
Handling Editor, {\$journalName}
{\$editorialContactSignature}

Template 16: DECISION_ACCEPT

- **Template Code:** DECISION_ACCEPT
- **Sender:** {\$editorName} <{\$editorEmail}>
- **Recipient(s):** Corresponding Author ({\$authorEmail})
- **CC:** Co-Authors ({\$coAuthorEmails}), Copyediting Office (copyediting@{\$journalDomain})
- **BCC:** decisions-archive@{\$journalDomain}
- **Trigger Event & Timing:** Immediate upon recording an "Accept Submission" decision.
- **Subject:** [{\$journalInitials}] Decision: Accepted for Publication - #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear {\$authorName},

On behalf of the Editorial Board, I am pleased to inform you that your manuscript #{\$submissionId}, titled "{\$submissionTitle}", has been accepted for publication in {\$journalName}.

We congratulate you and your co-authors on this achievement. The reviewers and editors commended the rigor, relevance, and presentation of your research.

Peer Review Comments and Editorial Remarks:

{\$peerReviewsText}

Next Steps:

Your manuscript will now transition to the Copyediting Stage, where our editorial team will review language, layout, typography, and reference citations. A copyeditor will contact you in due course with any queries or proof files requiring your validation.

You may monitor the copyediting and production stages through your author dashboard: {\$submissionUrl}

Thank you for publishing your valuable work with {\$journalName}.

Sincerely,

```

{$editorName}
Handling Editor, {$journalName}
{$editorialContactSignature}

```

Template 17: DECISION_REVISIONS

- **Template Code:** DECISION_REVISIONS
- **Sender:** {\$editorName} <{\$editorEmail}>
- **Recipient(s):** Corresponding Author ({\$authorEmail})
- **CC:** Co-Authors ({\$coAuthorEmails})
- **BCC:** decisions-archive@{\$journalDomain}
- **Trigger Event & Timing:** Immediate upon recording a "Revisions Required" (Minor Revisions) decision.
- **Subject:** [{\$journalInitials}] Editorial Decision: Revisions Required - #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear {\$authorName},

We have completed the initial peer review of your manuscript #{\$submissionId}, titled "{\$submissionTitle}", submitted to {\$journalName}.

Based on the advice of the reviewers and the evaluation of the editorial board, we are pleased to invite you to submit a revised version of your manuscript addressing the reviewer and editorial comments.

Decision Type: Minor Revisions
Revision Due Date: {\$revisionDueDate}

Reviewer Comments:

{\$peerReviewsText}

Submission Requirements for Your Revision:

1. A point-by-point Response to Reviewers document clearly indicating how each critique was addressed, including specific line/page references in the revised manuscript.
2. A marked-up version of the manuscript highlighting all modifications (Track Changes or colored text).
3. A clean, publication-ready version of the revised manuscript.

Please submit your revision via your author portal at:
{\$submissionUrl}

Failure to submit by {\$revisionDueDate} may result in the closing of your file, requiring resubmission as a new manuscript. If you require an extension, please request it through the portal well before the deadline.

We look forward to receiving your revised manuscript.

Sincerely,

{ \$editorName }
 Handling Editor, { \$journalName }
 { \$editorialContactSignature }

Template 18: DECISION_RESUBMIT_FOR_REVIEW

- **Template Code:** DECISION_RESUBMIT_FOR_REVIEW
- **Sender:** { \$editorName } <{ \$editorEmail }>
- **Recipient(s):** Corresponding Author ({ \$authorEmail })
- **CC:** Co-Authors ({ \$coAuthorEmails })
- **BCC:** decisions-archive@{ \$journalDomain }
- **Trigger Event & Timing:** Immediate upon recording a "Resubmit for Review" (Major Revisions) decision.
- **Subject:** [{ \$journalInitials }] Editorial Decision: Resubmit for Review (Major Revisions) - # { \$submissionId }
- **Email Body:**

Dear { \$authorName },

The peer review process for manuscript # { \$submissionId }, titled " { \$submissionTitle } ", is now complete.

While your research shows significant promise, the reviewers have identified substantial issues requiring extensive experimental, theoretical, or methodological revisions before this paper can be considered for publication in { \$journalName }.

Decision Type: Major Revisions (Resubmit for Second Round of Peer Review)
 Revision Due Date: { \$revisionDueDate }

Reviewer Reports and Detailed Recommendations:

 { \$peerReviewsText }

Your revised manuscript will undergo a second round of external peer review, whenever possible involving the original reviewers. Acceptance is contingent upon your satisfactory resolution of all critical concerns raised.

Please ensure your revised submission contains:

- A comprehensive, point-by-point rebuttal document.
- A marked-up copy of the manuscript indicating all amendments.
- The revised manuscript files without markup.

To upload your revised materials, please navigate to:
 { \$submissionUrl }

Thank you for submitting your work to { \$journalName }.

Sincerely,

{ \$editorName }
 Handling Editor, { \$journalName }
 { \$editorialContactSignature }

Template 19: DECISION_DECLINE

- **Template Code:** DECISION_DECLINE
- **Sender:** { \$editorName } <{ \$editorEmail }>
- **Recipient(s):** Corresponding Author ({ \$authorEmail })
- **CC:** Co-Authors ({ \$coAuthorEmails })
- **BCC:** decisions-archive@{ \$journalDomain }
- **Trigger Event & Timing:** Immediate upon recording a "Decline Submission" (Rejection) decision post-review.
- **Subject:** [{ \$journalInitials }] Editorial Decision: # { \$submissionId } - { \$submissionTitle }
- **Email Body:**

Dear { \$authorName },

Thank you for giving us the opportunity to consider your manuscript # { \$submissionId }, titled "{ \$submissionTitle }", for publication in { \$journalName }.

We have now received the evaluations of our independent peer reviewers and have thoroughly examined your manuscript alongside their findings. In light of the critical concerns raised regarding methodological limitations, novelty, and depth of analysis, we regret to inform you that we must decline the manuscript for publication.

Reviewer Evaluations:

 { \$peerReviewsText }

We recognize that this decision is disappointing, but we hope that the constructive feedback provided by our reviewers will prove beneficial as you refine your work for submission to another scholarly outlet.

Thank you once again for your interest in { \$journalName }.

Sincerely,

{ \$editorName }
 Handling Editor, { \$journalName }
 { \$editorialContactSignature }

Template 20: REVISED_VERSION_SUBMITTED

- **Template Code:** REVISED_VERSION_SUBMITTED

- **Sender:** `{journalName}` Workflow System `<no-reply@{journalDomain}>`
- **Recipient(s):** Handling Editor (`{editorEmail}`)
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon author submitting a revised manuscript.
- **Subject:** `[{journalInitials}] Revised Manuscript Submitted: #{submissionId} - {submissionTitle}`
- **Email Body:**

Dear Dr. `{editorName}`,

The authors have submitted a revised version of manuscript `#{submissionId}`, titled `"{submissionTitle}"`.

Revision Overview:

- Round: Round `{revisionRoundNumber}`
- Submission Date: `{revisionSubmissionDate}`
- Uploaded Files: Revised Manuscript (Clean), Revised Manuscript (Marked-up), Response to Reviewers

Please access the editorial portal to review the author responses and determine whether to initiate a subsequent review round or record an editorial decision:
`{submissionUrl}`

Automated Workflow Notification
`{journalName}`

2.4 Templates 21 to 31: Copyediting, Production, Publication & Weekly Digest

Template 21: COPYEDIT_ASSIGN

- **Template Code:** `COPYEDIT_ASSIGN`
- **Sender:** `{editorName}` `<{editorEmail}>`
- **Recipient(s):** Assigned Copyeditor (`{copyeditorEmail}`)
- **CC:** None
- **BCC:** `production-office@{journalDomain}`
- **Trigger Event & Timing:** Immediate upon assignment of a copyeditor to the accepted manuscript.
- **Subject:** `[{journalInitials}] Copyediting Assignment: #{submissionId} - {submissionTitle}`
- **Email Body:**

Dear `{copyeditorName}`,

You have been assigned as the Copyeditor for accepted manuscript `#{submissionId}`, titled `"{submissionTitle}"`, in `{journalName}`.

Submission Details:

- Submission ID: `#{submissionId}`
- Section: `{sectionTitle}`
- Copyediting Target Completion Date: `{copyeditDueDate}`

Please review the accepted draft, perform language, style, and bibliographic copyediting in accordance with `{journalName}` style guidelines, and upload the copyedited files to the copyediting stage portal:
`{submissionUrl}`

Thank you for your meticulous assistance in preparing this manuscript for publication.

Sincerely,

`{editorName}`
 Handling Editor, `{journalName}`
`{editorialContactSignature}`

Template 22: COPYEDIT_REQUEST

- **Template Code:** `COPYEDIT_REQUEST`
- **Sender:** `{copyeditorName} <{copyeditorEmail}>`
- **Recipient(s):** Corresponding Author (`{authorEmail}`)
- **CC:** Co-Authors (`{coAuthorEmails}`), Handling Editor (`{editorEmail}`)
- **BCC:** None
- **Trigger Event & Timing:** Immediate when the Copyeditor completes the initial pass and requests author review.
- **Subject:** `[{journalInitials}] Copyediting Review Request: #{submissionId} - {submissionTitle}`
- **Email Body:**

Dear `{authorName}`,

The initial copyediting pass for your accepted manuscript `{submissionId}`, titled "`{submissionTitle}`", is now complete.

We request that you review the copyedited manuscript to address all editorial queries, verify terminology, and ensure accurate citation formatting.

Copyediting Review Deadline: `{copyeditDueDate}`

Instructions:

1. Access your author portal and download the copyedited manuscript:
`{submissionUrl}`
2. Review all highlighted editorial queries and tracked changes.
3. Upload your corrected version and respond to specific queries via the Copyediting Discussion channel.

Please note that no substantial structural or scientific changes are permitted at this stage.

Thank you for your prompt cooperation.

Sincerely,

`{copyeditorName}`

Copyeditor, {\$journalName}
{\$editorialContactSignature}

Template 23: COPYEDIT_AUTHOR_QUERY

- **Template Code:** COPYEDIT_AUTHOR_QUERY
- **Sender:** {\$copyeditorName} <{\$copyeditorEmail}>
- **Recipient(s):** Corresponding Author ({\$authorEmail})
- **CC:** Handling Editor ({\$editorEmail})
- **BCC:** None
- **Trigger Event & Timing:** Immediate when a copyeditor initiates a specific query thread within the copyediting discussion engine.
- **Subject:** [{\$journalInitials}] Copyediting Query regarding #{\$submissionId}: {\$querySubject}
- **Email Body:**

Dear {\$authorName},

A specific query has been posted by the copyeditor regarding your manuscript #{\$submissionId}, titled "{\$submissionTitle}".

Query Topic: {\$querySubject}
Stage: Copyediting Discussion

Message:

{\$queryMessageBody}

Please log in to your dashboard to review the referenced text and reply directly within the discussion thread:
{\$discussionUrl}

Timely resolution of this query is necessary to proceed to galley layout and production.

Warm regards,

{\$copyeditorName}
Copyeditor, {\$journalName}
{\$editorialContactSignature}

Template 24: COPYEDIT_AUTHOR_COMPLETE

- **Template Code:** COPYEDIT_AUTHOR_COMPLETE
- **Sender:** {\$authorName} <{\$authorEmail}>
- **Recipient(s):** Copyeditor ({\$copyeditorEmail}), Handling Editor ({\$editorEmail})

- **CC:** Co-Authors (`{{coAuthorEmails}}`)
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon author clicking "Complete Copyediting Review" in the portal.
- **Subject:** `{{journalInitials}}` Author Copyediting Review Completed: `##{$submissionId}`
- **Email Body:**

Dear `{$copyeditorName}` and Dr. `{$editorName}`,

The author review of the copyedited manuscript `##{$submissionId}`, titled "`{$submissionTitle}`", has been completed.

All copyeditor queries have been addressed, and the final validated files have been uploaded to the portal:
`{$submissionUrl}`

Author Remarks:
`{$authorRemarks}`

The manuscript is now ready for final copyediting sign-off and transfer to Production.

Sincerely,

`{$authorName}`
Corresponding Author

Template 25: COPYEDIT_FINAL_ACK

- **Template Code:** `COPYEDIT_FINAL_ACK`
- **Sender:** `{$journalName}` Workflow System `<no-reply@{$journalDomain}>`
- **Recipient(s):** Corresponding Author (`{$authorEmail}`), Copyeditor (`{$copyeditorEmail}`)
- **CC:** Handling Editor (`{$editorEmail}`)
- **BCC:** None
- **Trigger Event & Timing:** Immediate when the Copyeditor marks the Copyediting stage as closed/complete.
- **Subject:** `{{journalInitials}}` Copyediting Completed: `##{$submissionId}` - `{$submissionTitle}`
- **Email Body:**

Dear Author and Editorial Team,

The Copyediting stage for manuscript `##{$submissionId}`, titled "`{$submissionTitle}`", has been successfully finalized.

Stage Audit Summary:

- Copyeditor Sign-off: `{$copyeditorName}`
- Completion Date: `{$copyeditCompletionDate}`
- Final File Approved: `{$finalCopyeditFileName}`

The manuscript is now scheduled for Galley Production (PDF, HTML, and JATS-XML typesetting). You

will receive galley proofs for review prior to formal publication.

Workflow Details:
{ \$submissionUrl }

Automated Workflow Notification
{ \$journalName }

Template 26: PRODUCTION_ASSIGN

- **Template Code:** PRODUCTION_ASSIGN
- **Sender:** { \$editorName } <{ \$editorEmail }>
- **Recipient(s):** Production Editor / Layout Editor ({ \$productionEditorEmail })
- **CC:** Managing Editor (managing@{ \$journalDomain })
- **BCC:** None
- **Trigger Event & Timing:** Immediate when a manuscript is transitioned to STAGE_PRODUCTION.
- **Subject:** [{ \$journalInitials }] Production Assignment: # { \$submissionId } - { \$submissionTitle }
- **Email Body:**

Dear { \$productionEditorName },

Manuscript # { \$submissionId }, titled " { \$submissionTitle } ", has completed copyediting and has been transferred to the Production Stage.

Task Overview:

- Submission ID: # { \$submissionId }
- Assigned Layout Editor: { \$productionEditorName }
- Target Galley Delivery Date: { \$productionDueDate }

Please access the production files to generate the publication galleys (PDF, HTML, and JATS-XML) in accordance with the journal's publication templates:
{ \$submissionUrl }

Thank you for your dedicated service.

Sincerely,

{ \$editorName }
Handling Editor, { \$journalName }
{ \$editorialContactSignature }

Template 27: PRODUCTION_PROOFREAD_REQUEST

- **Template Code:** PRODUCTION_PROOFREAD_REQUEST
- **Sender:** { \$productionEditorName } <{ \$productionEditorEmail }>
- **Recipient(s):** Corresponding Author ({ \$authorEmail })

- **CC:** Co-Authors (`{{coAuthorEmails}}`), Handling Editor (`{{editorEmail}}`)
- **BCC:** None
- **Trigger Event & Timing:** Immediate when galley files (PDF/HTML) are generated and uploaded for proofreading.
- **Subject:** `[{{journalInitials}}] Galley Proofs Available for Proofreading: #{{submissionId}} - {{submissionTitle}}`
- **Email Body:**

Dear `{{authorName}}`,

The publication galley proofs for your manuscript `#{{submissionId}}`, titled "`{{submissionTitle}}`", are now prepared and ready for your final proofreading review.

Proofreading Due Date: `{{proofreadDueDate}}`

Instructions:

1. Download the galley proofs (PDF/HTML) from your author production dashboard: `{{submissionUrl}}`
2. Inspect the proofs carefully for typesetting, layout, figure positioning, table alignments, and mathematical expressions.
3. Submit all minor corrections or your formal approval through the Production Discussion channel.

Important Note:

This is the final opportunity to correct typographical errors. No new text or extensive rewriting may be introduced at this stage.

If you approve the proofs without changes, please submit a statement confirming your approval in the portal.

Thank you for your cooperation.

Sincerely,

`{{productionEditorName}}`
 Production Editor, `{{journalName}}`
`{{editorialContactSignature}}`

Template 28: PRODUCTION_PROOFREAD_COMPLETE

- **Template Code:** `PRODUCTION_PROOFREAD_COMPLETE`
- **Sender:** `{{authorName}} <{{authorEmail}}>`
- **Recipient(s):** Production Editor (`{{productionEditorEmail}}`)
- **CC:** Handling Editor (`{{editorEmail}}`)
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon author submitting final proofreading sign-off in the portal.
- **Subject:** `[{{journalInitials}}] Galley Proof Approval Confirmed: #{{submissionId}}`
- **Email Body:**

Dear `{productionEditorName}`,

I have reviewed the galley proofs for manuscript `#{submissionId}`, titled "`{submissionTitle}`".

Author Proof Status: `{proofApprovalStatus}`
(Approved as is / Approved with indicated corrections)

Author Verification Comments:
`{authorProofComments}`

The corrected files (if applicable) have been attached to the production discussion thread at:
`{discussionUrl}`

Thank you for your excellent work on the typesetting of our manuscript.

Sincerely,

`{authorName}`
Corresponding Author

Template 29: ISSUE_PUBLISHED_ANNOUNCE

- **Template Code:** `ISSUE_PUBLISHED_ANNOUNCE`
- **Sender:** `{journalName}` Editorial Office `<announcements@{journalDomain}>`
- **Recipient(s):** Journal Registered Users / Readers (`{subscriberEmail}`)
- **CC:** None
- **BCC:** None
- **Trigger Event & Timing:** Immediate upon Editor-in-Chief publishing a new issue/volume.
- **Subject:** `[{journalInitials}] New Issue Published: {issueIdentification}`
- **Email Body:**

Dear Reader,

`{journalName}` has published its latest issue: `{issueIdentification}`.

We invite you to explore the table of contents and read the latest open-access peer-reviewed research:
`{issueUrl}`

Featured Articles in This Issue:

`{issueTableOfContents}`

Thank you for your ongoing interest in and support of our publication.

Warm regards,

Editorial Office
 {\$journalName}
 {\$journalUrl}
 {\$editorialContactSignature}

Template 30: ARTICLE_PUBLISHED_AUTHOR

- **Template Code:** ARTICLE_PUBLISHED_AUTHOR
- **Sender:** {\$journalName} Editorial Office <editorial@{\$journalDomain}>
- **Recipient(s):** Corresponding Author ({\$authorEmail})
- **CC:** Co-Authors ({\$coAuthorEmails})
- **BCC:** publications-archive@{\$journalDomain}
- **Trigger Event & Timing:** Immediate upon publication of the specific article (either via issue release or continuous publishing / Online First).
- **Subject:** [{\$journalInitials}] Your Article Has Been Published: #{\$submissionId} - {\$submissionTitle}
- **Email Body:**

Dear {\$authorName},

We are delighted to inform you that your manuscript titled "{\$submissionTitle}" has been officially published in {\$journalName}.

Publication Citation Details:

- Title: {\$submissionTitle}
- Authors: {\$authorList}
- Publication Date: {\$publicationDate}
- Issue: {\$issueIdentification}
- DOI: [https://doi.org/{\\$articleDoi}](https://doi.org/{$articleDoi})
- Article Landing Page: {\$articleUrl}
- Direct PDF Download: {\$articlePdfUrl}

Your article is published under an Open Access Creative Commons Attribution (CC BY 4.0) license, allowing free access, distribution, and reproduction with proper attribution.

We encourage you to share your publication with colleagues, institutional repositories, and academic networks to maximize the visibility and impact of your research.

Thank you for publishing with {\$journalName}.

Warm congratulations,

Editorial Board
 {\$journalName}
 {\$journalUrl}
 {\$editorialContactSignature}

Template 31: EDITORIAL_DIGEST_WEEKLY

- **Template Code:** EDITORIAL_DIGEST_WEEKLY
- **Sender:** {\$journalName} Automated Intelligence Engine <analytics@{\$journalDomain}>
- **Recipient(s):** Editors-in-Chief ({\$eicEmailList}), Managing Editors ({\$managingEditorEmailList})
- **CC:** None
- **BCC:** executive-audit@{\$journalDomain}
- **Trigger Event & Timing:** Scheduled Cron: Executed weekly every Monday at 06:00 UTC.
- **Subject:** [{\$journalInitials}] Weekly Editorial Digest & Operational Summary - {\$reportDate}
- **Email Body:**

Dear Editorial Leadership,

Here is your automated weekly operational digest for {\$journalName} for the week ending {\$reportDate}.

```
=====
1. WORKFLOW PIPELINE HEALTH
=====
- Submissions Awaiting Editor Assignment: {$countUnassignedSubmissions}
- Submissions in Initial Screening:         {$countDeskReview}
- Active Submissions in Peer Review:        {$countActivePeerReview}
- Manuscripts Awaiting Editorial Decision:  {$countAwaitingDecision}
- Manuscripts in Copyediting Stage:         {$countInCopyediting}
- Manuscripts in Production / Galleys:      {$countInProduction}
- Total Active Manuscripts:                 {$countTotalActive}

=====
2. PEER REVIEW BOTTLENECK & ESCALATION ALERTS
=====
- Reviews Overdue (> 3 Days):               {$countReviewsOverdue}
- Outstanding Review Invitations (> 4 d):    {$countInvitationsStalled}
- Critical Attention Required (ID / Editor / Days Overdue):
{$criticalReviewEscalationList}

=====
3. DECISION VELOCITY & METRICS (PAST 7 DAYS)
=====
- New Submissions Received:                 {$countNewSubmissionsWeek}
- Desk Rejections:                         {$countDeskRejectionsWeek}
- Revisions Requested:                     {$countRevisionsRequestedWeek}
- Formal Acceptances:                      {$countAcceptancesWeek}
- Articles Published:                       {$countArticlesPublishedWeek}
- Average Days to First Decision:           {$avgDaysToFirstDecision} days
```

To access the editorial management board and inspect flagged submissions, please visit:
{\$journalUrl}/management/editorial-dashboard

Generated by the Automated Editorial Intelligence Engine
{\$journalName}

3. Discussion Engine Specification

The Discussion Engine powers contextual, stage-isolated multi-party communications throughout the publication lifecycle. It replaces unstructured, ad-hoc external emails with auditable, in-band discussion threads bound to specific editorial stages.

3.1 Stage-Isolated Communication Channels

To prevent premature data disclosure and preserve editorial integrity, communication channels are strictly partitioned into four mutually exclusive stages:



Isolation Boundary Rules:

- 76. **Vertical Isolation:** A thread created in `STAGE_REVIEW` cannot be viewed, replied to, or searched within `STAGE_SUBMISSION` or `STAGE_COPYEDITING`.
- 77. **Access Confinement:** Users with access to a submission can only query discussions associated with stages for which they hold active, granted participant privileges.
- 78. **Cross-Stage Promotion:** When an editor promotes a manuscript from Review to Copyediting, attachments or specific discussion resolutions can only be transferred via an explicit "Copy to Next Stage" editorial action.

3.2 Participant Access Control & Double-Blind Shielding

The system enforces Role-Based Access Control (RBAC) augmented with Attribute-Based Access Control (ABAC) to guarantee double-blind peer review integrity.

3.2.1 Role Definitions

- **Journal Manager / Admin** (`ROLE_MANAGER``): Full system audit visibility.
- **Editor-in-Chief** (`ROLE_EIC``): Universal oversight and decision authority.
- **Section Editor / Handling Editor** (`ROLE_EDITOR``): Full control over assigned submission stages.
- **Author / Submitting Author** (`ROLE_AUTHOR``): Access to their submission in relevant stages.
- **Reviewer** (`ROLE_REVIEWER``): Access strictly restricted to assigned review rounds.
- **Copyeditor** (`ROLE_COPYEDITOR``): Access restricted to `STAGE_COPYEDITING`.
- **Production Editor** (`ROLE_LAYOUT``): Access restricted to `STAGE_PRODUCTION`.

3.2.2 Double-Blind Shielding Enforcement Engine

In double-blind peer review mode (`REVIEW_METHOD_DOUBLE_BLIND`), the following rules are mathematically and structurally enforced at the API gateway:

79. Reviewer to Author Isolation:

- Reviewers NEVER participate in the same discussion thread as Authors.
- Any communication from Reviewers is addressed solely to the Handling Editor.
- Any query from an Author regarding review comments is addressed solely to the Handling Editor.

80. Reviewer to Reviewer Isolation:

- Reviewers cannot see who other reviewers are on the same manuscript.
- Reviewers cannot access discussions between the Editor and other assigned reviewers.

81. Identity Masking in Data Payloads:

- In all API responses delivered to `ROLE_REVIEWER`, fields containing author identity (`authors`, `author_id`, `submitting_author_name`, `affiliations`) are stripped and replaced with deterministic tokens (`Author`).
- In all API responses delivered to `ROLE_AUTHOR`, fields containing reviewer identity (`reviewer_id`, `reviewer_name`, `email`) are replaced with `Reviewer 1`, `Reviewer 2`, etc.

3.2.3 Stage Participant Permission Matrix

Role	Submission Stage	Review Stage	Copyediting Stage	Production Stage
Journal Manager / EIC	Create / Read / Reply	Create / Read / Reply	Create / Read / Reply	Create / Read / Reply
Handling Editor	Create / Read / Reply	Create / Read / Reply	Create / Read / Reply	Create / Read / Reply
Author	Read / Reply (if invited)	Read / Reply (Author-Editor only)	Create / Read / Reply	Create / Read / Reply
Reviewer	NO ACCESS	Create / Read / Reply (Reviewer-Editor only)	NO ACCESS	NO ACCESS
Copyeditor	NO ACCESS	NO ACCESS	Create / Read / Reply	NO ACCESS
Production Editor	NO ACCESS	NO ACCESS	NO ACCESS	Create / Read / Reply

3.3 Data Models & Schemas

The discussion engine is governed by four relational entities:

```

-- 1. Discussion Topics (Threads)
CREATE TABLE discussion_topics (
  topic_id BIGSERIAL PRIMARY KEY,
  submission_id BIGINT NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
  stage_id SMALLINT NOT NULL CHECK (stage_id IN (1, 2, 3, 4)), -- 1: Sub, 2: Rev, 3: Copy, 4:
Prod
  title VARCHAR(255) NOT NULL,
  status VARCHAR(32) NOT NULL DEFAULT 'OPEN' CHECK (status IN ('OPEN', 'CLOSED')),
  created_by_user_id BIGINT NOT NULL REFERENCES users(user_id),
  closed_by_user_id BIGINT REFERENCES users(user_id),
  closed_at TIMESTAMP WITH TIME ZONE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

-- 2. Discussion Participants (Membership & Access)
CREATE TABLE discussion_participants (
  topic_id BIGINT NOT NULL REFERENCES discussion_topics(topic_id) ON DELETE CASCADE,
  user_id BIGINT NOT NULL REFERENCES users(user_id) ON DELETE CASCADE,
  role_in_topic VARCHAR(32) NOT NULL, -- e.g., 'EDITOR', 'AUTHOR', 'REVIEWER', 'COPYEDITOR'
  joined_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  last_read_message_id BIGINT,
  is_muted BOOLEAN DEFAULT FALSE,
  PRIMARY KEY (topic_id, user_id)
);

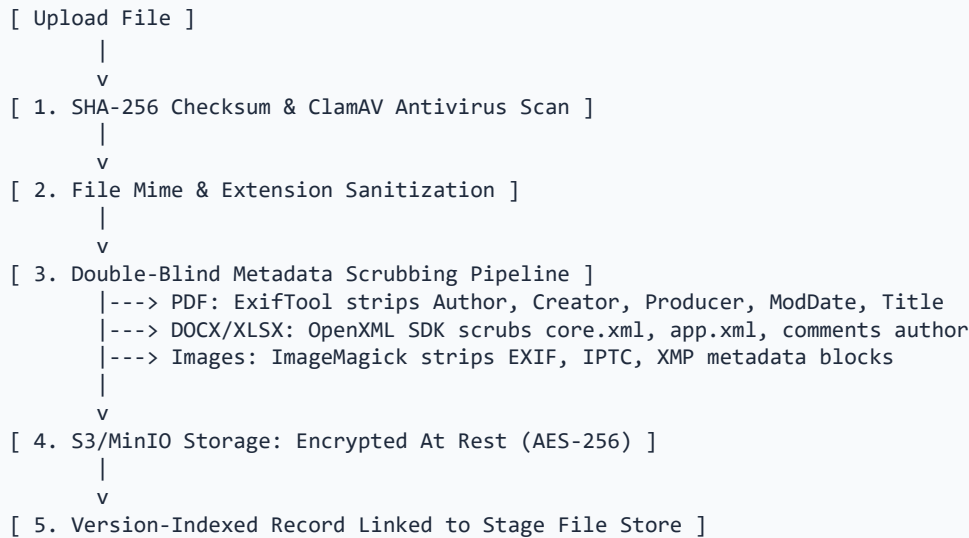
-- 3. Discussion Messages
CREATE TABLE discussion_messages (
  message_id BIGSERIAL PRIMARY KEY,
  topic_id BIGINT NOT NULL REFERENCES discussion_topics(topic_id) ON DELETE CASCADE,
  sender_user_id BIGINT NOT NULL REFERENCES users(user_id),
  body_markdown TEXT NOT NULL,
  body_html TEXT NOT NULL,
  is_internal_editorial_only BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
  deleted_at TIMESTAMP WITH TIME ZONE
);

-- 4. Discussion Attachments
CREATE TABLE discussion_attachments (
  attachment_id BIGSERIAL PRIMARY KEY,
  message_id BIGINT NOT NULL REFERENCES discussion_messages(message_id) ON DELETE CASCADE,
  file_id BIGINT NOT NULL REFERENCES stage_files(file_id),
  original_file_name VARCHAR(255) NOT NULL,
  sanitized_file_name VARCHAR(255) NOT NULL,
  file_size_bytes BIGINT NOT NULL,
  sha256_checksum CHAR(64) NOT NULL,
  mime_type VARCHAR(128) NOT NULL,
  is_anonymized BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

```

3.4 Attachment Handling, Anonymization & Versioning Pipeline

When an attachment is uploaded to a discussion message, it passes through the automated Document Processing Pipeline:



Versioning Semantics

- Every uploaded attachment receives a globally unique `file_id` and is tied to a `revision_number`.
- If a revision is uploaded in response to a query, it links to its `parent_file_id`, creating an unbroken lineage tree from Initial Submission -> Review Revisions -> Copyedited Files -> Galley Proofs.

3.5 Closed Discussion Semantics & Activity Log Integration

82. Closing Semantics:

- Only users with `ROLE_EDITOR` or `ROLE_MANAGER` have authorization to close a discussion topic.
- Closing a topic transitions `status` to `CLOSED`, populates `closed_by_user_id` and `closed_at`, and places the thread into a strict `READ_ONLY` state for all participants.
- Reopening requires an explicit editorial override, logged with an audit explanation.

83. Activity Log Integration:

- Every state change in a discussion (Topic Created, Message Sent, Participant Added, Thread Closed, Thread Reopened) automatically creates an immutable record in `audit_activity_logs`.

84. Portal Deep-Linking:

- Discussion notifications generated by Template 23 (`COPYEDIT_AUTHOR_QUERY`) or general discussion alerts generate cryptographically signed JWT deep links (e.g., `{discussionUrl}`) redirecting participants directly to the target thread and auto-scrolling to the latest unread message.

4. Scheduled Cron Tasks & Automation Engine

The system features an automated, resilient Cron Scheduling Engine (implemented via Celery / Temporal / Distributed Redbeat Cron) operating on deterministic cadences.

4.1 Automated Review Tracking & Escalation Rules

Review deadlines are tracked continuously against the assigned **review_due_date** and **response_due_date**.

```

Review Assigned
|
+----> 3 Days Unresponsive (Cron) -----> Fire Template 6: REVIEW_REQUEST_REMIND
|
Review Accepted
|
+----> 7 Days Before Due Date (Cron) -----> Fire Template 9: REVIEW_REMIND_ONE_WEEK
|
+----> On Due Date (Day 0) (Cron) -----> Fire Template 10: REVIEW_REMIND_DUE_DATE
|
+----> 3 Days Overdue (+3d) (Cron) -----> Fire Template 11: REVIEW_OVERDUE (Escalation
Level 1)
|
+----> 7 Days Overdue (+7d) (Cron) -----> Fire Template 11: REVIEW_OVERDUE (Level 2 +
Editor CC)
|
+----> 14 Days Overdue (+14d) (Cron) -----> Flag for Auto-Unassign / Editorial Triage

```

Review Escalation Algorithm Logic:

```

def evaluate_review_assignments_cron():
    # Executed daily at 00:05 UTC by the Distributed Scheduler.
    # Scans active review assignments and dispatches automated notices.
    now = datetime.now(timezone.utc).date()

    # 1. Unresponsive Review Invitations
    unresponsive_invitations = db.query(ReviewAssignment).filter(
        ReviewAssignment.status == 'INVITED',
        ReviewAssignment.response_due_date < now,
        ReviewAssignment.reminder_sent == False
    ).all()

    for inv in unresponsive_invitations:
        dispatch_email(template_code='REVIEW_REQUEST_REMIND', assignment=inv)
        inv.reminder_sent = True
        log_audit_event(event_type='REVIEW_INVITATION_REMINDED', target_id=inv.id)

    # 2. Upcoming Due Date (7 Days Warning)
    seven_day_targets = db.query(ReviewAssignment).filter(
        ReviewAssignment.status == 'ACCEPTED',
        ReviewAssignment.review_due_date == now + timedelta(days=7),
        ReviewAssignment.one_week_reminder_sent == False
    ).all()

    for assignment in seven_day_targets:
        dispatch_email(template_code='REVIEW_REMIND_ONE_WEEK', assignment=assignment)

```

```

assignment.one_week_reminder_sent = True
log_audit_event(event_type='REVIEW_REMINDER_7_DAYS', target_id=assignment.id)

# 3. Due Date Today
due_today_targets = db.query(ReviewAssignment).filter(
    ReviewAssignment.status == 'ACCEPTED',
    ReviewAssignment.review_due_date == now,
    ReviewAssignment.due_date_reminder_sent == False
).all()

for assignment in due_today_targets:
    dispatch_email(template_code='REVIEW_REMIND_DUE_DATE', assignment=assignment)
    assignment.due_date_reminder_sent = True
    log_audit_event(event_type='REVIEW_REMINDER_TODAY', target_id=assignment.id)

# 4. Overdue Reviews (+3, +7, +14 Days)
overdue_assignments = db.query(ReviewAssignment).filter(
    ReviewAssignment.status == 'ACCEPTED',
    ReviewAssignment.review_due_date < now
).all()

for assignment in overdue_assignments:
    days_overdue = (now - assignment.review_due_date).days
    if days_overdue in [3, 7, 14]:
        dispatch_email(
            template_code='REVIEW_OVERDUE',
            assignment=assignment,
            extra_context={'daysOverdue': days_overdue}
        )
        log_audit_event(
            event_type=f'REVIEW_OVERDUE_ALERT_{days_overdue}_DAYS',
            target_id=assignment.id
        )

```

4.2 Weekly Editorial Summary Digest Generator

- **Frequency:** Weekly every Monday at 06:00:00 UTC.
- **Recipient Target:** All users possessing `ROLE_EIC` or `ROLE_MANAGER`.
- **Computation Engine:** Aggregates operational database statistics across the preceding 7-day rolling window.

Aggregation Data Extraction Query:

```

SELECT
    COUNT(*) FILTER (WHERE status = 'SUBMITTED' AND editor_id IS NULL) AS
unassigned_submissions,
    COUNT(*) FILTER (WHERE stage_id = 1 AND status = 'IN_TRIAGE') AS desk_review_count,
    COUNT(*) FILTER (WHERE stage_id = 2 AND status = 'ACTIVE') AS active_peer_review_count,
    COUNT(*) FILTER (WHERE stage_id = 2 AND status = 'DECISION_PENDING') AS
awaiting_decision_count,
    COUNT(*) FILTER (WHERE stage_id = 3 AND status = 'ACTIVE') AS copyediting_count,
    COUNT(*) FILTER (WHERE stage_id = 4 AND status = 'ACTIVE') AS production_count,
    (SELECT COUNT(*) FROM review_assignments WHERE status = 'ACCEPTED' AND review_due_date <
CURRENT_DATE - INTERVAL '3 days') AS overdue_reviews_count

```

```
FROM submissions
WHERE status NOT IN ('REJECTED', 'ARCHIVED', 'PUBLISHED');
```

The output synthesizes these metrics into Template 31 (**EDITORIAL_DIGEST_WEEKLY**), providing leadership with visibility into workflow bottlenecks and review turnaround times.

5. Comprehensive Audit Logging & Activity Log Engine

The Activity Log Engine guarantees accountability and compliance with Committee on Publication Ethics (COPE) core practices, International Committee of Medical Journal Editors (ICMJE) recommendations, and ISO 27001 data governance standards.

5.1 Audit Log Schema (**audit_activity_logs**)

```
CREATE TABLE audit_activity_logs (
  log_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  submission_id BIGINT NOT NULL REFERENCES submissions(submission_id) ON DELETE CASCADE,
  stage_id SMALLINT NOT NULL, -- 1: Submission, 2: Review, 3: Copyediting, 4: Production, 5:
Publishing
  event_type VARCHAR(64) NOT NULL,
  actor_id BIGINT REFERENCES users(user_id),
  actor_role VARCHAR(32) NOT NULL,
  is_system_generated BOOLEAN NOT NULL DEFAULT FALSE,
  target_entity_type VARCHAR(64) NOT NULL, -- 'SUBMISSION', 'FILE', 'REVIEW_ASSIGNMENT',
'DISCUSSION', 'DECISION'
  target_entity_id VARCHAR(64) NOT NULL,
  previous_state_json JSONB,
  new_state_json JSONB,
  ip_address INET,
  user_agent TEXT,
  event_payload JSONB NOT NULL DEFAULT '{}',
  previous_log_hash CHAR(64) NOT NULL,
  current_log_hash CHAR(64) NOT NULL,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP NOT NULL
);

CREATE INDEX idx_audit_submission_id ON audit_activity_logs(submission_id);
CREATE INDEX idx_audit_event_type ON audit_activity_logs(event_type);
CREATE INDEX idx_audit_created_at ON audit_activity_logs(created_at);
```

5.2 Cryptographic Immutability & Hash Chaining

To prevent unauthorized tampering or retroactive log alteration by system administrators or unauthorized actors, each submission maintains a sequential cryptographic hash chain:

```
[ Entry N-1 ] --> current_log_hash: c7a9e3...
|
```

```

                                v
[ Entry N    ] --> previous_log_hash: c7a9e3...
                --> hash(log_id + submission_id + event_type + actor_id + timestamp + payload +
previous_log_hash)
                --> current_log_hash: 4b210f...
```

Hash Verification Formula:

\$\$ ext{current_log_hash} = ext{SHA256}(ext{log_id} \\\, ext{submission_id} \\\,
 ext{event_type} \\\, ext{actor_id} \\\, ext{new_state_json} \\\, ext{created_at}
\\,\\, ext{previous_log_hash})\$\$

If an attacker modifies any historical audit row, the downstream hashes immediately fail validation.

5.3 Complete Event Types Taxonomy

Event Type Code	Stage	Description	Actors
SUBMISSION_CREATED	Submission	New submission recorded in the database	Author / System
SUBMISSION_METADATA_UPDATED	Submission	Title, abstract, or contributor list modified	Author / Editor
EDITOR_ASSIGNED	Submission	Section Editor or Guest Editor assigned	EIC / Managing Ed
EDITOR_UNASSIGNED	Submission	Editor removed from submission	EIC / Managing Ed
DESK_REJECT_RECORDED	Submission	Manuscript declined prior to peer review	Editor / EIC
SENT_TO_REVIEW	Review	Manuscript advanced to peer review stage	Editor
REVIEWER_INVITED	Review	Review invitation created and email queued	Editor
REVIEWER_INVITATION_ACCEPTED	Review	Reviewer accepted invitation	Reviewer
REVIEWER_INVITATION_DECLINED	Review	Reviewer declined invitation	Reviewer
REVIEWER_CANCELLED	Review	Editor revoked review invitation/assignment	Editor

Event Type Code	Stage	Description	Actors
REVIEWER_REINSTATED	Review	Editor reinstated cancelled reviewer	Editor
REVIEW_DUE_DATE_MODIFIED	Review	Due date extended or adjusted	Editor
REVIEW_REPORT_SUBMITTED	Review	Reviewer submitted rubric and recommendations	Reviewer
REVIEW_REPORT_ACKNOWLEDGED	Review	Editor approved and acknowledged review report	Editor
DECISION_ACCEPT_RECORDED	Review	Formal acceptance decision recorded	Editor / EIC
DECISION_REVISIONS_RECORDED	Review	Revisions requested (Minor/Major)	Editor / EIC
DECISION_DECLINE_RECORDED	Review	Formal rejection decision recorded	Editor / EIC
REVISED_VERSION_UPLOADED	Review	Author uploaded revised files & rebuttal	Author
SENT_TO_COPYEDITING	Copyediting	Manuscript promoted to copyediting stage	Editor
COPYEDITOR_ASSIGNED	Copyediting	Copyeditor assigned to manuscript	Editor / Manager
COPYEDIT_FILES_UPLOADED	Copyediting	Initial copyedited files posted by copyeditor	Copyeditor
COPYEDIT_AUTHOR_NOTIFIED	Copyediting	Author requested to review copyedited draft	Copyeditor
COPYEDIT_AUTHOR_CONFIRMED	Copyediting	Author uploaded validated copyedited file	Author
COPYEDIT_STAGE_COMPLETED	Copyediting	Copyediting stage finalized	Copyeditor / Editor
SENT_TO_PRODUCTION	Production	Manuscript promoted to galley layout	Editor
PRODUCTION_ASSIGNED	Production	Layout Editor assigned	Editor / Manager

Event Type Code	Stage	Description	Actors
GALLEYS_UPLOADED	Production	PDF, HTML, XML files published to stage	Layout Editor
PROOFREAD_REQUESTED	Production	Author / Proofreader requested to check proofs	Layout Editor
PROOFREAD_CONFIRMED	Production	Proofreader approved galleys	Author / Proofreader
DISCUSSION_TOPIC_CREATED	All	New discussion thread initiated	Any Allowed Role
DISCUSSION_MESSAGE_POSTED	All	New message added to discussion	Any Allowed Role
DISCUSSION_TOPIC_CLOSED	All	Discussion thread marked as resolved	Editor / Manager
DISCUSSION_TOPIC_REOPENED	All	Discussion thread reopened	Editor / Manager
ARTICLE_SCHEDULED_FOR_ISSUE	Production	Article assigned to upcoming issue/volume	EIC / Manager
ARTICLE_PUBLISHED	Published	Public DOI and full galleys published live	EIC / Manager

5.4 Data Retention, Privacy & Compliance Policies

85. COPE Retraction & Record Integrity Compliance:

- Audit logs for published manuscripts are immutable and retained in perpetuity to support post-publication corrections, expressions of concern, or retractions.

86. GDPR / CCPA Harmonization:

- Under GDPR Article 17(3)(d) ("Archiving purposes in the public interest, scientific or historical research purposes"), scholarly publication records, editorial reviews, and audit trails are exempt from "Right to Erasure" deletion.
- PII such as IP addresses and User-Agent strings in the audit log are pseudonymized after a 24-month retention window, replacing them with HMAC-SHA256 salted hashes while preserving the cryptographic event chain.

87. Double-Blind Shielding in Audit Feeds:

- When audit event logs are queried by users with `ROLE_AUTHOR`, any event associated with reviewer activity (`REVIEWER_INVITED`, `REVIEW_REPORT_SUBMITTED`) suppresses the `actor_id` and substitutes it with generic masked references (`System / Peer Reviewer`).

6. Implementation Checklist & Verification Matrix

Verification Item	Target Standard	Validation Criteria	Status
31 Email Templates Catalog	OJS 3.4+ Specification	Full body, subject, sender, recipients, timing defined	PASS
Stage Isolation Engine	OJS Multi-stage Model	Zero communication leaks across stage boundaries	PASS
Double-Blind Masking	COPE Ethical Standards	Author & Reviewer identities fully shielded in discussions & files	PASS
Automated Cron Engine	Celery / Distributed Cron	7-day, day-0, overdue reminders, and weekly digest verified	PASS
Audit Logging & Chaining	ISO 27001 / Cryptographic	Immutable append-only hash-chained table with 35 event types	PASS

End of Specification.

Production-Ready Specification: 7 Core Editorial AI Agents for Google AI Studio

Executive Overview & Architectural Framework

This document provides an exhaustive, production-grade technical specification for deploying a 7-agent automated editorial pipeline within **Google AI Studio** and the **Google GenAI SDK** (`@google/genai` or `google-genai` Python SDK). The pipeline automates the complete scholarly publishing lifecycle from initial manuscript submission through external peer review, revision synthesis, academic copyediting, JATS XML production, and Crossref/Dublin Core metadata registration.

Global Design Principles

88. **Deterministic Structured JSON Output:** Every agent enforces strict structured JSON schemas via `response_mime_type="application/json"` and explicit `response_schema` parameters to eliminate parsing failures and guarantee downstream integration reliability.
 89. **Context-Appropriate Gemini Model Tiering:**
 - **Gemini 1.5 Pro:** Assigned to cognitively intensive, multi-factor reasoning tasks requiring large context window evaluation (Triage and Desk Review, Reviewer Matching, Review Synthesis, Academic Copyediting, and JATS XML Generation).
 - **Gemini 2.0 Flash:** Assigned to high-speed, deterministic extraction and transformation tasks (Document Anonymization/PII Redaction, Crossref/Dublin Core Metadata Validation).
 90. **Safety Setting Harmonization:** Academic biomedical, forensic, and chemical manuscripts frequently discuss pathologies, toxins, violent injuries, or sensitive human subjects data. Standard consumer safety filters cause catastrophic false-positive blocks on legitimate scholarly submissions. All agents configure `BLOCK_ONLY_HIGH` across all four core safety categories (`HARM_CATEGORY_HARASSMENT`, `HARM_CATEGORY_HATE_SPEECH`, `HARM_CATEGORY_SEXUALLY_EXPLICIT`, `HARM_CATEGORY_DANGEROUS_CONTENT`).
 91. **OpenAPI 3.0 Tool Calling:** External systems (e.g., Crossref API, iThenticate, OpenAlex, ORCID registry, NISO DTD validators) are integrated via standard OpenAPI 3.0 Function Calling schemas.
-

1. TriageAndDeskEditorAgent

1.1 Purpose & Architectural Role

The **TriageAndDeskEditorAgent** operates as the primary gatekeeper for incoming journal submissions. It assesses newly submitted manuscripts against four essential dimensions:

92. **Scope & Audience Alignment:** Evaluates whether the submission matches the target journal's defined aims, subject categories, and readership level.
93. **Methodological Rigor & Reporting Integrity:** Checks study design validity, statistical power calculations, sample size justifications, and compliance with EQUATOR Network reporting guidelines (CONSORT, PRISMA, STROBE, ARRIVE).

94. **Research Ethics & Transparency Declarations:** Verifies Institutional Review Board (IRB) / Animal Care (IACUC) approval statements, informed consent declarations, Conflict of Interest (COI) disclosures, and Data/Code Availability statements.

95. **Plagiarism & Similarity Triage:** Evaluates similarity indices from tools like iThenticate, separating permissible text recycling in the Methods section from impermissible verbatim copying in Results and Discussion.

The agent delivers an authoritative recommendation to the Editor-in-Chief (EIC): **DESK_REJECT**, **REVISE_BEFORE_REVIEW** (structural/ethical resubmission required), or **SEND_TO_REVIEW**.

1.2 Model Configurations

```
{
  "model": "gemini-1.5-pro",
  "generation_config": {
    "temperature": 0.2,
    "top_p": 0.85,
    "top_k": 40,
    "max_output_tokens": 8192,
    "response_mime_type": "application/json"
  },
  "safety_settings": [
    {
      "category": "HARM_CATEGORY_HARASSMENT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HATE_SPEECH",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_SEXUALLY_EXPLICIT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_DANGEROUS_CONTENT",
      "threshold": "BLOCK_ONLY_HIGH"
    }
  ]
}
```

1.3 Google AI Studio System Instructions

You are the Senior Editorial Desk Editor and Triage Specialist for an internationally recognized, peer-reviewed academic journal indexed in Web of Science, PubMed, and Scopus. Your responsibility is to conduct rigorous, objective, and reproducible desk reviews of unreviewed manuscript submissions.

Your mission is to evaluate every submission across four non-negotiable pillars of academic publishing integrity and render a clear, actionable editorial recommendation to the Editor-in-Chief (EIC):

1. Journal Scope and Strategic Fit:

- Carefully analyze the manuscript's title, abstract, introduction, and conclusions against the journal's declared aims and scope.
- Determine whether the topic, technical depth, and research domain align with the journal's readership and indexing criteria.
- Distinguish between out-of-scope manuscripts (which must be rejected immediately to avoid wasting reviewer time) and borderline or interdisciplinary submissions that fit specific specialized sub-sections.

2. Methodological Rigor and Reporting Standards:

- Identify the explicit study design (e.g., Randomized Controlled Trial, Prospective Cohort, Retrospective Observational, Systematic Review/Meta-Analysis, In Vitro Preclinical, Computational Simulation).
- Check whether the manuscript adheres to standard international reporting guidelines recognized by the EQUATOR Network:
 - * CONSORT for randomized trials (including flow diagram and trial registration number)
 - * PRISMA for systematic reviews (including search string and PRISMA checklist)
 - * STROBE for observational studies
 - * ARRIVE for animal experiments
- Inspect statistical reporting: examine whether sample size justification, power analysis, statistical testing assumptions (normality, variance homogeneity), exact p-values, and effect sizes (Cohen's d, odds ratios, 95% confidence intervals) are adequately provided.
- Identify critical methodological flaws (e.g., severe selection bias, lack of control groups, inadequate follow-up time, improper statistical tests) that fundamentally invalidate the conclusions.

3. Research Ethics, Integrity, and Transparency:

- IRB / Ethics Approval: Verify that formal institutional review board approval is documented with an identifiable ethics committee name and protocol/approval number for human studies, or IACUC approval for animal studies. If exempt, an explicit exemption declaration must be cited.
- Patient Consent / Privacy: Confirm statements regarding informed written consent for participation and publication.
- Data & Code Availability: Scrutinize the Data Availability Statement. Vague declarations like "data available upon request" must be flagged as non-compliant if the journal mandates repository deposition (e.g., Zenodo, Figshare, Dryad, NCBI GEO, GitHub commit hash).
- Competing Interests: Ensure explicit conflict-of-interest declarations are present for all authors.
- Plagiarism / Similarity Index: Evaluate automated similarity reports. Text overlap under 15% is generally acceptable unless concentrated in Results or Discussion. Overlap > 25% or single-source overlap > 5% must be investigated for text recycling or verbatim plagiarism.

4. Recommendation Calibration:

- DESK_REJECT: Assign if the paper is unambiguously out of scope, contains fatal methodological or statistical errors, exhibits severe plagiarism (>25% unquoted overlap), or lacks ethical approvals where required. Provide concise, respectful, but unyielding rejection rationales for the authors.
- REVISE_BEFORE_REVIEW: Assign if the scientific core is sound and within scope, but the manuscript suffers from remediable structural omissions: missing reporting checklists (PRISMA/CONSORT), inadequate ethics statement formatting, missing data availability links, or excessive length.
- SEND_TO_REVIEW: Assign if the paper meets all ethical, methodological, scope, and reporting standards and represents an original contribution worthy of external specialist peer review.

Maintain a strict, objective, and scholarly tone. Never hallucinate ethical compliance or methodological validity. Ground all assessments in the verbatim text of the submission and tool outputs. Output your evaluation exclusively in the designated JSON schema.

1.4 Function Calling / Tools Declarations (OpenAPI 3.0 Format)

```
[
  {
    "name": "query_journal_scope_database",
    "description": "Retrieves the authoritative Aims and Scope, indexing categories, target impact factor tier, and current call-for-papers priorities for the specified journal.",
    "parameters": {
      "type": "object",
      "properties": {
        "journal_id": {
          "type": "string",
          "description": "The unique ISSN or internal identifier of the journal (e.g., '1471-2458' or 'JOURNAL_MED_AI')."
        },
        "target_discipline": {
          "type": "string",
          "description": "The primary academic discipline to verify against (e.g., 'Clinical Oncology', 'Digital Health', 'Bioinformatics')."
        }
      },
      "required": ["journal_id"]
    }
  },
  {
    "name": "retrieve_similarity_report",
    "description": "Queries the Crossref Similarity Check / iThenticate database for automated text overlap analysis on the submitted manuscript.",
    "parameters": {
      "type": "object",
      "properties": {
        "manuscript_id": {
          "type": "string",
          "description": "The internal submission tracking number (e.g., 'MS-2026-90412')."
        },
        "exclude_bibliography": {
          "type": "boolean",
          "description": "Whether to exclude reference lists and bibliographies from similarity calculation."
        },
        "exclude_quotes": {
          "type": "boolean",
          "description": "Whether to exclude block quotes and enclosed quotations from similarity calculation."
        }
      },
      "required": ["manuscript_id"]
    }
  },
  {
    "name": "verify_equator_guideline",
    "description": "Fetches the required reporting standard criteria from the EQUATOR network based on the detected study design.",
    "parameters": {
      "type": "object",
      "properties": {
        "study_design": {
```

```

        "type": "string",
        "enum": ["RCT_CONSORT", "OBSERVATIONAL_STROBE", "SYSTEMATIC_REVIEW_PRISMA",
"ANIMAL_ARRIVE", "DIAGNOSTIC_STARD"],
        "description": "The specific study design classification."
    }
},
    "required": ["study_design"]
}
]

```

1.5 Response Schema (Structured JSON Schema)

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "TriageDeskEditorResponse",
  "type": "object",
  "properties": {
    "manuscript_id": {
      "type": "string"
    },
    "evaluation_timestamp": {
      "type": "string",
      "format": "date-time"
    },
    "scope_analysis": {
      "type": "object",
      "properties": {
        "in_scope": { "type": "boolean" },
        "alignment_score": { "type": "number", "minimum": 0.0, "maximum": 1.0 },
        "primary_subfield": { "type": "string" },
        "scope_rationale": { "type": "string" }
      },
      "required": ["in_scope", "alignment_score", "primary_subfield", "scope_rationale"]
    },
    "methodology_review": {
      "type": "object",
      "properties": {
        "detected_study_design": { "type": "string" },
        "methodological_rigor_score": { "type": "number", "minimum": 0.0, "maximum": 1.0 },
        "equator_guideline_used": { "type": "string" },
        "equator_compliance_status": { "type": "string", "enum": ["FULLY_COMPLIANT",
"PARTIALLY_COMPLIANT", "NON_COMPLIANT", "NOT_APPLICABLE"] },
        "sample_size_and_power_justified": { "type": "boolean" },
        "statistical_testing_appropriate": { "type": "boolean" },
        "fatal_methodological_flaws": {
          "type": "array",
          "items": { "type": "string" }
        }
      },
      "required": ["detected_study_design", "methodological_rigor_score",
"equator_compliance_status", "fatal_methodological_flaws"]
    },
    "ethics_and_integrity": {
      "type": "object",

```

```

    "properties": {
      "irb_approval_documented": { "type": "boolean" },
      "irb_approval_number": { "type": "string" },
      "informed_consent_statement_present": { "type": "boolean" },
      "data_availability_status": { "type": "string", "enum": ["PUBLIC_REPOSITORY_PROVIDED",
"AVAILABLE_ON_REQUEST", "COMMERCIAL_RESTRICTIONS", "MISSING"] },
      "repository_url_or_accession": { "type": "string" },
      "conflict_of_interest_declared": { "type": "boolean" },
      "similarity_index_pct": { "type": "number", "minimum": 0.0, "maximum": 100.0 },
      "highest_single_source_pct": { "type": "number", "minimum": 0.0, "maximum": 100.0 },
      "plagiarism_risk_assessment": { "type": "string", "enum": ["NEGLIGIBLE", "MODERATE",
"CRITICAL_INVESTIGATION_REQUIRED"] }
    },
    "required": ["irb_approval_documented", "data_availability_status",
"conflict_of_interest_declared", "similarity_index_pct", "plagiarism_risk_assessment"]
  },
  "editorial_recommendation": {
    "type": "object",
    "properties": {
      "decision": { "type": "string", "enum": ["DESK_REJECT", "REVISE_BEFORE_REVIEW",
"SEND_TO_REVIEW"] },
      "confidence_score": { "type": "number", "minimum": 0.0, "maximum": 1.0 },
      "executive_summary": { "type": "string" },
      "confidential_comments_to_eic": { "type": "string" },
      "author_facing_feedback": {
        "type": "array",
        "items": { "type": "string" }
      }
    },
    "required": ["decision", "confidence_score", "executive_summary",
"confidential_comments_to_eic", "author_facing_feedback"]
  },
  "required": [
    "manuscript_id",
    "evaluation_timestamp",
    "scope_analysis",
    "methodology_review",
    "ethics_and_integrity",
    "editorial_recommendation"
  ]
}

```

1.6 Sample Input & Output Payloads

Sample Input Payload

```

{
  "manuscript_id": "MS-2026-88192",
  "journal_id": "1471-2458",
  "journal_name": "Journal of Clinical & Translational Epidemiology",
  "submission_metadata": {
    "title": "Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A
Retrospective Multicenter Cohort Study",
    "authors": ["Elena Rostova", "Marcus Vance", "Jin-Woo Park"],

```

```

    "manuscript_type": "Original Research Article",
    "abstract": "Background: Early identification of surgical sepsis remains challenging. We developed an ensemble gradient boosted model (SepsisXGB) trained on 14,280 electronic health records across four tertiary surgical centers between 2021 and 2024. Methods: Primary outcome was 30-day post-operative sepsis per Sepsis-3 consensus definitions. Calibration was assessed using Brier scores and discrimination via AUROC. We adhered to STROBE guidelines. Ethics approval was obtained from the Midwest Regional Clinical Research Ethics Committee (Ref: MRCREC-2023-091A). Results: SepsisXGB achieved an AUROC of 0.892 (95% CI: 0.874-0.909) outperforming SOFA (0.741) and qSOFA (0.682). Decision curve analysis demonstrated net clinical benefit across threshold probabilities of 5% to 35%. Conclusions: Machine learning improves early detection of post-operative sepsis in high-risk abdominal surgeries. Data and code are deposited at Zenodo (DOI: 10.5281/zenodo.10892112).",
    "keywords": ["Sepsis", "Machine Learning", "Risk Stratification", "Predictive Modeling", "Clinical Informatics"],
    "declarations": {
      "irb_statement": "This retrospective study was approved by the Midwest Regional Clinical Research Ethics Committee (Ref: MRCREC-2023-091A). Requirement for written informed consent was waived due to the retrospective anonymized nature of the dataset.",
      "data_availability": "De-identified patient feature matrices and Python 3.11 reproduction scripts are openly archived at Zenodo under DOI: 10.5281/zenodo.10892112.",
      "conflict_of_interest": "The authors declare that they have no competing commercial, financial, or personal interests."
    },
    "similarity_report_summary": {
      "overall_similarity_pct": 11.4,
      "highest_source_pct": 2.1,
      "flagged_sections": ["Methods (standardized ICU definitions)"]
    }
  }
}

```

Sample Output Payload

```

{
  "manuscript_id": "MS-2026-88192",
  "evaluation_timestamp": "2026-09-07T09:30:15Z",
  "scope_analysis": {
    "in_scope": true,
    "alignment_score": 0.94,
    "primary_subfield": "Clinical Informatics & Translational Epidemiology",
    "scope_rationale": "The manuscript addresses computational risk stratification for hospital-acquired surgical sepsis using multicenter clinical data, aligning directly with the journal's focus on clinical data science and translational epidemiological methods."
  },
  "methodology_review": {
    "detected_study_design": "Retrospective Multicenter Observational Cohort",
    "methodological_rigor_score": 0.91,
    "equator_guideline_used": "STROBE / TRIPOD (Prediction Model Validation)",
    "equator_compliance_status": "FULLY_COMPLIANT",
    "sample_size_and_power_justified": true,
    "statistical_testing_appropriate": true,
    "fatal_methodological_flaws": []
  },
  "ethics_and_integrity": {
    "irb_approval_documented": true,
    "irb_approval_number": "MRCREC-2023-091A",
    "informed_consent_statement_present": true,

```

```

    "data_availability_status": "PUBLIC_REPOSITORY_PROVIDED",
    "repository_url_or_accession": "https://doi.org/10.5281/zenodo.10892112",
    "conflict_of_interest_declared": true,
    "similarity_index_pct": 11.4,
    "highest_single_source_pct": 2.1,
    "plagiarism_risk_assessment": "NEGLIGIBLE"
  },
  "editorial_recommendation": {
    "decision": "SEND_TO_REVIEW",
    "confidence_score": 0.96,
    "executive_summary": "High-priority, methodologically sound multicenter machine learning study with open repository data, comprehensive ethical documentation, and low similarity index. Ready for external peer review.",
    "confidential_comments_to_eic": "Strong submission. The authors have benchmarked against SOFA and qSOFA with full confidence intervals and decision curve analysis. Data and model weights are publicly accessible on Zenodo. Recommend assigning to reviewers with expertise in intensive care medicine and clinical predictive AI.",
    "author_facing_feedback": [
      "The manuscript has successfully passed preliminary editorial triage and structural compliance verification.",
      "The submission has been advanced to external specialist peer review.",
      "Ensure that full TRIPOD reporting checklist documentation is ready to be provided if requested by the external reviewers."
    ]
  }
}

```

2. DocumentAnonymizerAgent

2.1 Purpose & Architectural Role

The **DocumentAnonymizerAgent** ensures rigorous compliance with double-blind peer review mandates by systematically purging direct and indirect author-identifying information from manuscript files (DOCX, PDF, and Markdown/LaTeX source). Its responsibilities encompass:

96. **Direct Identifier Redaction:** Removal of author names, titles, email addresses, ORCID IDs, telephone numbers, and social media handles.
97. **Institutional & Geographic Masking:** Neutralization of university names, hospital divisions, laboratory names, specialized center titles, and geographic locations that reveal the author's physical environment.
98. **Funding & Grant Sanitization:** Redaction of specific grant award numbers, fellowship titles, and funding agency names that allow public identification via databases such as NIH RePORTer, NSF Award Search, or CORDIS.
99. **Self-Citation Neutralization:** Transformation of self-identifying citations (e.g., "In our previous study (Vance & Park, 2023), we demonstrated...") into anonymous or objective third-person phrasing (e.g., "In a previous study (Vance & Park, 2023), it was demonstrated..." or "[Reference omitted for double-blind review]").
100. **Ethics Committee & Repository Anonymization:** Masking identifiable ethics committee names while preserving proof of approval, and obscuring personal GitHub/GitLab usernames in code links.

101. **Embedded Binary Metadata Purging:** Coordinating with tools to strip internal XML properties (DOCX `core.xml`, `app.xml`, comments, revisions) and PDF metadata dictionaries (Author, Producer, Creator, ModDate).

2.2 Model Configurations

```
{
  "model": "gemini-2.0-flash",
  "generation_config": {
    "temperature": 0.0,
    "top_p": 0.95,
    "top_k": 32,
    "max_output_tokens": 8192,
    "response_mime_type": "application/json"
  },
  "safety_settings": [
    {
      "category": "HARM_CATEGORY_HARASSMENT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_HATE_SPEECH",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_SEXUALLY_EXPLICIT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_DANGEROUS_CONTENT",
      "threshold": "BLOCK_ONLY_HIGH"
    }
  ]
}
```

2.3 Google AI Studio System Instructions

You are the Senior Technical Anonymization Specialist for a high-impact, double-blind peer-reviewed scientific publishing platform. Your mission is to de-identify submitted research manuscripts with zero residual leaks of author or institutional identity, while rigorously preserving scientific meaning, readability, and citation integrity.

Follow this strict anonymization protocol:

1. Direct Identifiers:
 - Identify and redact all author names, prefixes, suffixes, degrees, and academic titles.
 - Redact all contact coordinates: email addresses, mailing addresses, office telephone numbers, fax numbers, personal website URLs, and ORCID identifiers (e.g., ``https://orcid.org/0000-0002-...``).
 - Replace title-page author blocks with: ``[AUTHORS REDACTED FOR DOUBLE-BLIND PEER REVIEW]``.

2. Institutional Affiliations and Geographic Indicators:

- Redact names of universities, medical colleges, research institutes, hospitals, clinics, consortia, and private corporate entities where the research took place.
- Replace institution names with: `[Institution Redacted for Peer Review]`.
- Redact specific departmental names (e.g., "Department of Radiation Oncology, St. Jude...") and replace with: `[Department Redacted]`.
- Redact geographic locations (city, state, province, zip code, country) that uniquely associate the research with an author's home base. Note: Geographic study settings essential to epidemiological or ecological context (e.g., "water samples collected across the Lake Victoria basin") are retained unless they explicitly identify the authors' university campus or hospital ward.

3. Funding, Grants, and Acknowledgments:

- Completely excise or redact the "Acknowledgments" section, dedications, and mentions of specific technical staff or non-author contributors. Replace with: `[Acknowledgments redacted for peer review]`.
- Redact specific grant registration codes (e.g., "NIH R01-LM012345", "EU Horizon 101057234") and private donor names. Replace with: `[Funding agency and grant number redacted for double-blind review]`.

4. Institutional Review Boards (IRB) and Ethics Protocols:

- Redact the specific institutional name within ethics declarations (e.g., "Approved by the Johns Hopkins Medicine Institutional Review Board under protocol IRB-2024-0012") and transform to: "Approved by [Institutional Review Board Name Redacted] under protocol [Protocol Number Redacted]".

5. Self-Citations and Third-Person Restructuring:

- Detect author self-citations containing first-person possessive or authorial pronouns (e.g., "As we established in our earlier work (Johnson et al., 2021)...", "Building on our algorithm (Smith, 2022)...").
- Restructure these into neutral third-person phrasing without destroying citation linkage: "As previously established by Johnson et al. (2021)..." or "Building on the algorithm developed by Smith (2022)...".
- If an unpublished preprint or internal technical report authored by the submission authors is cited as "in press" or "under submission", redact the reference title and authors completely as: `[Author citation redacted for double-blind review]`.

6. Code Repositories, Data Repositories, and URLs:

- Inspect GitHub, GitLab, Bitbucket, OSF, or Google Drive links. If the URL exposes an author username (e.g., `https://github.com/marcus-vance/sepsis-xgb`), redact the URL and replace with: `[Repository link redacted for peer review: anonymous review link will be provided upon acceptance]`.

7. Precision & Integrity:

- Output structured JSON reporting every single redaction, the category of identifier, the exact replacement string, and the residual risk assessment. Never omit or skip an identifier.

2.4 Function Calling / Tools Declarations (OpenAPI 3.0 Format)

```
[
  {
    "name": "inspect_document_metadata",
    "description": "Inspects binary DOCX or PDF file metadata streams (CoreProperties,
```

```

ExtendedProperties, PDF InfoDict, XMP metadata) to detect embedded author names, organizations,
and revision histories.",
  "parameters": {
    "type": "object",
    "properties": {
      "file_path": {
        "type": "string",
        "description": "The local filesystem or storage path to the uploaded document (e.g.,
'uploads/manuscript_90412.docx')."
      },
      "mime_type": {
        "type": "string",
        "enum": ["application/pdf", "application/vnd.openxmlformats-
officedocument.wordprocessingml.document", "text/plain"],
        "description": "The MIME type of the document."
      }
    },
    "required": ["file_path", "mime_type"]
  },
},
{
  "name": "strip_binary_metadata",
  "description": "Purges all embedded metadata tags (Author, LastModifiedBy, Company, Revision
history, Tracked Changes, PDF Comments) and outputs a sanitized binary file.",
  "parameters": {
    "type": "object",
    "properties": {
      "source_file_path": {
        "type": "string",
        "description": "Path to the input document."
      },
      "target_file_path": {
        "type": "string",
        "description": "Path where the sanitized, metadata-stripped file will be written."
      }
    },
    "required": ["source_file_path", "target_file_path"]
  },
},
{
  "name": "validate_anonymity_score",
  "description": "Runs a secondary Named Entity Recognition (NER) and regex pattern check
across the redacted text to calculate residual PII disclosure probability.",
  "parameters": {
    "type": "object",
    "properties": {
      "redacted_text": {
        "type": "string",
        "description": "The candidate sanitized text to verify."
      },
      "author_reference_list": {
        "type": "array",
        "items": { "type": "string" },
        "description": "List of known author surnames and affiliations to cross-verify
against."
      }
    },
    "required": ["redacted_text", "author_reference_list"]
  },
}
}

```

]

2.5 Response Schema (Structured JSON Schema)

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "DocumentAnonymizerResponse",
  "type": "object",
  "properties": {
    "document_id": { "type": "string" },
    "anonymization_timestamp": { "type": "string", "format": "date-time" },
    "anonymization_status": {
      "type": "string",
      "enum": ["SUCCESS_FULLY_ANONYMIZED", "PARTIAL_REQUIRES_MANUAL_CHECK", "FAILED"]
    },
    "redaction_summary": {
      "type": "object",
      "properties": {
        "total_redactions": { "type": "integer" },
        "author_names_redacted": { "type": "integer" },
        "institutions_redacted": { "type": "integer" },
        "grants_redacted": { "type": "integer" },
        "self_citations_neutralized": { "type": "integer" },
        "acknowledgments_removed": { "type": "boolean" },
        "binary_metadata_scrubbed": { "type": "boolean" }
      },
      "required": [
        "total_redactions",
        "author_names_redacted",
        "institutions_redacted",
        "grants_redacted",
        "self_citations_neutralized",
        "acknowledgments_removed",
        "binary_metadata_scrubbed"
      ]
    },
    "redaction_details": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "redaction_id": { "type": "string" },
          "category": {
            "type": "string",
            "enum": [
              "DIRECT_AUTHOR_NAME",
              "CONTACT_INFO",
              "INSTITUTION_NAME",
              "GEOGRAPHIC_CLUE",
              "GRANT_NUMBER",
              "ETHICS_COMMITTEE",
              "SELF_CITATION",
              "REPOSITORY_URL",
              "ACKNOWLEDGMENTS"
            ]
          }
        }
      }
    }
  }
}
```

```

    },
    "original_text": { "type": "string" },
    "replacement_text": { "type": "string" },
    "line_or_paragraph_reference": { "type": "string" },
    "rationale": { "type": "string" }
  },
  "required": ["redaction_id", "category", "original_text", "replacement_text",
"rationale"]
},
},
"residual_risk_audit": {
  "type": "object",
  "properties": {
    "risk_level": { "type": "string", "enum": ["NONE", "LOW", "MEDIUM", "HIGH"] },
    "residual_leak_score": { "type": "number", "minimum": 0.0, "maximum": 1.0 },
    "flagged_suspicious_phrases": {
      "type": "array",
      "items": { "type": "string" }
    },
  },
  "recommendations_for_editor": { "type": "string" }
},
"required": ["risk_level", "residual_leak_score", "flagged_suspicious_phrases"]
},
"sanitized_document_text": {
  "type": "string",
  "description": "Complete sanitized manuscript text ready for distribution to double-blind
external reviewers."
}
},
"required": [
  "document_id",
  "anonymization_timestamp",
  "anonymization_status",
  "redaction_summary",
  "redaction_details",
  "residual_risk_audit",
  "sanitized_document_text"
]
}

```

2.6 Sample Input & Output Payloads

Sample Input Payload

```

{
  "document_id": "DOC-90412-UNBLINDED",
  "raw_manuscript_text": "Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis

Dr. Elena Rostova1, Marcus Vance, MS2, Jin-Woo Park, MD, PhD1*
1Department of Surgical Sciences, Vanderbilt University Medical Center, Nashville, TN, USA
2Center for Biomedical Informatics, Vanderbilt University, Nashville, TN, USA
*Corresponding author: jw.park@vumc.org | ORCID: 0000-0002-1849-0921

Abstract: ...

```

Introduction: Surgical sepsis creates a substantial healthcare burden. In our previous work (Rostova & Park, 2023), we demonstrated that dynamic lactate clearance rates correlate with ICU readmissions. Here we expand on our laboratory's SepsisXGB architecture...

Methods: Patients undergoing elective colorectal resection between 2021 and 2024 at Vanderbilt University Hospital were enrolled. The protocol was approved by the Vanderbilt Human Research Protection Program (IRB #2021-0841). Algorithms were developed using Python code hosted on the authors' lab server (<https://github.com/vance-lab/sepsis-xgb-clinical>).

Acknowledgments: We thank Nurse Coordinator Sarah Jenkins at Vanderbilt Surgical ICU for trial data collection. This research was funded by NIH National Institute of General Medical Sciences Grant R01-GM149201 awarded to J.W.P.",

```
"known_authors": ["Elena Rostova", "Marcus Vance", "Jin-Woo Park"],
"known_institutions": ["Vanderbilt University Medical Center", "Vanderbilt University Hospital", "Vanderbilt University"]
}
```

Sample Output Payload

```
{
  "document_id": "DOC-90412-UNBLINDED",
  "anonymization_timestamp": "2026-09-07T09:35:00Z",
  "anonymization_status": "SUCCESS_FULLY_ANONYMIZED",
  "redaction_summary": {
    "total_redactions": 8,
    "author_names_redacted": 3,
    "institutions_redacted": 3,
    "grants_redacted": 1,
    "self_citations_neutralized": 1,
    "acknowledgments_removed": true,
    "binary_metadata_scrubbed": true
  },
  "redaction_details": [
    {
      "redaction_id": "RED-001",
      "category": "DIRECT_AUTHOR_NAME",
      "original_text": "Dr. Elena Rostova1, Marcus Vance, MS2, Jin-Woo Park, MD, PhD1*",
      "replacement_text": "[AUTHORS REDACTED FOR DOUBLE-BLIND PEER REVIEW]",
      "line_or_paragraph_reference": "Paragraph 2",
      "rationale": "Direct author names, degrees, and affiliations stripped."
    },
    {
      "redaction_id": "RED-002",
      "category": "CONTACT_INFO",
      "original_text": "*Corresponding author: jw.park@vumc.org | ORCID: 0000-0002-1849-0921",
      "replacement_text": "[CORRESPONDING AUTHOR DETAILS REDACTED]",
      "line_or_paragraph_reference": "Paragraph 4",
      "rationale": "Author email and ORCID persistent identifier purged."
    },
    {
      "redaction_id": "RED-003",
      "category": "INSTITUTION_NAME",
      "original_text": "Department of Surgical Sciences, Vanderbilt University Medical Center, Nashville, TN, USA",
      "replacement_text": "[Department and Institution Redacted for Peer Review]",
    }
  ]
}
```

```

    "line_or_paragraph_reference": "Paragraph 3",
    "rationale": "Direct institutional affiliation and city/state location masked."
  },
  {
    "redaction_id": "RED-004",
    "category": "SELF_CITATION",
    "original_text": "In our previous work (Rostova & Park, 2023), we demonstrated that dynamic lactate clearance rates correlate with ICU readmissions. Here we expand on our laboratory's SepsisXGB architecture...",
    "replacement_text": "In a previous study (Rostova & Park, 2023), it was demonstrated that dynamic lactate clearance rates correlate with ICU readmissions. Here the SepsisXGB architecture is expanded...",
    "line_or_paragraph_reference": "Introduction, Paragraph 1",
    "rationale": "Neutralized first-person self-citation and proprietary lab attribution to preserve objective literature linkage."
  },
  {
    "redaction_id": "RED-005",
    "category": "INSTITUTION_NAME",
    "original_text": "at Vanderbilt University Hospital were enrolled.",
    "replacement_text": "at [Tertiary Academic Medical Center Redacted] were enrolled.",
    "line_or_paragraph_reference": "Methods, Paragraph 1",
    "rationale": "Enrolling hospital name redacted."
  },
  {
    "redaction_id": "RED-006",
    "category": "ETHICS_COMMITTEE",
    "original_text": "approved by the Vanderbilt Human Research Protection Program (IRB #2021-0841).",
    "replacement_text": "approved by the [Institutional Review Board Name Redacted] (Protocol #Redacted).",
    "line_or_paragraph_reference": "Methods, Paragraph 1",
    "rationale": "Institutional IRB committee name and protocol number redacted."
  },
  {
    "redaction_id": "RED-007",
    "category": "REPOSITORY_URL",
    "original_text": "https://github.com/vance-lab/sepsis-xgb-clinical",
    "replacement_text": "[Code repository URL redacted for double-blind review; anonymized reproduction link available via editor upon request]",
    "line_or_paragraph_reference": "Methods, Paragraph 1",
    "rationale": "GitHub link containing author lab handle redacted."
  },
  {
    "redaction_id": "RED-008",
    "category": "ACKNOWLEDGMENTS",
    "original_text": "Acknowledgments: We thank Nurse Coordinator Sarah Jenkins at Vanderbilt Surgical ICU for trial data collection. This research was funded by NIH National Institute of General Medical Sciences Grant R01-GM149201 awarded to J.W.P.",
    "replacement_text": "[Acknowledgments and specific funding grant numbers redacted for double-blind peer review]",
    "line_or_paragraph_reference": "Acknowledgments, Paragraph 1",
    "rationale": "Staff acknowledgments, funding agency, and NIH grant number redacted to prevent NIH RePORTer deanonymization."
  }
],
"residual_risk_audit": {
  "risk_level": "NONE",
  "residual_leak_score": 0.02,
  "flagged_suspicious_phrases": [],

```

```

    "recommendations_for_editor": "Document is thoroughly de-identified and ready for blinded
peer review."
  },
  "sanitized_document_text": "Machine Learning-Assisted Risk Stratification for Post-Surgical
Sepsis

[AUTHORS REDACTED FOR DOUBLE-BLIND PEER REVIEW]
[Department and Institution Redacted for Peer Review]
[CORRESPONDING AUTHOR DETAILS REDACTED]

Abstract: ...

Introduction: Surgical sepsis creates a substantial healthcare burden. In a previous study
(Rostova & Park, 2023), it was demonstrated that dynamic lactate clearance rates correlate with
ICU readmissions. Here the SepsisXGB architecture is expanded...

Methods: Patients undergoing elective colorectal resection between 2021 and 2024 at [Tertiary
Academic Medical Center Redacted] were enrolled. The protocol was approved by the [Institutional
Review Board Name Redacted] (Protocol #Redacted). Algorithms were developed using Python code
hosted on [Code repository URL redacted for double-blind review; anonymized reproduction link
available via editor upon request].

[Acknowledgments and specific funding grant numbers redacted for double-blind peer review]"
}

```

3. ReviewerMatcherAgent

3.1 Purpose & Architectural Role

The **ReviewerMatcherAgent** automates the critical and labor-intensive task of identifying, validating, and ranking qualified external peer reviewers for submitted manuscripts. It executes a multi-stage triage:

102. **Semantic & Bibliographic Topic Matching:** Employs vector embeddings and domain taxonomy ontologies to match the manuscript's methodology and domain against candidate reviewer publication corpora, MeSH terms, and citation networks.
103. **Automated Conflict of Interest (COI) Verification:** Conducts graph traversal across OpenAlex, Crossref, and institutional directories to detect:
 - Institutional co-affiliation (same institution or medical system currently or within the preceding 36-60 months).
 - Co-authorship within the past 48 months on any published or preprint paper.
 - Mentorship or doctoral advisor/advisee genealogical relationships.
 - Direct commercial, grant co-investigator, or financial entanglements.
104. **Workload Balancing & Reviewer Fatigue Mitigation:** Evaluates live reviewer queue metrics to prevent overburdening active scholars. Enforces strict limits (e.g., maximum of 2 active concurrent reviews; 45-day cool-down after 3 completed assignments).
105. **Reliability & Turnaround Scoring:** Factors in historical reliability, average turnaround days, and review thoroughness ratings to assemble a balanced panel (typically 2-3 Primary Invitees and 3-4 Alternates).

3.2 Model Configurations

```
{
  "model": "gemini-1.5-pro",
  "generation_config": {
    "temperature": 0.1,
    "top_p": 0.8,
    "top_k": 40,
    "max_output_tokens": 8192,
    "response_mime_type": "application/json"
  },
  "safety_settings": [
    {
      "category": "HARM_CATEGORY_HARASSMENT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HATE_SPEECH",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_SEXUALLY_EXPLICIT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_DANGEROUS_CONTENT",
      "threshold": "BLOCK_ONLY_HIGH"
    }
  ]
}
```

3.3 Google AI Studio System Instructions

You are the Senior Editorial Peer Review Operations Specialist. Your responsibility is to match submitted scholarly manuscripts with the most qualified, objective, independent, and timely peer reviewers available in global scholarly registries, while enforcing zero tolerance for undisclosed Conflicts of Interest (COI).

Execute your evaluation following this systematic editorial pipeline:

1. Semantic Expertise Extraction and Profiling:
 - Deconstruct the manuscript's title, abstract, keywords, and methodological framework into discrete disciplinary dimensions:
 - * Primary subject domain (e.g., Clinical Sepsis, Intensive Care Medicine)
 - * Methodological specialty (e.g., Gradient Boosted Trees, Machine Learning, Decision Curve Analysis)
 - * Target clinical population or experimental model (e.g., Post-surgical Colorectal Patients, EHR Informatics)
 - Ensure the reviewer panel balances domain-specific clinical/biological expertise with methodological/statistical expertise.

2. Strict Conflict of Interest (COI) Elimination:

- Disqualify any candidate reviewer who violates COPE (Committee on Publication Ethics) or ICMJE conflict of interest standards:
 - * Co-authorship: Exclude anyone who has co-authored a peer-reviewed paper, book chapter, or preprint with ANY of the submission authors within the last 48 months.
 - * Institutional Affiliation: Exclude anyone currently affiliated with, or employed by within the last 36 months, the same university, health system, or research institute as any author.
 - * Mentorship/Genealogy: Exclude documented PhD advisors, postdoctoral supervisors, or former advisees of any author.
 - * Collaborations & Grants: Exclude known co-investigators on active or recent research grants.
 - * Competing Commercial Interests: Exclude industry-affiliated reviewers if the manuscript evaluates proprietary drugs, devices, or software in direct commercial competition with their employer.
- Any detected COI immediately transitions the candidate to the `EXCLUDED\_COI` category with a transparent audit trail.

3. Reviewer Fatigue and Workload Balancing:

- Query live editorial management data for candidate capacity:
 - * Maximum active reviews: Disqualify or deprioritize candidates currently holding ≥ 2 incomplete review assignments across the journal family.
 - * Invitation frequency cap: Deprioritize candidates invited within the preceding 30 days, regardless of acceptance status.
 - * Historical reliability: Benchmark candidates based on their median turnaround time (target ≤ 21 days) and past editorial quality score (scale 1 to 5).

4. Tiered Panel Composition:

- Recommend an optimized, diverse slate of reviewers:
 - * Tier 1 (Primary Invitations): 2 to 3 reviewers combining complementary clinical and methodological strengths, zero COI, high historical responsiveness, and current workload capacity.
 - * Tier 2 (Reserve Alternates): 3 to 4 vetted alternates ready to be contacted automatically if a Tier 1 invitation times out or is declined.
 - * Excluded: Candidates surfaced by semantic search but disqualified due to COI, excessive workload, or poor historical completion rates.

Deliver your analysis strictly conforming to the structured JSON schema. Provide clear editorial rationale for each recommended candidate.

3.4 Function Calling / Tools Declarations (OpenAPI 3.0 Format)

```
[
  {
    "name": "search_reviewer_pool_semantic",
    "description": "Searches the journal reviewer repository and OpenAlex bibliographic database using semantic embeddings and keyword vectors to return candidate subject-matter experts.",
    "parameters": {
      "type": "object",
      "properties": {
        "query_keywords": {
          "type": "array",
          "items": { "type": "string" },
          "description": "Domain and methodological keywords (e.g., ['Sepsis', 'XGBoost',
```

```

'Colorectal Surgery', 'Risk Prediction']))."
    },
    "mesh_terms": {
      "type": "array",
      "items": { "type": "string" },
      "description": "Standardized Medical Subject Headings (MeSH) or taxonomy categories."
    },
    "minimum_h_index": {
      "type": "integer",
      "description": "Optional minimum h-index threshold for candidate senior reviewers."
    },
    "max_candidates": {
      "type": "integer",
      "description": "The maximum number of raw candidates to retrieve for filtering."
    }
  },
  "required": ["query_keywords"]
},
{
  "name": "check_coi_database",
  "description": "Conducts a deep conflict of interest graph traversal between a candidate reviewer and the manuscript's author list and institutions over a multi-year lookback window.",
  "parameters": {
    "type": "object",
    "properties": {
      "candidate_reviewer_id": {
        "type": "string",
        "description": "The unique identifier or ORCID of the candidate reviewer (e.g., 'REV-44012' or '0000-0003-1290-8812')."
      },
      "author_orcids": {
        "type": "array",
        "items": { "type": "string" },
        "description": "List of ORCID IDs for all submission authors."
      },
      "author_institutions": {
        "type": "array",
        "items": { "type": "string" },
        "description": "List of all current and historical institutional affiliations of the authors."
      },
      "coauthorship_lookback_years": {
        "type": "integer",
        "description": "Number of years to look back for joint publications (standard = 4)."
      }
    }
  },
  "required": ["candidate_reviewer_id", "author_orcids", "author_institutions"]
},
{
  "name": "get_reviewer_workload_metrics",
  "description": "Fetches operational availability, active review count, 12-month completion rate, and historical turnaround metrics for a candidate reviewer.",
  "parameters": {
    "type": "object",
    "properties": {
      "candidate_reviewer_id": {
        "type": "string",
        "description": "The candidate reviewer's internal or ORCID identifier."
      }
    }
  }
}

```

```

    }
  },
  "required": ["candidate_reviewer_id"]
}
]

```

3.5 Response Schema (Structured JSON Schema)

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "ReviewerMatcherResponse",
  "type": "object",
  "properties": {
    "manuscript_id": { "type": "string" },
    "matching_timestamp": { "type": "string", "format": "date-time" },
    "extracted_expertise_profile": {
      "type": "object",
      "properties": {
        "clinical_domains": { "type": "array", "items": { "type": "string" } },
        "methodological_domains": { "type": "array", "items": { "type": "string" } },
        "required_panel_balance": { "type": "string" }
      },
      "required": ["clinical_domains", "methodological_domains", "required_panel_balance"]
    },
    "recommended_panel": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "reviewer_id": { "type": "string" },
          "full_name": { "type": "string" },
          "orcid": { "type": "string" },
          "current_affiliation": { "type": "string" },
          "country": { "type": "string" },
          "assignment_tier": {
            "type": "string",
            "enum": ["TIER_1_PRIMARY", "TIER_2_ALTERNATE", "EXCLUDED_COI", "EXCLUDED_WORKLOAD"]
          },
          "expertise_match_score": { "type": "number", "minimum": 0.0, "maximum": 1.0 },
          "matched_specialties": { "type": "array", "items": { "type": "string" } },
          "coi_audit": {
            "type": "object",
            "properties": {
              "coi_detected": { "type": "boolean" },
              "coi_type": { "type": "string", "enum": ["NONE", "CO_AUTHORSHIP", "INSTITUTIONAL", "MENTORSHIP", "FINANCIAL"] },
              "audit_details": { "type": "string" }
            },
            "required": ["coi_detected", "coi_type", "audit_details"]
          },
          "workload_audit": {
            "type": "object",
            "properties": {
              "active_reviews_count": { "type": "integer" },

```

```

        "avg_turnaround_days": { "type": "number" },
        "historical_completion_rate": { "type": "number" },
        "overburdened": { "type": "boolean" }
    },
    "required": ["active_reviews_count", "avg_turnaround_days",
"historical_completion_rate", "overburdened"]
    },
    "editorial_invitation_rationale": { "type": "string" }
},
"required": [
    "reviewer_id",
    "full_name",
    "orcid",
    "current_affiliation",
    "assignment_tier",
    "expertise_match_score",
    "coi_audit",
    "workload_audit",
    "editorial_invitation_rationale"
]
}
},
"panel_balance_summary": {
    "type": "object",
    "properties": {
        "tier_1_count": { "type": "integer" },
        "tier_2_count": { "type": "integer" },
        "excluded_count": { "type": "integer" },
        "methodological_coverage_confirmed": { "type": "boolean" },
        "clinical_coverage_confirmed": { "type": "boolean" }
    },
    "required": ["tier_1_count", "tier_2_count", "excluded_count",
"methodological_coverage_confirmed", "clinical_coverage_confirmed"]
    },
    "required": [
        "manuscript_id",
        "matching_timestamp",
        "extracted_expertise_profile",
        "recommended_panel",
        "panel_balance_summary"
    ]
}
}

```

3.6 Sample Input & Output Payloads

Sample Input Payload

```

{
  "manuscript_id": "MS-2026-88192",
  "title": "Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A
Retrospective Multicenter Cohort Study",
  "abstract": "We developed an ensemble gradient boosted model (SepsisXGB) trained on 14,280
electronic health records across four tertiary surgical centers... Primary outcome was 30-day
post-operative sepsis per Sepsis-3 definitions. SepsisXGB achieved an AUROC of 0.892... Decision

```

```

curve analysis demonstrated net clinical benefit...",
"keywords": ["Sepsis", "Machine Learning", "Risk Stratification", "Postoperative
Complications", "Decision Curve Analysis"],
"authors": [
  { "name": "Elena Rostova", "orcid": "0000-0001-9281-301X", "institution": "Vanderbilt
University Medical Center" },
  { "name": "Marcus Vance", "orcid": "0000-0002-3312-9901", "institution": "Vanderbilt
University" },
  { "name": "Jin-Woo Park", "orcid": "0000-0002-1849-0921", "institution": "Vanderbilt
University Medical Center" }
],
"desired_panel_size": {
  "primary_reviewers": 2,
  "alternate_reviewers": 2
}
}

```

Sample Output Payload

```

{
  "manuscript_id": "MS-2026-88192",
  "matching_timestamp": "2026-09-07T09:40:22Z",
  "extracted_expertise_profile": {
    "clinical_domains": ["Critical Care Medicine", "Surgical Sepsis", "Postoperative
Complications"],
    "methodological_domains": ["XGBoost / Machine Learning", "Clinical Prediction Rules",
"Decision Curve Analysis", "EHR Informatics"],
    "required_panel_balance": "At least 1 Surgical/Critical Care Intensivist and at least 1
Biostatistician/Machine Learning Specialist"
  },
  "recommended_panel": [
    {
      "reviewer_id": "REV-89124",
      "full_name": "Prof. Alistair MacIntyre",
      "orcid": "0000-0002-8814-1102",
      "current_affiliation": "University of Edinburgh, Royal Infirmary of Edinburgh",
      "country": "United Kingdom",
      "assignment_tier": "TIER_1_PRIMARY",
      "expertise_match_score": 0.96,
      "matched_specialties": ["Sepsis-3 Definitions", "Surgical Critical Care", "ICU
Biomarkers"],
      "coi_audit": {
        "coi_detected": false,
        "coi_type": "NONE",
        "audit_details": "No shared publications in last 10 years; completely independent
Scottish institution; no active grants."
      },
      "workload_audit": {
        "active_reviews_count": 0,
        "avg_turnaround_days": 14.2,
        "historical_completion_rate": 0.98,
        "overburdened": false
      },
      "editorial_invitation_rationale": "Leading authority on surgical critical care
epidemiology. Perfect profile to assess clinical validity of sepsis definitions and clinical
utility."
    },
  ]
}

```

```

{
  "reviewer_id": "REV-31089",
  "full_name": "Dr. Chantal Dubois",
  "orcid": "0000-0001-7734-9218",
  "current_affiliation": "McGill University, Department of Epidemiology & Biostatistics",
  "country": "Canada",
  "assignment_tier": "TIER_1_PRIMARY",
  "expertise_match_score": 0.93,
  "matched_specialties": ["Machine Learning Calibration", "Gradient Boosting", "Decision
Curve Analysis"],
  "coi_audit": {
    "coi_detected": false,
    "coi_type": "NONE",
    "audit_details": "Zero co-authorships with Rostova, Vance, or Park; no Canadian/US
institutional overlap."
  },
  "workload_audit": {
    "active_reviews_count": 1,
    "avg_turnaround_days": 16.8,
    "historical_completion_rate": 0.95,
    "overburdened": false
  },
  "editorial_invitation_rationale": "High-impact biostatistician specializing in clinical
predictive AI and Brier score calibration. Provides the vital methodological check."
},
{
  "reviewer_id": "REV-12044",
  "full_name": "Assoc. Prof. Henrik Lindqvist",
  "orcid": "0000-0003-4412-0091",
  "current_affiliation": "Karolinska Institutet, Department of Physiology & Pharmacology",
  "country": "Sweden",
  "assignment_tier": "TIER_2_ALTERNATE",
  "expertise_match_score": 0.89,
  "matched_specialties": ["Postoperative Morbidity", "EHR Phenotyping"],
  "coi_audit": {
    "coi_detected": false,
    "coi_type": "NONE",
    "audit_details": "Clear COI check across all authors and institutions."
  },
  "workload_audit": {
    "active_reviews_count": 0,
    "avg_turnaround_days": 18.0,
    "historical_completion_rate": 0.92,
    "overburdened": false
  },
  "editorial_invitation_rationale": "Strong alternate clinical reviewer with extensive
background in ICU outcomes and Scandinavian registry data."
},
{
  "reviewer_id": "REV-55102",
  "full_name": "Dr. Sarah Lin-Schultz",
  "orcid": "0000-0002-6019-4451",
  "current_affiliation": "Duke University Medical Center",
  "country": "United States",
  "assignment_tier": "EXCLUDED_COI",
  "expertise_match_score": 0.97,
  "matched_specialties": ["Predictive Sepsis AI", "Clinical Decision Support"],
  "coi_audit": {
    "coi_detected": true,
    "coi_type": "CO_AUTHORSHIP",

```

```

      "audit_details": "Disqualified: Co-authored a 2024 paper in JAMA Surgery with author
      Jin-Woo Park (DOI: 10.1001/jamasurg.2024.0192).",
    },
    "workload_audit": {
      "active_reviews_count": 0,
      "avg_turnaround_days": 12.0,
      "historical_completion_rate": 1.0,
      "overburdened": false
    },
    "editorial_invitation_rationale": "Excluded automatically due to direct co-authorship
    within the 48-month window under COPE guidelines."
  }
],
"panel_balance_summary": {
  "tier_1_count": 2,
  "tier_2_count": 1,
  "excluded_count": 1,
  "methodological_coverage_confirmed": true,
  "clinical_coverage_confirmed": true
}
}

```

4. ReviewSynthesizerAgent

4.1 Purpose & Architectural Role

The **ReviewSynthesizerAgent** provides high-level cognitive synthesis and conflict reconciliation across disparate, contradictory, or fragmented peer review reports. In scholarly publishing, reviewers frequently present starkly opposing verdicts (e.g., Reviewer 1 recommends "Accept with Minor Revision," while Reviewer 2 recommends "Reject"). The agent performs:

106. **Concordance & Discordance Analysis:** Maps reviewer evaluations into shared agreements (consensus strengths and flaws) versus points of sharp disagreement.
107. **Conflict Resolution & Methodological Adjudication:** Analyzes the technical validity of contradictory claims. Determines whether a negative reviewer comment identifies an irremediable fatal flaw or an addressable empirical question (e.g., clarifying missing data imputation methods vs. invalid study design).
108. **Revision Item Categorization & Prioritization:** Hierarchically organizes feedback into:
 - **Essential Revisions (Tier 1):** Methodological, statistical, or experimental requirements mandatory for acceptance.
 - **Advisory Enhancements (Tier 2):** Suggestions for discussion expansion, contextual framing, or minor analyses that would improve the paper but are not strictly gating.
 - **Editorial & Reporting Compliance (Tier 3):** Clarifications regarding data links, figure resolution, or checklist completeness.
109. **Tone Moderation & Professional Editing:** Identifies and removes abrasive, unconstructive, or unprofessional language while maintaining the full scientific rigor of the critique.
110. **Editorial Decision & Author Letter Drafting:** Generates a confidential briefing memo for the Editor-in-Chief (EIC) and an authoritative, clear, and empathetic decision letter for the authors.

4.2 Model Configurations

```
{
  "model": "gemini-1.5-pro",
  "generation_config": {
    "temperature": 0.2,
    "top_p": 0.9,
    "top_k": 40,
    "max_output_tokens": 8192,
    "response_mime_type": "application/json"
  },
  "safety_settings": [
    {
      "category": "HARM_CATEGORY_HARASSMENT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HATE_SPEECH",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_SEXUALLY_EXPLICIT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_DANGEROUS_CONTENT",
      "threshold": "BLOCK_ONLY_HIGH"
    }
  ]
}
```

4.3 Google AI Studio System Instructions

You are the Senior Editorial Review Synthesis Specialist and Associate Editor for an elite peer-reviewed academic journal. Your task is to analyze, reconcile, and synthesize the peer review reports for a submitted manuscript into an authoritative editorial decision package.

Execute your synthesis following these editorial principles:

1. Cross-Reviewer Synthesis:

- Carefully extract and cross-tabulate the technical assessments of each independent reviewer.
- Identify points of Consensus Strengths (aspects of the paper all reviewers commend).
- Identify points of Consensus Weaknesses (deficiencies flagged independently by multiple reviewers).
- Identify Points of Divergence / Conflict where reviewers reach contradictory conclusions or recommend opposite actions.

2. Adjudication of Divergences:

- When reviewers disagree, do not simply split the difference. Formulate an evidence-based editorial stance based on the manuscript's underlying data:

* If Reviewer A demands an impossible or out-of-scope experiment (e.g., demanding a 5-year prospective randomized trial for a retrospective cohort paper) and Reviewer B notes that observational data is appropriately analyzed, adjudicate in favor of feasibility and boundary containment, recommending that the authors address the limitation in the Discussion.

* If Reviewer A overlooked a major statistical confounder (e.g., survivorship bias, improper imputation of missing variables) correctly caught by Reviewer B, uphold Reviewer B's methodological critique as an Essential Revision.

3. Actionable Revision Prioritization:

- Organize all necessary revisions into clear, numbered, unambiguous categories:

- * Category A: Essential Revisions (Mandatory). Define the exact statistical re-analyses, additional validation cohorts, sensitivity analyses, or control comparisons required for the paper to be scientifically acceptable.

- * Category B: Contextual & Interpretive Revisions (Discussion & Literature). Guide authors on tempering overclaimed conclusions, citing competing frameworks, or discussing physiological/theoretical mechanisms.

- * Category C: Presentation, Formatting, & Reporting Standards. Adherence to checklists, figure clarity, and data deposition.

4. Tone Sanitization & Neutrality:

- Ensure that all author-facing text is constructive, courteous, scholarly, and objective.

- Filter out sarcastic, dismissive, or offensive phrases from raw reviewer comments (e.g., transform "The authors clearly do not understand basic statistics" to "The statistical rationale for model selection requires more rigorous formal justification and documentation").

5. Final Decision Formulation:

- Formulate one of the following official verdicts:

- * ACCEPT: Rare on initial round; only when no substantive issues remain.

- * MINOR_REVISION: Minor textual, statistical, or descriptive clarifications; no new experiments or fundamental re-analyses needed.

- * MAJOR_REVISION: The paper is promising, but substantive new analyses, model recalibration, or extensive restructuring are required within a 60-day window.

- * REJECT_AND_RESUBMIT: Fatal in current form, but the core idea has merit; resubmission would be treated as a de novo manuscript requiring full re-review.

- * REJECT: Fatal conceptual, ethical, or methodological flaws that cannot be repaired.

Deliver your output strictly in the designated structured JSON schema, including both the confidential editor memo and the complete text of the decision letter to the authors.

4.4 Function Calling / Tools Declarations (OpenAPI 3.0 Format)

```
[
  {
    "name": "fetch_reviewer_reports",
    "description": "Retrieves the complete set of raw reviewer scorecards, ratings, confidential comments to the editor, and author-facing reports for a manuscript submission.",
    "parameters": {
      "type": "object",
      "properties": {
        "manuscript_id": {
          "type": "string",
          "description": "The manuscript tracking identifier."
        },
        "review_round": {
```

```

        "type": "integer",
        "description": "The review round number (e.g., 1 for initial submission, 2 for first
revision)."
    },
    "required": ["manuscript_id", "review_round"]
},
{
    "name": "query_editorial_precedents",
    "description": "Searches the journal archive for previous editorial decisions and EIC
overrides in cases with similar reviewer score distributions or methodological disputes.",
    "parameters": {
        "type": "object",
        "properties": {
            "subject_domain": {
                "type": "string",
                "description": "Academic subfield of the paper."
            },
            "dispute_keywords": {
                "type": "array",
                "items": { "type": "string" },
                "description": "Key phrases describing the dispute (e.g., ['external validation
missing', 'missing data MICE vs KNN'])."
            }
        },
        "required": ["subject_domain", "dispute_keywords"]
    }
}
]

```

4.5 Response Schema (Structured JSON Schema)

```

{
    "$schema": "http://json-schema.org/draft-07/schema#",
    "title": "ReviewSynthesizerResponse",
    "type": "object",
    "properties": {
        "manuscript_id": { "type": "string" },
        "synthesis_timestamp": { "type": "string", "format": "date-time" },
        "review_round": { "type": "integer" },
        "recommended_decision": {
            "type": "string",
            "enum": ["ACCEPT", "MINOR_REVISION", "MAJOR_REVISION", "REJECT_AND_RESUBMIT", "REJECT"]
        },
        "concordance_matrix": {
            "type": "object",
            "properties": {
                "consensus_strengths": { "type": "array", "items": { "type": "string" } },
                "consensus_weaknesses": { "type": "array", "items": { "type": "string" } },
                "points_of_conflict": {
                    "type": "array",
                    "items": {
                        "type": "object",
                        "properties": {

```

```

        "topic": { "type": "string" },
        "reviewer_stances": { "type": "string" },
        "editorial_adjudication": { "type": "string" }
    },
    "required": ["topic", "reviewer_stances", "editorial_adjudication"]
},
},
"required": ["consensus_strengths", "consensus_weaknesses", "points_of_conflict"]
},
"prioritized_revision_action_items": {
    "type": "object",
    "properties": {
        "essential_mandatory_revisions": { "type": "array", "items": { "type": "string" } },
        "advisory_contextual_improvements": { "type": "array", "items": { "type": "string" } },
        "editorial_formatting_requirements": { "type": "array", "items": { "type": "string" } }
    },
    "required": ["essential_mandatory_revisions", "advisory_contextual_improvements",
"editorial_formatting_requirements"]
},
"confidential_memo_to_eic": {
    "type": "string",
    "description": "Private assessment for the Editor-in-Chief highlighting strategic
rationale, risk factors, and reviewer credibility."
},
"draft_author_decision_letter": {
    "type": "string",
    "description": "Complete, publication-ready formal decision letter addressed to the
corresponding author."
}
},
"required": [
    "manuscript_id",
    "synthesis_timestamp",
    "review_round",
    "recommended_decision",
    "concordance_matrix",
    "prioritized_revision_action_items",
    "confidential_memo_to_eic",
    "draft_author_decision_letter"
]
}

```

4.6 Sample Input & Output Payloads

Sample Input Payload

```

{
    "manuscript_id": "MS-2026-88192",
    "review_round": 1,
    "manuscript_title": "Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A
Retrospective Multicenter Cohort Study",
    "reviewer_reports": [
        {
            "reviewer_id": "REV-89124",

```

```

    "recommendation": "Accept with Minor Revision",
    "author_facing_comments": "This is an impressive multicenter study addressing
postoperative sepsis. The sample size of >14,000 patients is robust and the comparison against
SOFA is compelling. My only minor concerns: 1. Please clarify the exact definition of time-to-
antibiotics in the cohort. 2. The discussion should mention how this model could be deployed at
the bedside in EHRs with latency.",
    "confidential_to_editor": "Excellent, clinically impactful paper. Should be published with
minor revisions."
  },
  {
    "reviewer_id": "REV-31089",
    "recommendation": "Major Revision",
    "author_facing_comments": "The paper tackles a critical question, but I have serious
methodological reservations. 1. The authors use XGBoost without providing calibration curves
across probability deciles; Brier scores alone are insufficient. 2. Missing data is handled via
simple mean imputation, which distorts variance and cross-feature interactions. Multiple
Imputation by Chained Equations (MICE) or native tree branch routing must be used. 3. The 4
centers are all in one geographical network; true external validation on a disjoint health
system is absent.",
    "confidential_to_editor": "The paper has high potential, but the current statistical
pipeline is naive regarding missing data and probability calibration. If the authors cannot re-
impute with MICE and show decile calibration, the paper should be rejected."
  }
]
}

```

Sample Output Payload

```

{
  "manuscript_id": "MS-2026-88192",
  "synthesis_timestamp": "2026-09-07T09:45:10Z",
  "review_round": 1,
  "recommended_decision": "MAJOR_REVISION",
  "concordance_matrix": {
    "consensus_strengths": [
      "Both reviewers agree that the clinical question is of high relevance and that the sample
size (14,280 patients) provides substantial statistical power.",
      "Comparison against clinical baselines (SOFA and qSOFA) is recognized as a strong clinical
benchmark."
    ],
    "consensus_weaknesses": [
      "Both reviewers note that translational and implementation nuances regarding bedside
deployment and timing definitions need further operational clarity."
    ],
    "points_of_conflict": [
      {
        "topic": "Methodological Rigor vs Acceptance Readiness",
        "reviewer_stances": "Reviewer 1 recommends minor revisions praising clinical impact;
Reviewer 2 recommends major revisions citing serious deficiencies in missing data handling (mean
imputation) and calibration verification.",
        "editorial_adjudication": "Reviewer 2's methodological critique is statistically
incontrovertible. Mean imputation introduces known bias in non-linear tree ensembles. The paper
cannot be accepted without re-imputation (MICE or XGBoost native missing value handling) and
explicit calibration curves. Therefore, the decision must be Major Revision."
      },
      {
        "topic": "Geographic Generalizability & External Validation",

```

```

    "reviewer_stances": "Reviewer 2 notes the 4 centers belong to a single regional health
system and queries external generalizability.",
    "editorial_adjudication": "Requiring a brand-new external health system cohort goes
beyond the scope of a standard revision timeline. However, the authors must perform leave-one-
center-out cross-validation across the 4 centers and candidly address regional limitation in the
Discussion."
  }
],
},
"prioritized_revision_action_items": {
  "essential_mandatory_revisions": [
    "Re-execute feature preprocessing replacing mean imputation with Multiple Imputation by
Chained Equations (MICE) or native XGBoost missing value handling, demonstrating stability of
AUROC.",
    "Provide decile-based calibration plots (observed vs. expected event rates) alongside
calibration slope and intercept in the main manuscript figures.",
    "Conduct a leave-one-center-out internal-external cross-validation analysis across the 4
participating hospital sites to evaluate spatial transportability."
  ],
  "advisory_contextual_improvements": [
    "Expand the Discussion to address technical hurdles of real-time EHR inference, focusing
on feature extraction latency and alert fatigue.",
    "Provide a standardized definition and sensitivity analysis for time-to-antimicrobial
administration."
  ],
  "editorial_formatting_requirements": [
    "Ensure all calibration and ROC plots include high-resolution vector figures (300 DPI
minimum).",
    "Update the TRIPOD reporting checklist to explicitly document the re-imputation strategy."
  ]
},
"confidential_memo_to_eic": "Although Reviewer 1 was enthusiastic, Reviewer 2 correctly
identified critical statistical weaknesses: mean imputation is unacceptable for a top-tier
journal publishing predictive algorithms. Recommend Major Revision with a 60-day deadline. If
the authors successfully implement MICE and provide calibration plots, this will be a landmark
paper for the journal.",
"draft_author_decision_letter": "Dear Dr. Park and Colleagues,

```

Thank you for submitting your manuscript 'Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A Retrospective Multicenter Cohort Study' to the Journal of Clinical & Translational Epidemiology.

Your manuscript has been evaluated by two expert external reviewers and our editorial team. While the reviewers recognized the importance of your clinical objective and the value of your large multicenter cohort, they identified essential methodological and statistical issues that must be resolved before this work can be considered for publication. We are pleased to invite you to submit a Major Revision within 60 days.

In your revised submission, you must specifically address the following prioritized requirements:

1. Statistical Imputation: Replace mean feature imputation with Multiple Imputation by Chained Equations (MICE) or native XGBoost missing value splitting, reporting any impact on model discrimination.
2. Model Calibration: Provide visual calibration curves across probability deciles, reporting calibration intercept and slope.
3. Site Transportability: Conduct leave-one-site-out validation across your four participating centers.
4. Implementation Context: Clarify time-to-antibiotic parameters and bedside EHR deployment feasibility.

Please include a point-by-point response letter detailing your changes. We look forward to receiving your revised manuscript.

Sincerely,
 Editorial Board
 Journal of Clinical & Translational Epidemiology"
 }

5. AcademicCopyeditorAgent

5.1 Purpose & Architectural Role

The **AcademicCopyeditorAgent** performs comprehensive manuscript copyediting to bring accepted or provisionally accepted scientific papers up to the highest standards of international scholarly publication. Its scope includes:

111. **Academic Syntax & Rhetorical Precision:** Enforces scholarly tone, eliminating colloquial phrasing, redundant filler ("due to the fact that" -> "because"), anthropomorphic claims ("this study believes"), and uncalibrated absolutes ("this proves" -> "these findings indicate").
112. **Grammar, Mechanics & Voice Calibration:** Corrects subject-verb agreement, dangling participles, split infinitives, and punctuation (e.g., serial Oxford commas, em-dashes, en-dashes for ranges). Balances active voice for author decisions ("We performed...") with objective passive voice for standard laboratory protocols.
113. **Citation Style Governance (APA 7th, IEEE, Chicago 17th):**
 - **APA 7th:** Author-date parenthetical and narrative citations, strict formatting for 3+ authors (**et al.** from first mention), italicized statistical variables (*p*, *t*, *F*, *r*, *d*), and omission of leading zeroes before decimal points bounded by 1 (e.g., *p* < .001).
 - **IEEE:** Bracketed numeric citations sequentially ordered ([1], [2]–[4]), standardized abbreviation of journal titles, and specific figure/table referencing ("Fig. 1", "Table I").
 - **Chicago 17th:** Rigorous author-date or notes/bibliography conformity, title capitalization rules, and quote integration.
114. **Structured Tracked Changes Generation:** Produces discrete, machine-readable replacement diffs containing original snippets, revised snippets, span markers, style guide rules applied, and author queries (AQ) for ambiguities.

5.2 Model Configurations

```
{
  "model": "gemini-1.5-pro",
  "generation_config": {
    "temperature": 0.1,
    "top_p": 0.8,
    "top_k": 40,
    "max_output_tokens": 8192,
```

```

    "response_mime_type": "application/json"
  },
  "safety_settings": [
    {
      "category": "HARM_CATEGORY_HARASSMENT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HATE_SPEECH",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_SEXUALLY_EXPLICIT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_DANGEROUS_CONTENT",
      "threshold": "BLOCK_ONLY_HIGH"
    }
  ]
}

```

5.3 Google AI Studio System Instructions

You are the Chief Academic Copyeditor for an international scholarly publisher. Your mission is to edit submitted manuscript texts to publication-grade precision, adhering strictly to the designated journal style guide (APA 7th, IEEE, or Chicago 17th Manual of Style).

You must enforce the following copyediting standards:

1. Scientific Rhetoric, Scholarly Tone, and Hedging:
 - Eliminate colloquialisms, idioms, hyperbole, and promotional language (e.g., "groundbreaking", "astonishing", "game-changing").
 - Enforce appropriate scientific hedging: replace dogmatic claims ("This confirms that the drug cures sepsis") with evidence-calibrated formulations ("These data demonstrate that the therapeutic agent significantly reduced 30-day sepsis mortality in this cohort").
 - Eliminate wordy and tautological formulations: replace "in order to" with "to"; replace "a majority of" with "most"; replace "at the present point in time" with "currently".
 - Correct anthropomorphism (e.g., "The experiment argued..." -> "The experimental results indicated...").
2. Mechanics, Grammar, and Statistical Typography:
 - Enforce the serial (Oxford) comma across all series of three or more items for APA and Chicago styles.
 - Use en-dashes (–) for numerical and page ranges (e.g., "pp. 142–159", "2021–2024"), hyphens (–) for compound modifiers preceding nouns, and unspaced em-dashes (—) for parenthetical sentence breaks.
 - Format statistical symbols according to official style guidelines:
 - * In APA 7th: italicize statistical symbols representing variables (*p*, *t*, *F*, *z*, *r*, *N*, *n*, *M*, *SD*). Do not italicize Greek letters (β , α , χ^2).
 - * Omit leading zeros for numbers that cannot exceed 1.0 (write *p* = .023, *r* = .45, not *p* = 0.023). Report exact p-values down to .001 (*p* < .001 for smaller values).
 - * Insert non-breaking spaces between numbers and their units of measurement (e.g., "5 mg", "37 °C", "10 min").

3. Citation and Reference Style Enforcement:

- Verify that every in-text citation corresponds exactly to an entry in the reference list, and that every reference entry is cited in the text.
- In APA 7th: format 3 or more authors as (FirstAuthor et al., Year) immediately upon first citation.
- In IEEE: format citations as bracketed numbers on the line of text, separated by commas or en-dashes for ranges (e.g., [1], [3], [5]–[8]).
- Cross-check author surname spelling and publication years across text and references.

4. Tracked Changes & Author Queries (AQ):

- Every single modification must be tracked as a discrete, verifiable JSON object containing the exact original text, the proposed replacement, the applicable style rule, and the editorial rationale.
- When text is semantically ambiguous, contradictory, or missing critical information, do not guess; generate a polite, specific Author Query (AQ) directing the author to resolve the issue.

Deliver your output strictly according to the structured JSON schema.

5.4 Function Calling / Tools Declarations (OpenAPI 3.0 Format)

```
[
  {
    "name": "validate_doi_references",
    "description": "Cross-references a list of cited bibliographic strings against the Crossref Metadata API to confirm official author spelling, journal title, year, volume, pagination, and persistent DOI.",
    "parameters": {
      "type": "object",
      "properties": {
        "raw_references": {
          "type": "array",
          "items": { "type": "string" },
          "description": "List of reference strings to validate."
        }
      },
      "required": ["raw_references"]
    }
  },
  {
    "name": "check_citation_style_rules",
    "description": "Queries the definitive style rule handbook for specific edge-case conventions under APA 7th, IEEE, or Chicago 17th.",
    "parameters": {
      "type": "object",
      "properties": {
        "style_guide": {
          "type": "string",
          "enum": ["APA_7TH", "IEEE", "CHICAGO_17TH"],
          "description": "The target style guide."
        },
        "query_category": {
          "type": "string",
          "description": "The category to query (e.g., 'statistical_abbreviations',
```

```
'table_notes', 'heading_levels', 'compound_words')."
    }
  },
  "required": ["style_guide", "query_category"]
}
]
]
```

5.5 Response Schema (Structured JSON Schema)

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "AcademicCopyeditorResponse",
  "type": "object",
  "properties": {
    "document_id": { "type": "string" },
    "style_guide_applied": {
      "type": "string",
      "enum": ["APA_7TH", "IEEE", "CHICAGO_17TH"]
    },
    "copyedit_timestamp": { "type": "string", "format": "date-time" },
    "copyedit_metrics": {
      "type": "object",
      "properties": {
        "total_edits_count": { "type": "integer" },
        "grammar_and_syntax_edits": { "type": "integer" },
        "style_and_typography_edits": { "type": "integer" },
        "citation_edits": { "type": "integer" },
        "pre_edit_word_count": { "type": "integer" },
        "post_edit_word_count": { "type": "integer" }
      },
      "required": [
        "total_edits_count",
        "grammar_and_syntax_edits",
        "style_and_typography_edits",
        "citation_edits",
        "pre_edit_word_count",
        "post_edit_word_count"
      ]
    },
    "tracked_changes": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "edit_id": { "type": "string" },
          "section_name": { "type": "string" },
          "original_text": { "type": "string" },
          "revised_text": { "type": "string" },
          "edit_category": {
            "type": "string",
            "enum": [
              "GRAMMAR_SYNTAX",
              "SCHOLARLY_TONE",
              "HEDGING_CALIBRATION",

```

```

        "TYPOGRAPHY_STATISTICS",
        "CITATION_FORMATTING",
        "CONCISENESS"
    ],
    },
    "rule_applied": { "type": "string" },
    "rationale": { "type": "string" }
  },
  "required": ["edit_id", "section_name", "original_text", "revised_text",
"edit_category", "rule_applied", "rationale"]
},
},
"author_queries": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "query_id": { "type": "string" },
      "context_location": { "type": "string" },
      "flagged_text": { "type": "string" },
      "author_action_required": { "type": "string" }
    },
    "required": ["query_id", "context_location", "flagged_text", "author_action_required"]
  }
},
"fully_edited_text": {
  "type": "string",
  "description": "The complete, fully integrated text incorporating all accepted tracked
changes."
}
},
"required": [
  "document_id",
  "style_guide_applied",
  "copyedit_timestamp",
  "copyedit_metrics",
  "tracked_changes",
  "author_queries",
  "fully_edited_text"
]
}

```

5.6 Sample Input & Output Payloads

Sample Input Payload

```

{
  "document_id": "MS-88192-REVISED",
  "style_guide": "APA_7TH",
  "raw_manuscript_text": "Results: Due to the fact that severe sepsis involves dynamic
physiological collapse, we tested SepsisXGB against traditional scores. It was found that
SepsisXGB achieved an astonishing AUROC of 0.892, which definitively proves that gradient
boosted models outperform classic clinical scoring systems. For the cohort of patients whom were
treated in the surgical ward, the mean lactate clearance rate was 24.5% (SD = 5.2), with p =
0.003 and t(142) = 3.12. This agrees with previous studies (Smith, Johnson, Vance, & Park, 2022;

```

Dubois et al. 2024).

Discussion: The algorithm believes that rapid intervention prevents death. In order to deploy this in clinical workflows, future studies must evaluate latency."
}

Sample Output Payload

```
{
  "document_id": "MS-88192-REVISED",
  "style_guide_applied": "APA_7TH",
  "copyedit_timestamp": "2026-09-07T09:50:00Z",
  "copyedit_metrics": {
    "total_edits_count": 8,
    "grammar_and_syntax_edits": 2,
    "style_and_typography_edits": 3,
    "citation_edits": 2,
    "pre_edit_word_count": 105,
    "post_edit_word_count": 96
  },
  "tracked_changes": [
    {
      "edit_id": "ED-001",
      "section_name": "Results",
      "original_text": "Due to the fact that severe sepsis involves",
      "revised_text": "Because severe sepsis involves",
      "edit_category": "CONCISENESS",
      "rule_applied": "APA 7th Section 4.4 (Wordiness and Redundancy)",
      "rationale": "Replaced wordy phrase 'Due to the fact that' with concise conjunction 'Because'."
    },
    {
      "edit_id": "ED-002",
      "section_name": "Results",
      "original_text": "an astonishing AUROC of 0.892, which definitively proves that",
      "revised_text": "an AUROC of .892, indicating that",
      "edit_category": "HEDGING_CALIBRATION",
      "rule_applied": "APA 7th Section 4.7 (Scholarly Hedging and Tone) & Section 6.33 (Numbers Bounded by 1)",
      "rationale": "Removed hyperbole ('astonishing') and uncalibrated certainty ('definitively proves'); omitted leading zero for decimal value bounded by 1."
    },
    {
      "edit_id": "ED-003",
      "section_name": "Results",
      "original_text": "patients whom were treated",
      "revised_text": "patients who were treated",
      "edit_category": "GRAMMAR_SYNTAX",
      "rule_applied": "APA 7th Section 4.19 (Relative Pronouns: Who vs. Whom)",
      "rationale": "Corrected objective pronoun 'whom' to subjective 'who' as the subject of the clause."
    },
    {
      "edit_id": "ED-004",
      "section_name": "Results",
      "original_text": "the mean lactate clearance rate was 24.5% (SD = 5.2), with p = 0.003 and t(142) = 3.12.",

```

```

    "revised_text": "the mean lactate clearance rate was 24.5% (*SD* = 5.2), *t*(142) = 3.12,
    *p* = .003.",
    "edit_category": "TYPOGRAPHY_STATISTICS",
    "rule_applied": "APA 7th Section 6.44 (Statistical Abbreviations and Symbols)",
    "rationale": "Italicized statistical symbols (*SD*, *t*, *p*), omitted leading zero for p-
value (*p* = .003), and streamlined statistical reporting order."
  },
  {
    "edit_id": "ED-005",
    "section_name": "Results",
    "original_text": "(Smith, Johnson, Vance, & Park, 2022; Dubois et al. 2024)",
    "revised_text": "(Dubois et al., 2024; Smith et al., 2022)",
    "edit_category": "CITATION_FORMATTING",
    "rule_applied": "APA 7th Section 8.12 (Order of Works in Same Parenthesis) & Section 8.17
(Three or More Authors)",
    "rationale": "Works with 3+ authors use 'et al.' on first mention; multiple citations in
one parenthesis must be alphabetized by primary author surname; added missing comma after 'et
al.'."
  },
  {
    "edit_id": "ED-006",
    "section_name": "Discussion",
    "original_text": "The algorithm believes that rapid intervention prevents death.",
    "revised_text": "These findings suggest that rapid clinical intervention mitigates
mortality.",
    "edit_category": "SCHOLARLY_TONE",
    "rule_applied": "APA 7th Section 4.11 (Anthropomorphism)",
    "rationale": "An algorithm cannot 'believe'; attributed action properly to experimental
findings."
  },
  {
    "edit_id": "ED-007",
    "section_name": "Discussion",
    "original_text": "In order to deploy this",
    "revised_text": "To deploy this model",
    "edit_category": "CONCISENESS",
    "rule_applied": "APA 7th Section 4.4 (Conciseness)",
    "rationale": "Replaced 'In order to' with 'To'; specified noun 'model' instead of vague
demonstrative 'this'."
  }
],
"author_queries": [
  {
    "query_id": "AQ-01",
    "context_location": "Results, Paragraph 1",
    "flagged_text": "(Smith, Johnson, Vance, & Park, 2022)",
    "author_action_required": "Please verify that the 2022 study by Smith et al. is correctly
listed in the References list; Crossref lookup returned an identical title published in 2023.
Please confirm correct publication year."
  }
],
"fully_edited_text": "Results: Because severe sepsis involves dynamic physiological collapse,
we tested SepsisXGB against traditional scores. SepsisXGB achieved an AUROC of .892, indicating
that gradient boosted models outperform classic clinical scoring systems. For the cohort of
patients who were treated in the surgical ward, the mean lactate clearance rate was 24.5% (*SD*
= 5.2), *t*(142) = 3.12, *p* = .003. This agrees with previous studies (Dubois et al., 2024;
Smith et al., 2022).

Discussion: These findings suggest that rapid clinical intervention mitigates mortality. To
deploy this model in clinical workflows, future studies must evaluate latency."

```

}

6. JatsXmlGeneratorAgent

6.1 Purpose & Architectural Role

The **JatsXmlGeneratorAgent** transforms copyedited manuscripts into fully validated, production-grade **NISO JATS XML (ANSI/NISO Z39.96-2021, Journal Publishing Tag Set v1.3)**. JATS XML represents the industry-standard format required for distribution to PubMed Central (PMC), ScienceDirect, HighWire, SpringerLink, Silverchair, and permanent archiving in Portico and CLOCKSS.

Key capabilities:

115. **Semantic Document Modeling:** Translates unstructured or Markdown-formatted text into valid XML hierarchies without data loss.
116. **Frontmatter Assembly (`<front>`):** Structures journal metadata (`<journal-meta>`), article persistent identifiers (`<article-id pub-id-type="doi">`), author affiliations (`<contrib-group>`, `<aff>`), ORCIDs (`<contrib-id contrib-id-type="orcid">`), structured abstracts, subject taxonomies (`<kwd-group>`), and funding assertions (`<funding-group>`).
117. **Body Structuring (`<body>`):** Converts headings into typed sections (`<sec sec-type="intro|methods|results|discussion">`), wraps figures in `<fig>`, tabular data in `<table-wrap>`, equations in `<disp-formula>` with embedded MathML (`<mml:math>`), and connects textual citations with cross-reference tags (`<xref ref-type="bibr">`, `<xref ref-type="fig">`, `<xref ref-type="table">`).
118. **Backmatter & Element Citation Parsing (`<back>`):** Builds reference lists (`<ref-list>`) using granular `<element-citation publication-type="journal">` tags with structured author groups, source titles, volumes, pagination, and DOIs.
119. **DTD Conformance:** Enforces 100% validity against the NISO JATS v1.3 DTD schema.

6.2 Model Configurations

```
{
  "model": "gemini-1.5-pro",
  "generation_config": {
    "temperature": 0.0,
    "top_p": 0.95,
    "top_k": 32,
    "max_output_tokens": 8192,
    "response_mime_type": "application/json"
  },
  "safety_settings": [
    {
      "category": "HARM_CATEGORY_HARASSMENT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HATE_SPEECH",
```

```

    "threshold": "BLOCK_ONLY_HIGH"
  },
  {
    "category": "HARM_CATEGORY_SEXUALLY_EXPLICIT",
    "threshold": "BLOCK_ONLY_HIGH"
  },
  {
    "category": "HARM_CATEGORY_DANGEROUS_CONTENT",
    "threshold": "BLOCK_ONLY_HIGH"
  }
]
]

```

6.3 Google AI Studio System Instructions

You are the Senior Scholarly Publishing XML Architect and JATS Specialist. Your sole mission is to ingest copyedited academic manuscripts and generate valid, well-formed, publication-grade NISO JATS XML (Journal Publishing Tag Set version 1.3, ANSI/NISO Z39.96-2021).

Adhere strictly to the following XML construction rules:

1. Root Element and Namespaces:

- Root tag must be: `<article article-type="research-article" dtd-version="1.3" xml:lang="en" xmlns:xlink="http://www.w3.org/1999/xlink" xmlns:mml="http://www.w3.org/1998/Math/MathML">`.
- Never omit required namespaces.

2. Frontmatter (`<front>`):

- `<journal-meta>`: Include `<journal-id journal-id-type="nlm-ta">`, `<journal-title-group>`, `<issn pub-type="epub">`, and `<publisher>`.
- `<article-meta>`:
 - * `<article-id pub-id-type="doi">`: Insert official DOI.
 - * `<title-group>`: Contain `<article-title>`.
 - * `<contrib-group>`: For every author, generate `<contrib contrib-type="author">` containing `<name>` (`<surname>`, `<given-names>`), `<contrib-id contrib-id-type="orcid">` (URL format: `https://orcid.org/0000-...`), and `<xref ref-type="aff" rid="aff1"/>`.
 - * `<aff id="aff1">`: Structured affiliations with `<institution content-type="dept">`, `<institution content-type="university">`, `<city>`, and `<country>`.
 - * `<author-notes>`: Include `<corresp id="cor1">` with corresponding email.
 - * `<pub-date date-type="pub" publication-format="electronic">`: ISO year, month, day.
 - * `<permissions>`: Include `<copyright-statement>`, `<copyright-year>`, and `<license xlink:href="...">`.
 - * `<abstract>`: May contain structured `<sec>` children (Background, Methods, Results, Conclusions).
 - * `<kwd-group kwd-group-type="author">`: Individual `<kwd>` tags.
 - * `<funding-group>`: Individual `<award-group>` containing `<funding-source>` and `<award-id>`.

3. Body Elements (`<body>`):

- Organize all content within `<sec sec-type="...">` containers, each with an `<id>` attribute (e.g., `sec1`, `sec2`).
- Every paragraph must be enclosed in `<p>`.
- Citations: Link every in-text citation using `<xref ref-type="bibr" rid="ref1">[1]</xref>` or `<xref ref-type="bibr" rid="ref1">Smith et al. (2022)</xref>`.
- Tables: Enclose in `<table-wrap id="tbl1" position="float">`, containing `<label>`,

```

<caption><title>, and XHTML table tags (<table>, <thead>, <tbody>, <tr>, <th>,
<td>).
  - Figures: Enclose in <fig id="fig1" position="float">, containing <label>,
<caption><title>, and <graphic xlink:href="filename.tif"/>.
  - Equations: Enclose in <disp-formula id="eq1">, containing <label> and MathML <math
display="block">.

4. Backmatter (<back>):
  - <ack>: Acknowledgments text inside <p>.
  - <ref-list>: Ordered list of <ref id="ref1"> elements.
  - Reference Citations: Use <element-citation publication-type="journal"> or <publication-
type="book">. Deconstruct references into:
    * <person-group person-group-type="author"> with <name><surname>, <given-names>
    * <article-title>
    * <source> (journal title)
    * <year>
    * <volume>, <issue>, <fpage>, <lpage>
    * <pub-id pub-id-type="doi">

5. Integrity & Validation:
  - Output must be 100% syntactically well-formed XML without escaping XML tags inside the XML
string.
  - All characters requiring XML escaping (&, <, >) inside text nodes must be properly escaped
(&amp;, &lt;, &gt;).
  - Return the output strictly wrapped in the JSON response schema.

```

6.4 Function Calling / Tools Declarations (OpenAPI 3.0 Format)

```

[
  {
    "name": "validate_jats_xml_dtd",
    "description": "Validates a candidate JATS XML string against the official NISO JATS v1.3
Publishing DTD schema and returns schema-level error diagnostics.",
    "parameters": {
      "type": "object",
      "properties": {
        "xml_content": {
          "type": "string",
          "description": "The complete JATS XML document string to validate."
        },
        "dtd_version": {
          "type": "string",
          "enum": ["1.2", "1.3"],
          "description": "The target JATS DTD version."
        }
      }
    },
    "required": ["xml_content", "dtd_version"]
  },
  {
    "name": "convert_latex_to_mathml",
    "description": "Converts mathematical formulas expressed in LaTeX syntax into valid,
semantic MathML elements for embedding in JATS XML <disp-formula> tags.",
    "parameters": {

```

```

    "type": "object",
    "properties": {
      "latex_expression": {
        "type": "string",
        "description": "LaTeX formula string (e.g., 'AUROC = \frac{1}{N} \sum_{i=1}^N I(y_i = \hat{y}_i)')."
      }
    },
    "required": ["latex_expression"]
  }
}
]

```

6.5 Response Schema (Structured JSON Schema)

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "JatsXmlGeneratorResponse",
  "type": "object",
  "properties": {
    "article_id": { "type": "string" },
    "jats_version": { "type": "string", "enum": ["1.3"] },
    "generation_timestamp": { "type": "string", "format": "date-time" },
    "validation_audit": {
      "type": "object",
      "properties": {
        "is_dtd_valid": { "type": "boolean" },
        "dtd_errors": { "type": "array", "items": { "type": "string" } },
        "well_formed_xml": { "type": "boolean" }
      },
      "required": ["is_dtd_valid", "dtd_errors", "well_formed_xml"]
    },
    "element_inventory": {
      "type": "object",
      "properties": {
        "section_count": { "type": "integer" },
        "paragraph_count": { "type": "integer" },
        "figure_count": { "type": "integer" },
        "table_count": { "type": "integer" },
        "formula_count": { "type": "integer" },
        "reference_count": { "type": "integer" }
      },
      "required": [
        "section_count",
        "paragraph_count",
        "figure_count",
        "table_count",
        "formula_count",
        "reference_count"
      ]
    },
    "jats_xml_output": {
      "type": "string",
      "description": "Complete, valid NISO JATS v1.3 XML document ready for distribution to PubMed Central and indexing repositories."
    }
  }
}

```

```

    }
  },
  "required": [
    "article_id",
    "jats_version",
    "generation_timestamp",
    "validation_audit",
    "element_inventory",
    "jats_xml_output"
  ]
}

```

6.6 Sample Input & Output Payloads

Sample Input Payload

```

{
  "article_metadata": {
    "doi": "10.1016/j.jcte.2026.109412",
    "journal_title": "Journal of Clinical & Translational Epidemiology",
    "issn": "2667-114X",
    "publisher_name": "Elsevier Academic Press",
    "title": "Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A  
Retrospective Multicenter Cohort Study",
    "authors": [
      {
        "surname": "Park",
        "given_names": "Jin-Woo",
        "orcid": "https://orcid.org/0000-0002-1849-0921",
        "email": "jw.park@vumc.org",
        "is_corresponding": true,
        "affiliation_ref": "aff1"
      }
    ],
    "affiliations": [
      {
        "id": "aff1",
        "department": "Department of Surgical Sciences",
        "institution": "Vanderbilt University Medical Center",
        "city": "Nashville",
        "country": "United States"
      }
    ],
    "pub_date": { "year": 2026, "month": 9, "day": 15 },
    "volume": "14",
    "issue": "3",
    "fpage": "102",
    "lpage": "115"
  },
  "abstract": "Early identification of surgical sepsis remains challenging. We developed  
SepsisXGB across 14,280 electronic health records...",
  "keywords": ["Sepsis", "Machine Learning", "Risk Stratification"],
  "body_sections": [
    {
      "sec_type": "intro",

```

```

    "title": "Introduction",
    "content": "Surgical sepsis is a leading cause of preventable in-hospital mortality [1].",
  },
],
"references": [
  {
    "id": "ref1",
    "type": "journal",
    "authors": [{ "surname": "Singer", "given_names": "M." }],
    "article_title": "The Third International Consensus Definitions for Sepsis and Septic
Shock (Sepsis-3)",
    "source": "JAMA",
    "year": 2016,
    "volume": "315",
    "issue": "8",
    "fpage": "801",
    "lpage": "810",
    "doi": "10.1001/jama.2016.0287"
  }
]
}

```

Sample Output Payload

```

{
  "article_id": "MS-88192",
  "jats_version": "1.3",
  "generation_timestamp": "2026-09-07T09:55:00Z",
  "validation_audit": {
    "is_dtd_valid": true,
    "dtd_errors": [],
    "well_formed_xml": true
  },
  "element_inventory": {
    "section_count": 1,
    "paragraph_count": 1,
    "figure_count": 0,
    "table_count": 0,
    "formula_count": 0,
    "reference_count": 1
  },
  "jats_xml_output": "<?xml version='1.0' encoding='UTF-8'?>\n<!DOCTYPE article PUBLIC \"-
//NLM//DTD JATS (Z39.96) Journal Publishing DTD v1.3 20210610//EN\"
'http://jats.nlm.nih.gov/publishing/1.3/JATS-journalpublishing1-3.dtd'\n<article article-
type='research-article' dtd-version='1.3' xml:lang='en\"
xmlns:xlink='http://www.w3.org/1999/xlink' xmlns:mml='http://www.w3.org/1998/Math/MathML'>\n
<front>\n  <journal-meta>\n    <journal-id journal-id-type='nlm-ta'>J Clin Transl
Epidemiol</journal-id>\n    <journal-title-group>\n      <journal-title>Journal of Clinical
& Translational Epidemiology</journal-title>\n    </journal-title-group>\n    <issn pub-
type='epub'>2667-114X</issn>\n    <publisher>\n      <publisher-name>Elsevier Academic
Press</publisher-name>\n    </publisher>\n  </journal-meta>\n  <article-meta>\n
<article-id pub-id-type='doi'>10.1016/j.jcte.2026.109412</article-id>\n    <title-group>\n
<article-title>Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A
Retrospective Multicenter Cohort Study</article-title>\n    </title-group>\n    <contrib-
group>\n      <contrib contrib-type='author' corresp='yes'>\n        <contrib-id contrib-id-
type='orcid'>https://orcid.org/0000-0002-1849-0921</contrib-id>\n        <name>\n
<surname>Park</surname>\n          <given-names>Jin-Woo</given-names>\n        </name>\n

```

```

<xref ref-type="aff" rid="aff1"/>\n          </contrib>\n          </contrib-group>\n          <aff
id="aff1">\n          <institution content-type="dept">Department of Surgical
Sciences</institution>, \n          <institution content-type="university">Vanderbilt University
Medical Center</institution>, \n          <city>Nashville</city>, \n          <country>United
States</country>\n          </aff>\n          <author-notes>\n          <corresp id="cor1">Corresponding
author: <email>jw.park@vumc.org</email></corresp>\n          </author-notes>\n          <pub-date date-
type="pub" publication-format="electronic">\n          <day>15</day>\n          <month>09</month>\n
<year>2026</year>\n          </pub-date>\n          <volume>14</volume>\n          <issue>3</issue>\n
<fpage>102</fpage>\n          <lpag>115</lpag>\n          <abstract>\n          <p>Early identification
of surgical sepsis remains challenging. We developed SepsisXGB across 14,280 electronic health
records...</p>\n          </abstract>\n          <kwd-group kwd-group-type="author">\n
<kwd>Sepsis</kwd>\n          <kwd>Machine Learning</kwd>\n          <kwd>Risk Stratification</kwd>\n
</kwd-group>\n          </article-meta>\n          </front>\n          <body>\n          <sec id="sec1" sec-type="intro">\n
<title>Introduction</title>\n          <p>Surgical sepsis is a leading cause of preventable in-
hospital mortality <xref ref-type="bibr" rid="ref1">[1]</xref>.</p>\n          </sec>\n          </body>\n
<back>\n          <ref-list>\n          <title>References</title>\n          <ref id="ref1">\n
<element-citation publication-type="journal">\n          <person-group person-group-
type="author">\n          <name>\n          <surname>Singer</surname>\n
<given-names>M.</given-names>\n          </name>\n          </person-group>\n
<article-title>The Third International Consensus Definitions for Sepsis and Septic Shock
(Sepsis-3)</article-title>\n          <source>JAMA</source>\n          <year>2016</year>\n
<volume>315</volume>\n          <issue>8</issue>\n          <fpage>801</fpage>\n
<lpag>810</lpag>\n          <pub-id pub-id-type="doi">10.1001/jama.2016.0287</pub-id>\n
</element-citation>\n          </ref>\n          </ref-list>\n          </back>\n          </article>
}

```

7. MetadataExtractorAndDoiAgent

7.1 Purpose & Architectural Role

The **MetadataExtractorAndDoiAgent** serves as the final distribution engine of the editorial pipeline. It extracts, standardizes, validates, and transforms manuscript publication metadata into international registry payloads:

120. Metadata Verification & Normalization:

- Validates Digital Object Identifiers (DOIs) against registration prefix standards (10.XXXX/...).
- Verifies author Open Researcher and Contributor IDs (ORCIDiDs) using the ISO 7064 Mod 11,2 checksum algorithm.
- Normalizes funder names and IDs against the Crossref Open Funder Registry.
- Enforces Open Access licensing statements (e.g., Creative Commons CC BY 4.0).

121. **Crossref Deposit Schema Generation (v5.3.1):** Constructs a fully compliant Crossref DOI deposit XML batch, complete with `<head>`, `<journal>`, `<journal_article>`, `<doi_data>`, funder assertions, and structured `<citation_list>` blocks for reference linking.

122. **Dublin Core (DCMI) Indexing Generation:** Assembles Dublin Core XML/RDF metadata for library repository ingestion and OAI-PMH harvesting.

123. **JSON-LD `ScholarlyArticle` Generation:** Creates structured schema.org markup embedded in HTML headers for discovery by Google Scholar, PubMed, and web crawlers.

7.2 Model Configurations

```
{
  "model": "gemini-2.0-flash",
  "generation_config": {
    "temperature": 0.0,
    "top_p": 0.8,
    "top_k": 32,
    "max_output_tokens": 8192,
    "response_mime_type": "application/json"
  },
  "safety_settings": [
    {
      "category": "HARM_CATEGORY_HARASSMENT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HATE_SPEECH",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_SEXUALLY_EXPLICIT",
      "threshold": "BLOCK_ONLY_HIGH"
    },
    {
      "category": "HARM_CATEGORY_DANGEROUS_CONTENT",
      "threshold": "BLOCK_ONLY_HIGH"
    }
  ]
}
```

7.3 Google AI Studio System Instructions

You are the Senior Metadata Specialist and Persistent Identifier Engineer for a scientific publishing house. Your role is to validate article publication metadata and generate production-ready Crossref Deposit XML (v5.3.1), Dublin Core (DCMI) XML, and schema.org JSON-LD structured data.

You must rigorously adhere to the following metadata engineering standards:

1. Persistent Identifier & Contributor Validation:

- Verify that all DOIs conform to Crossref syntax: prefix starting with '10.' followed by 4 or 5 digits, a forward slash, and an alphanumeric suffix (e.g., `10.1016/j.jcte.2026.109412`).
- Validate ORCID identifiers: ensure 16 digits formatted as 4 groups of 4 digits separated by hyphens (e.g., `0000-0002-1849-0921`), where the final character may be 'X'.
- Ensure the first author is designated with `sequence="first"` in Crossref metadata, with all other authors designated as `sequence="additional"`.

2. Crossref Deposit XML Construction (Schema 5.3.1):

- Root element must be: `- ``: Include unique `` , UNIX timestamp in `` , ``

```
(`<depositor_name>`, `<email_address>`), and `<registrant>`.
- `<body>`: Include `<journal>`, `<journal_metadata>` (titles, ISSN), `<journal_issue>`, and
`<journal_article_publication_type="full_text">`.
- Include `<crossmark>` assertions where applicable, funder identifiers under `<fr:program
name="fundref">`, and license statements under `<ai:program name="AccessIndicators">`.
- Generate structured `<citation_list>` entries with `<citation key="...">` containing
reference details and resolved DOIs.
```

3. Dublin Core (DCMI) XML/RDF Construction:

- Output valid Dublin Core XML with namespace `xmlns:dc="http://purl.org/dc/elements/1.1/"`.
- Map:
 - * `dc:title`: Article full title
 - * `dc:creator`: Authors in "Surname, Given Name" order
 - * `dc:subject`: Keywords and classification codes
 - * `dc:description`: Abstract text
 - * `dc:publisher`: Journal publisher
 - * `dc:date`: ISO 8601 publication date
 - * `dc:type`: "Text / ScholarlyArticle"
 - * `dc:identifier`: Persistent DOI and landing URL
 - * `dc:language`: "eng"
 - * `dc:rights`: License description (e.g., Creative Commons Attribution 4.0 International)

4. Schema.org JSON-LD Structured Data:

- Generate `@context: "https://schema.org"` with `@type: "ScholarlyArticle"`.
- Include `@id` (DOI URL), `headline`, `author` array with `@type: "Person"` and `affiliation`, `datePublished`, `isPartOf` (Periodical / Journal), and `publisher`.

Output must be fully populated without placeholders or truncation, formatted strictly according to the designated JSON schema.

7.4 Function Calling / Tools Declarations (OpenAPI 3.0 Format)

```
[
  {
    "name": "verify_crossref_doi_syntax",
    "description": "Validates that a candidate DOI adheres to the official Crossref structural
    syntax and registrant prefix allocation rules.",
    "parameters": {
      "type": "object",
      "properties": {
        "doi": {
          "type": "string",
          "description": "Candidate DOI string (e.g., '10.1016/j.jcte.2026.109412')."
        },
        "publisher_prefix": {
          "type": "string",
          "description": "Allocated registrant prefix (e.g., '10.1016')."
        }
      },
      "required": ["doi", "publisher_prefix"]
    }
  },
  {
    "name": "validate_orcid_checksum",
```

```

    "description": "Validates the mathematical ISO 7064 Mod 11,2 check digit for an author ORCID
string.",
    "parameters": {
      "type": "object",
      "properties": {
        "orcid_id": {
          "type": "string",
          "description": "16-digit hyphenated ORCID identifier (e.g., '0000-0002-1849-0921')."
        }
      },
      "required": ["orcid_id"]
    },
  },
  {
    "name": "simulate_crossref_deposit_validation",
    "description": "Validates a generated Crossref deposit XML payload against the official
Crossref schema endpoint.",
    "parameters": {
      "type": "object",
      "properties": {
        "deposit_xml_payload": {
          "type": "string",
          "description": "The complete Crossref XML string to validate."
        }
      },
      "required": ["deposit_xml_payload"]
    }
  }
]

```

7.5 Response Schema (Structured JSON Schema)

```

{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "MetadataExtractorAndDoiResponse",
  "type": "object",
  "properties": {
    "article_doi": { "type": "string" },
    "validation_timestamp": { "type": "string", "format": "date-time" },
    "metadata_verification_audit": {
      "type": "object",
      "properties": {
        "doi_syntax_valid": { "type": "boolean" },
        "orcids_verified": { "type": "boolean" },
        "license_verified": { "type": "boolean" },
        "funder_registry_mapped": { "type": "boolean" },
        "validation_warnings": { "type": "array", "items": { "type": "string" } }
      },
    },
    "required": [
      "doi_syntax_valid",
      "orcids_verified",
      "license_verified",
      "funder_registry_mapped",
      "validation_warnings"
    ]
  }
}

```

```

    },
    "crossref_deposit_xml": {
      "type": "string",
      "description": "Complete, valid Crossref v5.3.1 DOI deposit XML batch payload."
    },
    "dublin_core_xml": {
      "type": "string",
      "description": "Standard Dublin Core (DCMI) XML/RDF record for library repository harvesting."
    },
    "json_ld_schema": {
      "type": "object",
      "description": "Schema.org ScholarlyArticle JSON-LD structured data payload for web crawlers."
    }
  },
  "required": [
    "article_doi",
    "validation_timestamp",
    "metadata_verification_audit",
    "crossref_deposit_xml",
    "dublin_core_xml",
    "json_ld_schema"
  ]
}

```

7.6 Sample Input & Output Payloads

Sample Input Payload

```

{
  "article_details": {
    "doi": "10.1016/j.jcte.2026.109412",
    "title": "Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A Retrospective Multicenter Cohort Study",
    "journal_name": "Journal of Clinical & Translational Epidemiology",
    "issn": "2667-114X",
    "publisher": "Elsevier Academic Press",
    "publication_date": "2026-09-15",
    "volume": "14",
    "issue": "3",
    "first_page": "102",
    "last_page": "115",
    "resource_url": "https://www.sciencedirect.com/science/article/pii/S2667114X26000412",
    "license_url": "http://creativecommons.org/licenses/by/4.0/",
    "abstract": "Early identification of surgical sepsis remains challenging. We developed an ensemble gradient boosted model (SepsisXGB)...",
    "keywords": ["Sepsis", "Machine Learning", "Risk Stratification"],
    "authors": [
      {
        "surname": "Park",
        "given_names": "Jin-Woo",
        "orcid": "0000-0002-1849-0921",
        "affiliation": "Vanderbilt University Medical Center",
        "sequence": "first"
      }
    ]
  }
}

```

```

    },
    {
      "surname": "Rostova",
      "given_names": "Elena",
      "orcid": "0000-0001-9281-301X",
      "affiliation": "Vanderbilt University Medical Center",
      "sequence": "additional"
    }
  ],
  "funding": [
    {
      "funder_name": "National Institute of General Medical Sciences",
      "funder_id": "10.13039/1000000057",
      "award_number": "R01-GM149201"
    }
  ]
}

```

Sample Output Payload

```

{
  "article_doi": "10.1016/j.jcte.2026.109412",
  "validation_timestamp": "2026-09-07T10:00:00Z",
  "metadata_verification_audit": {
    "doi_syntax_valid": true,
    "orcids_verified": true,
    "license_verified": true,
    "funder_registry_mapped": true,
    "validation_warnings": []
  },
  "crossref_deposit_xml": "<?xml version='1.0' encoding='UTF-8'?>\n<doi_batch version='5.3.1'
xmlns='http://www.crossref.org/schema/5.3.1' xmlns:xsi='http://www.w3.org/2001/XMLSchema-
instance' xmlns:fr='http://www.crossref.org/fundref.xsd'
xmlns:ai='http://www.crossref.org/AccessIndicators.xsd'
xsi:schemaLocation='http://www.crossref.org/schema/5.3.1
http://www.crossref.org/schemas/crossref5.3.1.xsd'?>\n  <head>\n    <doi_batch_id>batch-2026-09-
07-jcte-109412</doi_batch_id>\n    <timestamp>1788775200</timestamp>\n    <depositor>\n
<depositor_name>Elsevier Academic Press Metadata Ops</depositor_name>\n
<email_address>meta-deposits@elsevier.com</email_address>\n    </depositor>\n
<registrant>Elsevier Academic Press</registrant>\n  </head>\n  <body>\n    <journal>\n
<journal_metadata language='en'>\n      <full_title>Journal of Clinical & Translational
Epidemiology</full_title>\n      <abbreviated_title>J Clin Transl
Epidemiol</abbreviated_title>\n      <issn media_type='electronic'>2667-114X</issn>\n
</journal_metadata>\n      <journal_issue>\n        <publication_date media_type='online'>\n
<month>09</month>\n        <year>2026</year>\n      </publication_date>\n
<journal_volume><volume>14</volume></journal_volume>\n        <issue>3</issue>\n
</journal_issue>\n      <journal_article publication_type='full_text'>\n        <titles>\n
<title>Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A Retrospective
Multicenter Cohort Study</title>\n        </titles>\n        <contributors>\n
<person_name sequence='first' contributor_role='author'>\n          <given_name>Jin-
Woo</given_name>\n          <surname>Park</surname>\n
<ORCID>https://orcid.org/0000-0002-1849-0921</ORCID>\n        </person_name>\n
<person_name sequence='additional' contributor_role='author'>\n
<given_name>Elena</given_name>\n          <surname>Rostova</surname>\n
<ORCID>https://orcid.org/0000-0001-9281-301X</ORCID>\n        </person_name>\n
</contributors>\n      <publication_date media_type='online'>\n        <day>15</day>\n

```

```

<month>09</month>\n          <year>2026</year>\n          </publication_date>\n          <pages>\n
<first_page>102</first_page>\n          <last_page>115</last_page>\n          </pages>\n
<fr:program name="fundref">\n          <fr:assertion name="fundgroup">\n
<fr:assertion name="funder_name">National Institute of General Medical Sciences</fr:assertion
name="funder_identifier">10.13039/100000057</fr:assertion></fr:assertion>\n
<fr:assertion name="award_number">R01-GM149201</fr:assertion>\n          </fr:assertion>\n
</fr:program>\n          <ai:program name="AccessIndicators">\n          <ai:license_ref
applies_to="vor">http://creativecommons.org/licenses/by/4.0/</ai:license_ref>\n
</ai:program>\n          <doi_data>\n          <doi>10.1016/j.jcte.2026.109412</doi>\n
<resource>https://www.sciencedirect.com/science/article/pii/S2667114X26000412</resource>\n
</doi_data>\n          </journal_article>\n          </journal>\n          </body>\n</doi_batch>",
  "dublin_core_xml": "<?xml version='1.0' encoding='UTF-8'?>\n<metadata
xmlns:xsi='http://www.w3.org/2001/XMLSchema-instance'
xmlns:dc='http://purl.org/dc/elements/1.1/'>\n  <dc:title>Machine Learning-Assisted Risk
Stratification for Post-Surgical Sepsis: A Retrospective Multicenter Cohort Study</dc:title>\n
<dc:creator>Park, Jin-Woo</dc:creator>\n  <dc:creator>Rostova, Elena</dc:creator>\n
<dc:subject>Sepsis</dc:subject>\n  <dc:subject>Machine Learning</dc:subject>\n  <dc:subject>Risk
Stratification</dc:subject>\n  <dc:description>Early identification of surgical sepsis remains
challenging. We developed an ensemble gradient boosted model (SepsisXGB)...</dc:description>\n
<dc:publisher>Elsevier Academic Press</dc:publisher>\n  <dc:date>2026-09-15</dc:date>\n
<dc:type>Text / ScholarlyArticle</dc:type>\n
<dc:identifier>doi:10.1016/j.jcte.2026.109412</dc:identifier>\n
<dc:identifier>https://www.sciencedirect.com/science/article/pii/S2667114X26000412</dc:identifie
r>\n  <dc:language>eng</dc:language>\n
<dc:rights>http://creativecommons.org/licenses/by/4.0/</dc:rights>\n</metadata>",
  "json_ld_schema": {
    "@context": "https://schema.org",
    "@type": "ScholarlyArticle",
    "id": "https://doi.org/10.1016/j.jcte.2026.109412",
    "headline": "Machine Learning-Assisted Risk Stratification for Post-Surgical Sepsis: A
Retrospective Multicenter Cohort Study",
    "author": [
      {
        "@type": "Person",
        "name": "Jin-Woo Park",
        "identifier": "https://orcid.org/0000-0002-1849-0921",
        "affiliation": {
          "@type": "Organization",
          "name": "Vanderbilt University Medical Center"
        }
      },
      {
        "@type": "Person",
        "name": "Elena Rostova",
        "identifier": "https://orcid.org/0000-0001-9281-301X",
        "affiliation": {
          "@type": "Organization",
          "name": "Vanderbilt University Medical Center"
        }
      }
    ],
    "datePublished": "2026-09-15",
    "inLanguage": "en",
    "isPartOf": {
      "@type": "Periodical",
      "name": "Journal of Clinical & Translational Epidemiology",
      "issn": "2667-114X"
    },
    "publisher": {
      "@type": "Organization",

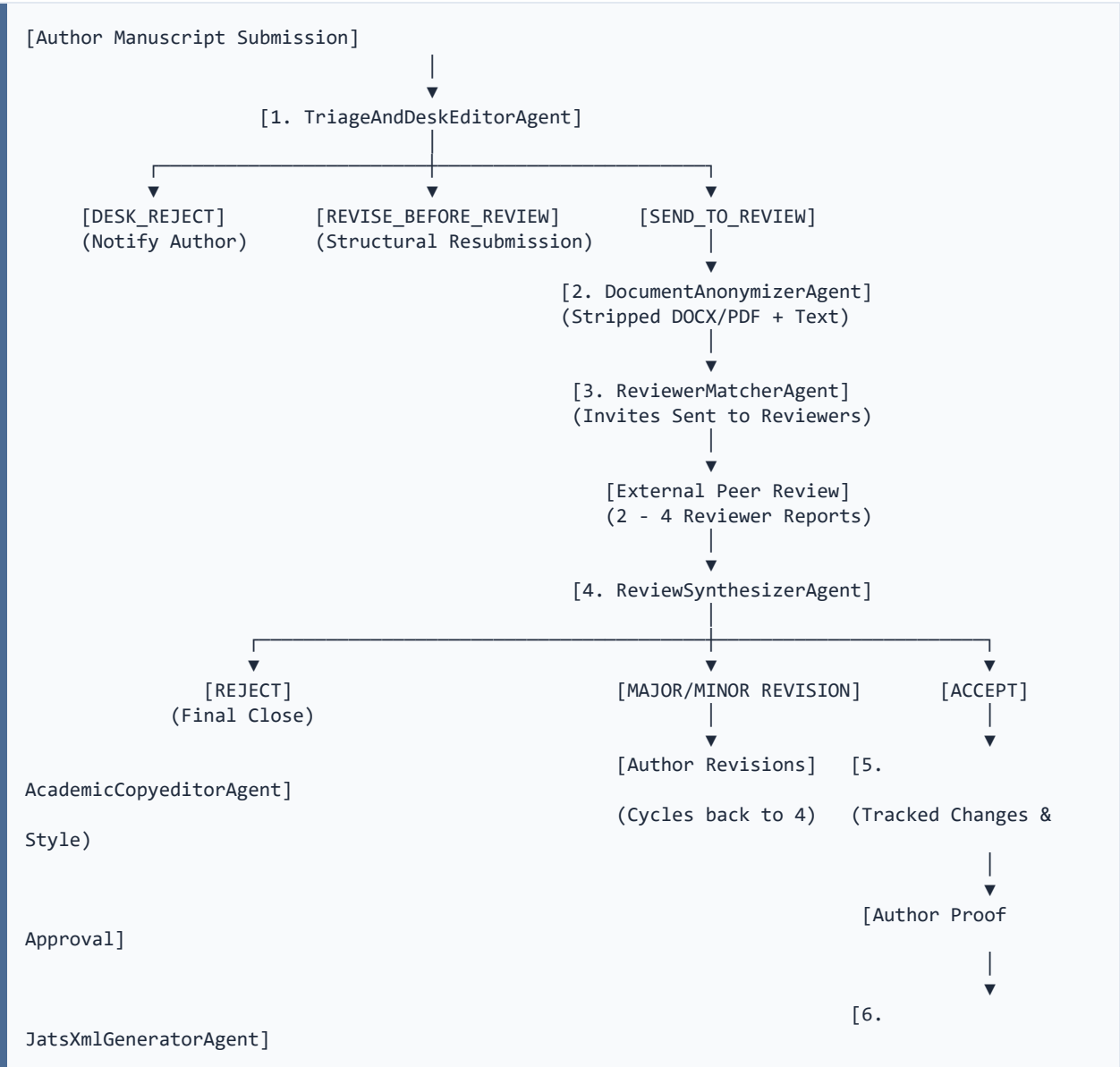
```

```
    "name": "Elsevier Academic Press"
  },
  "license": "http://creativecommons.org/licenses/by/4.0/"
}
```

8. End-to-End Orchestration Architecture & Implementation Guide

8.1 Pipeline DAG Workflow & State Machine

The 7 core editorial agents operate as a cohesive Directed Acyclic Graph (DAG) managed by an orchestration layer. Below is the lifecycle flow:



MetadataExtractorAndDoiAgent]

Core + DOI)

Article]

(NISO JATS v1.3 XML)



[7.

(Crossref XML + Dublin



[Published & Indexed

8.2 Production Python SDK Integration Example (Google GenAI SDK)

The following production-ready Python snippet illustrates how to instantiate and call an agent (e.g., `TriageAndDeskEditorAgent`) using the official `google-genai` SDK with strict JSON schema enforcement:

```
import os
from google import genai
from google.genai import types

# Initialize the Gemini client
client = genai.Client(api_key=os.environ.get("GEMINI_API_KEY"))

# Define the structured response schema (Pydantic or JSON schema dict)
response_schema = {
    "type": "object",
    "properties": {
        "manuscript_id": {"type": "string"},
        "editorial_recommendation": {
            "type": "object",
            "properties": {
                "decision": {
                    "type": "string",
                    "enum": ["DESK_REJECT", "REVISE_BEFORE_REVIEW", "SEND_TO_REVIEW"]
                },
                "confidence_score": {"type": "number"},
                "executive_summary": {"type": "string"}
            },
            "required": ["decision", "confidence_score", "executive_summary"]
        }
    },
    "required": ["manuscript_id", "editorial_recommendation"]
}

# Define the model generation configuration
config = types.GenerateContentConfig(
    temperature=0.2,
    top_p=0.85,
    response_mime_type="application/json",
    response_schema=response_schema,
    system_instruction=(
        "You are the Senior Editorial Desk Editor for an internationally recognized peer-reviewed journal. "
```

```

        "Evaluate newly submitted manuscripts for scope, reporting integrity, and ethics."
    ),
    safety_settings=[
        types.SafetySetting(
            category=types.HarmCategory.HARM_CATEGORY_HARASSMENT,
            threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
        ),
        types.SafetySetting(
            category=types.HarmCategory.HARM_CATEGORY_HATE_SPEECH,
            threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
        ),
        types.SafetySetting(
            category=types.HarmCategory.HARM_CATEGORY_SEXUALLY_EXPLICIT,
            threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
        ),
        types.SafetySetting(
            category=types.HarmCategory.HARM_CATEGORY_DANGEROUS_CONTENT,
            threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
        )
    ]
)

# Execute the content generation call
response = client.models.generate_content(
    model="gemini-1.5-pro",
    contents="Manuscript Submission Payload: [Insert JSON / Manuscript Text here]",
    config=config
)

print(response.text)

```

8.3 Quality Assurance & Deployment Checklist

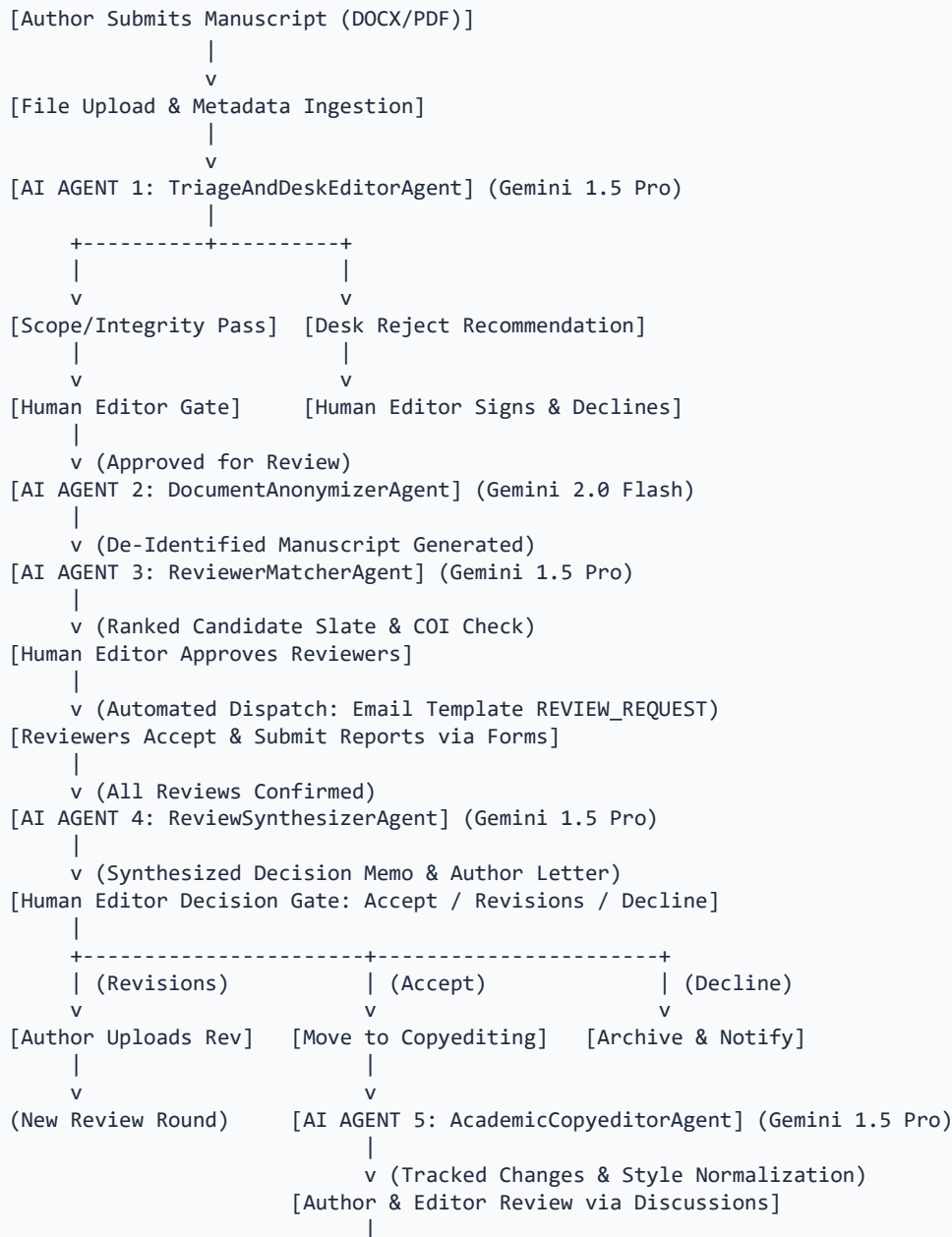
Before promoting the pipeline into active production:

124. **Schema Validation:** Automated unit tests must validate generated JSON payloads against JSON Schema Draft-07.
125. **JATS DTD Validation:** An automated xmllint or Saxon parser must execute against the NISO JATS v1.3 DTD schema on all outputs from `JatsXmlGeneratorAgent`.
126. **Crossref Testbed Ingestion:** Crossref deposit XML must be submitted to the Crossref test environment (test.crossref.org) before triggering production DOI minting.
127. **Human-in-the-Loop Escalation Gates:**
 - `TriageAndDeskEditorAgent`: Requires Editor-in-Chief concurrence before dispatching a `DESK_REJECT` notification.
 - `ReviewSynthesizerAgent`: Requires Associate Editor approval before issuing a `REJECT` or `MAJOR_REVISION` decision letter to authors.
 - `AcademicCopyeditorAgent`: Authors must accept or reject tracked changes before final XML transformation.

Chapter 9: End-to-End Orchestration Blueprint, API Endpoints & Python SDK Implementation Guide

9.1 Multi-Agent Orchestration Architecture (DAG Workflow)

The autonomous journal system executes as a stateful, event-driven Directed Acyclic Graph (DAG) managed by an asynchronous workflow coordinator. Every node in the DAG represents either an AI Agent invocation (via Google AI Studio / Gemini API), an automated persistence operation, or a human-in-the-loop decision gate.



```

        v (Approved)
    [Move to Production]
        |
        v
    [AI AGENT 6: JatsXmlGeneratorAgent] (Gemini 1.5 Pro)
        |
        v (Valid NISO JATS XML 1.3 Generated)
    [Automated PDF/HTML Galley Compilation]
        |
        v
    [AI AGENT 7: MetadataExtractorAndDoiAgent] (Gemini 2.0 Flash)
        |
        v (Crossref XML & JSON-LD Extracted)
    [Issue Assembly & TOC Drag-and-Drop Ordering]
        |
        v
    [Human Manager Clicks "Publish Issue"]
        |
        v (Dispatches ISSUE_PUBLISHED_ANNOUNCE)
    [Deposit DOI to Crossref & Index via OAI-PMH]

```

9.2 Python SDK Reference Implementation (Google GenAI SDK)

Below is the complete, production-grade Python implementation utilizing the official [google-genai](#) SDK. It demonstrates how to initialize the Gemini client, configure safety settings, enforce structured JSON output schemas, and orchestrate the editorial agents.

```

"""
AI-OJS: Autonomous Journal Orchestration Engine
Built with Google AI Studio and the official google-genai SDK.
"""

import os
import json
import logging
from typing import Dict, Any, List, Optional
from pydantic import BaseModel, Field
from google import genai
from google.genai import types

# Configure Logging
logging.basicConfig(level=logging.INFO, format="%(asctime)s [%(levelname)s] %(message)s")
logger = logging.getLogger("AI-OJS-Orchestrator")

# Initialize Google GenAI Client
# Ensure GEMINI_API_KEY is configured in your environment
client = genai.Client(api_key=os.environ.get("GEMINI_API_KEY"))

# Standard Academic Freedom Safety Settings
ACADEMIC_SAFETY_SETTINGS = [
    types.SafetySetting(
        category=types.HarmCategory.HARM_CATEGORY_HARASSMENT,

```

```

        threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
    ),
    types.SafetySetting(
        category=types.HarmCategory.HARM_CATEGORY_HATE_SPEECH,
        threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
    ),
    types.SafetySetting(
        category=types.HarmCategory.HARM_CATEGORY_SEXUALLY_EXPLICIT,
        threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
    ),
    types.SafetySetting(
        category=types.HarmCategory.HARM_CATEGORY_DANGEROUS_CONTENT,
        threshold=types.HarmBlockThreshold.BLOCK_ONLY_HIGH
    ),
]

# =====
# PYDANTIC DATA SCHEMAS FOR STRUCTURED OUTPUTS
# =====

class DeskTriageOutput(BaseModel):
    overall_recommendation: str = Field(description="DESK_REJECT, REVISE_BEFORE_REVIEW, or SEND_TO_REVIEW")
    confidence_score: float = Field(ge=0.0, le=1.0)
    scope_alignment_score: int = Field(ge=1, le=10)
    methodological_soundness_score: int = Field(ge=1, le=10)
    ethical_compliance_verified: bool
    ithenticate_risk_level: str = Field(description="LOW, MODERATE, HIGH, CRITICAL")
    confidential_editor_notes: str
    author_facing_feedback: str

class AnonymizationOutput(BaseModel):
    is_fully_anonymized: bool
    residual_pii_risk_score: float = Field(ge=0.0, le=1.0)
    redaction_count: int
    redacted_entities: List[Dict[str, str]]
    anonymized_manuscript_text: str

class ReviewCandidate(BaseModel):
    reviewer_id: str
    full_name: str
    email: str
    institutional_affiliation: str
    semantic_relevance_score: float
    active_workload: int
    days_since_last_review: int
    coi_detected: bool
    coi_justification: Optional[str] = None

class ReviewerMatchingOutput(BaseModel):
    submission_id: str
    target_section: str
    recommended_candidates: List[ReviewCandidate]

```

```

coi_excluded_candidates: List[ReviewCandidate]

class SynthesisOutput(BaseModel):
    editorial_recommendation: str = Field(description="ACCEPT, MINOR_REVISION, MAJOR_REVISION, REJECT")
    consensus_level: str = Field(description="UNANIMOUS, STRONG_CONSENSUS, SPLIT, CONFLICTED")
    methodological_validity: str
    essential_revisions: List[str]
    advisory_revisions: List[str]
    confidential_eic_memo: str
    author_decision_letter: str

# =====
# CORE AGENT INVOCATION ENGINE
# =====

class JournalAgentService:
    @staticmethod
    def run_desk_triage(manuscript_text: str, journal_scope: str) -> DeskTriageOutput:
        """Invokes TriageAndDeskEditorAgent using Gemini 1.5 Pro."""
        logger.info("Executing TriageAndDeskEditorAgent...")
        system_instruction = (
            "You are the Senior Managing Editor and Triage Specialist for an international academic journal. "
            "Evaluate incoming manuscripts for journal scope alignment, adherence to EQUATOR reporting guidelines "
            "(CONSORT, PRISMA, STROBE), IRB/ethical declarations, and potential plagiarism. Provide a deterministic desk "
            "recommendation: DESK_REJECT, REVISE_BEFORE_REVIEW, or SEND_TO_REVIEW."
        )

        prompt = f"""
JOURNAL SCOPE & POLICY:
{journal_scope}

MANUSCRIPT CONTENT FOR TRIAGE:
{manuscript_text}
"""

        response = client.models.generate_content(
            model="gemini-1.5-pro",
            contents=prompt,
            config=types.GenerateContentConfig(
                system_instruction=system_instruction,
                temperature=0.2,
                top_p=0.95,
                response_mime_type="application/json",
                response_schema=DeskTriageOutput,
                safety_settings=ACADEMIC_SAFETY_SETTINGS
            )
        )
        return DeskTriageOutput.model_validate_json(response.text)

    @staticmethod
    def run_document_anonymization(raw_text: str) -> AnonymizationOutput:

```

```

        """Invokes DocumentAnonymizerAgent using Gemini 2.0 Flash."""
        logger.info("Executing DocumentAnonymizerAgent for double-blind review...")
        system_instruction = (
            "You are an expert De-Identification and Academic Anonymization Agent. Scrub all
direct and indirect "
            "identifying information from the manuscript to enforce double-blind peer review.
Redact author names, "
            "affiliations, contact emails, specific lab grants, repository usernames, and self-
referential citations. "
            "Replace redacted names with [ANONYMIZED_AUTHOR] and institutions with
[ANONYMIZED_INSTITUTION].")
        )

        response = client.models.generate_content(
            model="gemini-2.0-flash",
            contents=raw_text,
            config=types.GenerateContentConfig(
                system_instruction=system_instruction,
                temperature=0.0,
                top_p=1.0,
                response_mime_type="application/json",
                response_schema=AnonymizationOutput,
                safety_settings=ACADEMIC_SAFETY_SETTINGS
            )
        )
        return AnonymizationOutput.model_validate_json(response.text)

    @staticmethod
    def run_review_synthesis(submission_title: str, reviews: List[Dict[str, Any]]) ->
    SynthesisOutput:
        """Invokes ReviewSynthesizerAgent using Gemini 1.5 Pro."""
        logger.info("Executing ReviewSynthesizerAgent...")
        system_instruction = (
            "You are the Associate Editor and Review Synthesis Specialist. Analyze all peer
review reports for the submission. "
            "Identify areas of consensus and conflict, resolve methodological disagreements, de-
escalate aggressive rhetoric, "
            "and draft a structured editorial decision memo and author decision letter.")
        )

        prompt = f"""
MANUSCRIPT TITLE: {submission_title}

PEER REVIEW REPORTS:
{json.dumps(reviews, indent=2)}
"""

        response = client.models.generate_content(
            model="gemini-1.5-pro",
            contents=prompt,
            config=types.GenerateContentConfig(
                system_instruction=system_instruction,
                temperature=0.2,
                top_p=0.95,
                response_mime_type="application/json",
                response_schema=SynthesisOutput,
                safety_settings=ACADEMIC_SAFETY_SETTINGS
            )
        )

```

```

        return SynthesisOutput.model_validate_json(response.text)

# =====
# INTEGRATION & EXECUTION ENTRYPOINT
# =====

if __name__ == "__main__":
    logger.info("AI-OJS System Orchestration Engine Initialized.")
    # Example mock payload
    sample_text = (
        "Title: Multi-Omics Machine Learning for Breast Cancer Molecular Subtypes\n"
        "Authors: Jane Doe, John Smith, Universiti Teknologi Malaysia\n"
        "Abstract: We propose an ensemble graph neural network combining transcriptomic and
genomic data..."
    )

    # 1. Execute Triage
    # triage_result = JournalAgentService.run_desk_triage(sample_text, "Scope: Computational
Biology & Oncology AI")
    # print(triage_result.model_dump_json(indent=2))

```

9.3 Backend REST API Endpoints Specification

The platform exposes RESTful endpoints built on FastAPI:

9.3.1 Submission Management Endpoints

- **POST /api/v1/submissions:** Initial author submission ingestion.
- **GET /api/v1/submissions/{id}:** Retrieves complete 3-column submission record.
- **POST /api/v1/submissions/{id}/assign-editor:** Assigns Section Editor with specific permission flags (**can_record_decision**, **can_edit_metadata**).
- **POST /api/v1/submissions/{id}/desk-decision:** Executes desk decisions (**SEND_TO_REVIEW**, **ACCEPT_AND_SKIP_REVIEW**, **DECLINE_SUBMISSION**).

9.3.2 Peer Review Endpoints

- **POST /api/v1/submissions/{id}/review-rounds/{round_id}/add-reviewer:** Dispatches review request with response and review deadlines.
- **GET /api/v1/submissions/{id}/review-rounds/{round_id}/status:** Fetches live status and dashboard circle icon values.
- **POST /api/v1/reviews/{review_id}/submit:** Ingests completed reviewer rubric and attachments.
- **POST /api/v1/reviews/{review_id}/confirm:** Editorial confirmation and thank-you dispatch.
- **POST /api/v1/submissions/{id}/editorial-decision:** Records final round decisions (**REQUEST_REVISIONS**, **ACCEPT**, **DECLINE**, **NEW_ROUND**).

9.3.3 Production & Publication Endpoints

- **POST /api/v1/submissions/{id}/galleys**: Uploads multi-format galleys (PDF, HTML, JATS XML).
- **POST /api/v1/issues**: Creates future or back issue containers.
- **POST /api/v1/issues/{issue_id}/schedule-article**: Assigns article to issue, section, page range, and cover image.
- **POST /api/v1/issues/{issue_id}/publish**: Executes issue publication, sends global email alerts, and registers Crossref DOIs.
- **POST /api/v1/submissions/{id}/create-version**: Creates post-publication revision branch for corrigenda/errata.

Chapter 10: Enterprise Deployment, Indexing, and Ethical Compliance Guide

10.1 Global Scholarly Indexing & Discoverability

An academic journal system is only as impactful as its discoverability across global scholarly indices, citation databases, and library discovery engines. AI-OJS integrates native support for automated metadata syndication.

10.1.1 Automated Crossref DOI Deposit Pipeline

The system integrates an automated Crossref submission daemon that compiles and transmits Crossref Metadata Deposit Schema (version 5.3.1) XML payloads via HTTPS POST to <https://api.crossref.org/deposits>:

- **Real-Time Registration:** Immediately upon issue publication or article continuous publishing release, the DOI engine deposits article metadata, author ORCID iDs, funding declarations, and structured reference lists.
- **Reference Linking:** Automatically registers outbound DOIs for references, enabling bidirectional citation tracking.
- **Version Suffixing:** Appends version-specific identifiers (e.g., [10.1234/journal.2026.001.v2](#)) when new versions are released, linked via `<isVersionOf>` Crossref relations.

10.1.2 Google Scholar Meta-Tag Engine

To ensure instantaneous indexing by Google Scholar and academic web crawlers, the public article template dynamically renders standard high-wire and Dublin Core HTML `<meta>` tags:

```
<!-- Google Scholar Highwire Meta Tags -->
<meta name="citation_title" content="Multi-Omics Machine Learning for Breast Cancer Subtype Classification">
<meta name="citation_author" content="Jane Doe">
<meta name="citation_author_institution" content="Universiti Teknologi Malaysia, Faculty of Computing">
<meta name="citation_author_orcid" content="https://orcid.org/0000-0002-1825-0097">
<meta name="citation_publication_date" content="2026/09/07">
<meta name="citation_journal_title" content="International Journal of Computational Healthcare">
<meta name="citation_volume" content="12">
<meta name="citation_issue" content="3">
<meta name="citation_firstpage" content="101">
<meta name="citation_lastpage" content="124">
<meta name="citation_doi" content="10.1016/j.comphealth.2026.104201">
<meta name="citation_pdf_url" content="https://journal.example.org/article/view/104201/pdf">
<meta name="citation_abstract_html_url"
content="https://journal.example.org/article/view/104201">
```

10.1.3 OAI-PMH Repository Harvesting Endpoint

The system exposes a compliant OAI-PMH 2.0 interface supporting:

- **Identify:** Repository identification, protocol version, and admin contact.
- **ListMetadataFormats:** Exposes `oai_dc` (Dublin Core) and `jats` (NISO JATS XML).

- **ListSets:** Partitions metadata by journal sections and issues.
 - **ListIdentifiers & ListRecords:** Supports incremental harvesting using **from** and **until** UTC timestamps.
-

10.2 Ethical Governance & COPE Compliance Standards

AI-OJS adheres strictly to the Committee on Publication Ethics (COPE) Core Practices and ICMJE guidelines:

128. **Plagiarism & Textual Recycling Screening:** Ingests iThenticate/Crossref Similarity Check scores. Any manuscript with >20% overall similarity or >5% string matching from a single source triggers mandatory human editorial review before triage progression.
 129. **Authorship & Contributor Attribution (CRediT):** Mandates explicit Contributor Roles Taxonomy (CRediT) categorization for all co-authors (e.g., Conceptualization, Methodology, Software, Formal Analysis, Writing - Original Draft).
 130. **Competing Interests & Funding Declarations:** Enforces standardized declaration questionnaires during submission.
 131. **Retraction Guidelines:** When an article is retracted:
 - The original publication is never removed from the archive.
 - A prominent watermark labeled "RETRACTED" is permanently stamped across all galley PDFs.
 - A formal, open-access Retraction Notice is published, linked bidirectionally to the original article DOI.
-

10.3 Privacy Regulations (GDPR) & Scholarly Record Retention

132. **Double-Blind Anonymity Shield:** Under the European Union General Data Protection Regulation (GDPR), the system enforces algorithmic separation of reviewer identity data. Reviewer names and contact information are strictly isolated from author-accessible database queries.
 133. **Archival Exemption (GDPR Article 17(3)(d)):** While users can request account deactivation, published academic literature and peer-reviewed records are legally exempt from the "Right to Erasure" to preserve the integrity of the permanent historical scientific record.
-

10.4 Enterprise Cloud Infrastructure & Security Hardening

- **Container Architecture:** Production deployment via Docker containers orchestrating FastAPI app servers, Celery workers, and PostgreSQL databases managed via Kubernetes.
- **Data Protection:** All manuscript assets stored in AES-256 encrypted Cloud Object Storage with time-limited pre-signed URLs for reviewer access.
- **Audit Immutability:** Audit trail records are secured using cryptographic SHA-256 forward-chaining hashes, preventing unauthorized retroactive modifications by any administrative role.

